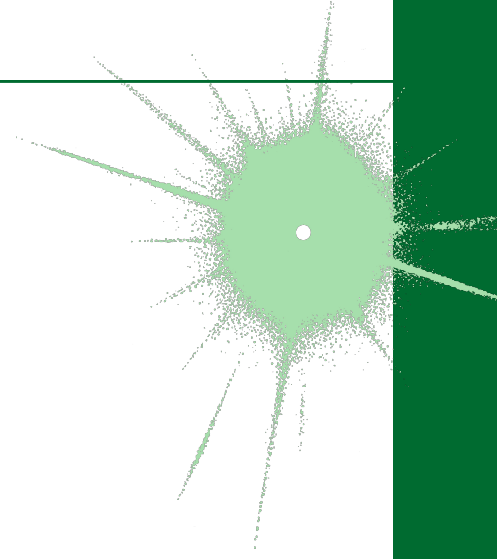


SECTION VII

Disorders of the Gastrointestinal System



QUESTIONS

VII-1. All of the following are true about specific functions of the gut EXCEPT:

- A. Gastric acid sterilizes the upper gut.
- B. Most nutrient absorption occurs in the small intestine.
- C. The pancreas secretes bicarbonate, which optimizes the pH in the intestine for enzyme activation.
- D. The stomach secretes intrinsic factor which is necessary for vitamin B₁₂ absorption.
- E. The terminal ileum is primarily responsible for stool dehydration, decreasing volumes from 1000–1500 mL to 100–200mL.

VII-2. All of the following diseases are associated with gastric acid hypersecretion EXCEPT:

- A. Duodenal ulcers
- B. G-cell hyperplasia
- C. Pernicious anemia
- D. Retained antrum syndrome
- E. Zollinger-Ellison syndrome

VII-3. Which of the following gastrointestinal conditions is characterized by noninflammatory visceral pain?

- A. Appendicitis
- B. Cholecystitis
- C. Inflammatory bowel disease
- D. Mesenteric ischemia
- E. Peptic ulcer

VII-4. Which of the following statements regarding gastrointestinal symptoms is TRUE?

- A. Diarrhea from malabsorption usually improves with fasting, whereas secretory diarrhea persists without oral intake.
- B. Sudden awakening from a sound sleep by pain suggests functional rather than organic disease.
- C. Symptoms from mechanical obstruction, ischemia, inflammatory bowel disease, and functional bowel disorders are alleviated by meals.
- D. Ulcer pain is typically acute in onset.
- E. Ulcer symptoms are usually made worse by eating.

VII-5. A 57-year-old female wishes to undergo a screening colonoscopy for colon cancer. She has no family history of colon cancer and currently has no symptoms referable to the gastrointestinal tract. Which of the following statements is true about colonoscopy as a screening test for colon cancer?

- A. Barium enema is as sensitive as colonoscopy for detecting colitis.
- B. Colonoscopy remains the gold standard for imaging the colonic mucosa.
- C. CT colonography has replaced traditional colonoscopy for many younger patients because of its ability to detect serrated polyps with greater sensitivity.
- D. Flexible sigmoidoscopy would be as effective as colonoscopy for detecting colon cancer in this patient.
- E. The cecum can only be reached in 75% of colonoscopies.

VII-6. Which of the following endoscopic procedures would warrant antibiotic prophylaxis?

- A. Endoscopic procedures in patients with prosthetic joints
- B. Endoscopic procedures in patients with synthetic vascular grafts
- C. Endoscopic ultrasound with fine-needle aspiration in a patient with a lower gastrointestinal tract lesion
- D. Endoscopic retrograde cholangiopancreatography with anticipated complete drainage in a patient with biliary obstruction in the absence of cholangitis
- E. Percutaneous endoscopic feeding tube placement

VII-7. All of the following are indications for urgent endoscopy for acute upper gastrointestinal bleeding EXCEPT:

- A. Bloody nasogastric aspirate that does not clear with large volume lavage
- B. Large volume initial bleed
- C. Orthostatic hypotension
- D. Resting hypotension
- E. Severe pain

VII-8. Mr. Tyler, a 45-year-old male with known chronic alcohol use, presents with two episodes of hematemesis. He recently sustained a fall and injured his knee. He has been taking ibuprofen several times a day for the last week for pain control. His initial hemoglobin is 10 g/dL. He receives 2 L of IV fluid and undergoes an urgent upper endoscopy. All of the following are true about his risk of rebleeding without intervention EXCEPT:

- A. A clean-based ulcer is associated with a low risk (3–5%) of rebleeding.
- B. Adherent clots covering the ulcer base have a 90% risk of rebleeding.
- C. Flat pigmented spots covering the ulcer base have a 10% risk of rebleeding.
- D. When a fibrin plug is seen protruding from a vessel wall in the base of an ulcer, (so-called sentinel clot or visible vessel), the risk of rebleeding from the ulcer is 40%.
- E. When active spurting from an ulcer is seen, there is a 90% risk of ongoing bleeding without therapy.

VII-9. The advantages of endoscopy over barium radiography in the evaluation of dysphagia include all of the following EXCEPT:

- A. Ability to assess function and morphology
- B. Ability to intervene as well as diagnose
- C. Ability to obtain biopsy specimens
- D. Increased sensitivity for the detection of abnormalities identified by color, e.g., Barrett metaplasia
- E. Increased sensitivity for the detection of mucosal lesions

VII-10. A 47-year-old man is evaluated in the emergency department for chest pain that developed at a restaurant after swallowing a piece of steak. He reports intermittent episodes of meat getting stuck in his lower chest over the

past 3 years, but none as severe as this event. He denies food regurgitation outside of these episodes or heartburn symptoms. He is able to swallow liquids without difficulty and has not had any weight loss. Which of the following is the most likely diagnosis?

- A. Achalasia
- B. Adenocarcinoma of the esophagus
- C. Esophageal diverticula
- D. Plummer-Vinson syndrome
- E. Schatzki ring

VII-11. A 36-year-old woman with AIDS and a CD4 count of 35/ μ L presents with odynophagia and progressive dysphagia. The patient reports daily fevers and a 20-lb weight loss. The patient has been treated with clotrimazole troches without relief. On physical examination, she is cachectic with a body mass index of 16 and a weight of 86 lb. Her temperature is 38.2°C (100.8°F). She is noted to be orthostatic by blood pressure and pulse. Examination of the oropharynx reveals no evidence of thrush. The patient undergoes esophagogastroduodenoscopy, which reveals serpiginous ulcers in the distal esophagus without vesicles. No yellow plaques are noted. Multiple biopsies are taken that show intranuclear and intracytoplasmic inclusions in large endothelial cells and fibroblasts. What is the best treatment for this patient's esophagitis?

- A. Ganciclovir
- B. Glucocorticoids
- C. Fluconazole
- D. Foscarnet
- E. Thalidomide

VII-12. A 43-year-old man presents with 6 months of worsening dysphagia and postprandial regurgitation. He reports difficulty and pain with swallowing both liquids and solids; he has no difficulty with the initial components of swallowing but reports pain in the mid-chest region. He frequently regurgitates undigested food 20–60 minutes after eating or drinking. In the past 2 months, he has lost 15 lb. He also has had one episode of presumed pneumonia 4 months ago notable for a right lower lobe infiltrate. He has no significant past medical history, takes no medications, and does not smoke cigarettes. He works as a service

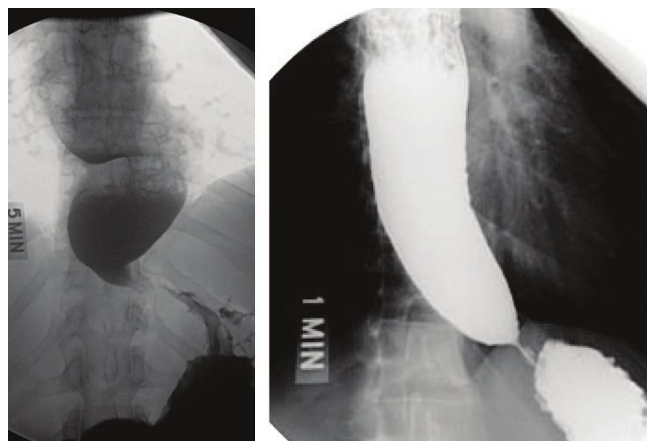


FIGURE VII-12

representative at a major electronics store and has never left the United States. Other than signs of recent weight loss, his physical examination is unremarkable. A barium swallow is performed and shown in Figure VII-12. Which of the following is the most likely cause of his disease?

- A. Autoimmune reaction to latent herpes virus
- B. Diffuse spasm on smooth muscle
- C. Infection by *Trypanosoma cruzi*
- D. Malignant growth of columnar epithelial cells
- E. Malignant growth squamous epithelial cells

VII-13. A 64-year-old man with a long history of abdominal pain, heartburn, and dyspepsia has an esophagogastroduodenoscopy to evaluate for peptic ulcer disease. No gastric or duodenal ulcers are found, but there are tongues of reddish mucosa extending proximally from the gastroesophageal junction into the esophagus. Biopsies of these areas demonstrate columnar metaplasia. All of the following statements regarding this diagnosis are true EXCEPT:

- A. Finding high-grade dysplasia mandates further intervention.
- B. High-dose proton pump inhibitor therapy will likely cause regression of the mucosal abnormalities.
- C. The incidence of these lesions has increased in the era of potent acid suppression.
- D. The patient has a high risk of coexisting cancer.
- E. The patient is at significant risk of esophageal adenocarcinoma.

VII-14. A 35-year-old man presents with progressive dysphagia over a 2-month period. He has noted that food has been getting stuck more frequently in the last 2 weeks. He has a history of intermittent gastroesophageal reflux disease for which he takes over-the-counter antacids every few weeks. He has a history of eczema as a child but is not

currently requiring treatment. He undergoes an esophagogastroduodenoscopy, which is shown in Figure VII-14A. The biopsy is shown in Figure VII-14B.

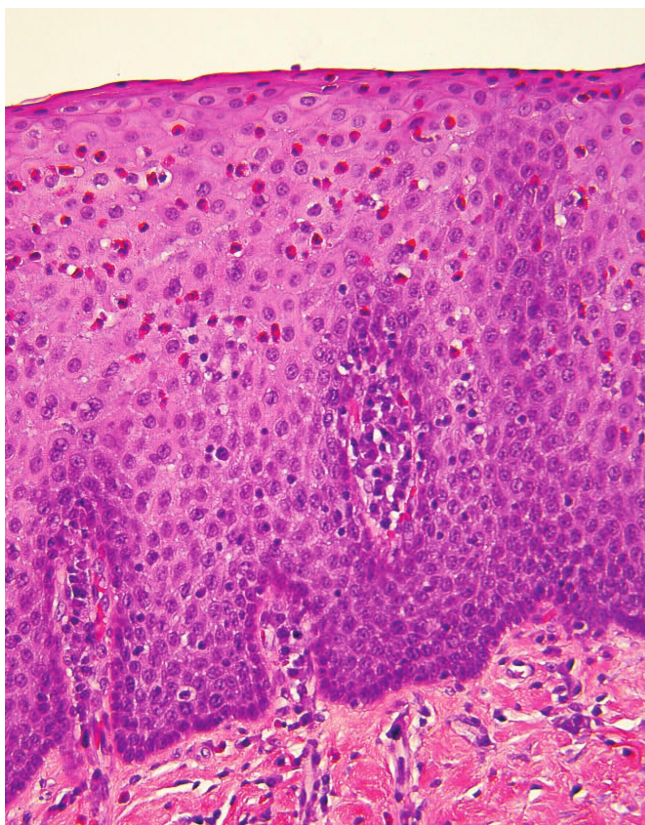


FIGURE VII-14B

Which of the following is the most appropriate first step in management?

- A. Elimination diet
- B. IV methylprednisolone
- C. Mepolizumab
- D. Trial of a proton pump inhibitor
- E. Topical corticosteroids

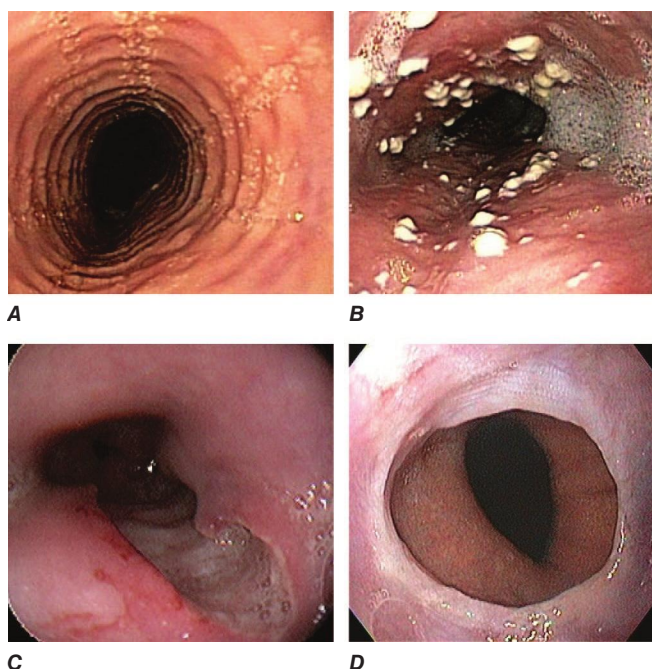


FIGURE VII-14A

VII-15. Ms. Pole is a 54-year-old woman who presents with recurrent episodes of chest discomfort at rest. She denies exertional dyspnea and reports that occasionally her pain is relieved with antacids. She works as a health care consultant and has frequent stressful battles with doctors and hospitals. An exercise cardiac stress test does not reveal any evidence of inducible ischemia. She has a barium swallow, which is shown in Figure VII-15. You suspect that she has diffuse esophageal spasm. Which of the following findings on esophageal manometry would support that diagnosis?

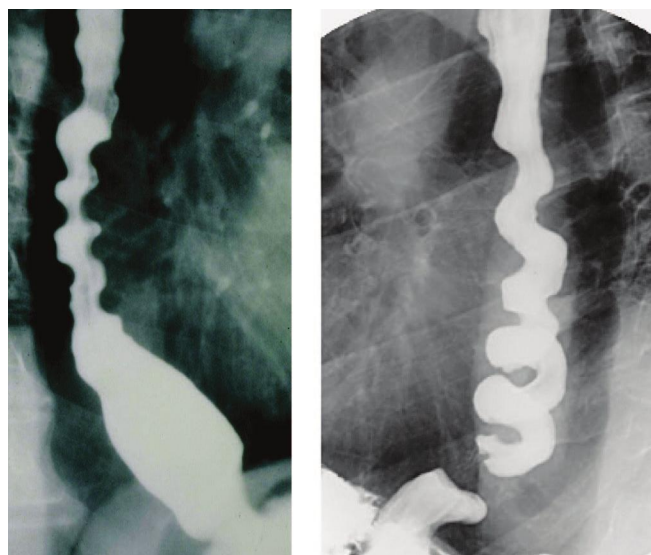


FIGURE VII-15

- A. Contractions in the distal esophagus with short latency relative to the time of the pharyngeal contraction
- B. Hypertensive peristalsis
- C. Impaired lower esophageal sphincter (LES) contraction and absent peristalsis with substantial fluid pressurization of the esophageal body
- D. Impaired LES contraction and absent peristalsis with no fluid pressurization of the esophageal body
- E. Impaired LES contraction and absent peristalsis with spastic esophageal contractions

VII-16. Which of the following statements regarding therapy for gastroesophageal reflux (GERD) therapy is TRUE?

- A. *Clostridium difficile* infection rates are decreased in patients receiving proton pump inhibitors.
- B. Histamine-2 receptor antagonists are more effective than proton pump inhibitors.
- C. The effectiveness of proton pump inhibitors increases substantially with increased dose.
- D. Proton pump inhibitors are a rare cause of interstitial nephritis.
- E. Weight loss has been proven in randomized controlled trials to ameliorate the symptoms of GERD.

VII-17. A 58-year-old man is evaluated for abdominal pain by his primary care physician. He reports severe stress at the job for the last 3 months and has since noted that he

has epigastric pain that is relieved by eating and drinking milk. He has not had food regurgitation, dysphagia, or bloody emesis or bowel movements. He denies any symptoms in his chest. Peptic ulcer disease is suspected. Which of the following statements regarding noninvasive testing for *Helicobacter pylori* is true?

- A. There is no reliable noninvasive method to detect *H. pylori*.
- B. Stool antigen testing is appropriate for proof of cure after therapy for *H. pylori*.
- C. Plasma antibodies to *H. pylori* offer the greatest sensitivity for diagnosis of infection.
- D. Exposure to low-dose radiation is a limitation to urea breath test.
- E. False-negative test results using the urea breath test may occur with recent use of nonsteroidal anti-inflammatory drugs.

VII-18. A 44-year-old woman complains of 6 months of epigastric pain that is worst between meals. She also reports symptoms of heartburn. The pain is typically relieved by over-the-counter antacid medications. She comes to the clinic after noting her stools darkening. She has no significant past medical history and takes no medications. Her physical examination is normal except for diffuse mid-epigastric pain. Her stools are heme positive. She undergoes esophagogastroduodenoscopy, which demonstrates a well-circumscribed, 2-cm duodenal ulcer that is positive for *Helicobacter pylori*. Which of the following is the recommended initial therapy given these findings?

- A. Lansoprazole, clarithromycin, and metronidazole for 14 days
- B. Pantoprazole and amoxicillin for 21 days
- C. Pantoprazole and clarithromycin for 14 days
- D. Omeprazole, bismuth, tetracycline, and metronidazole for 14 days
- E. Omeprazole, metronidazole, and clarithromycin for 7 days

VII-19. A 57-year-old man with peptic ulcer disease experiences transient improvement with *Helicobacter pylori* eradication. However, 3 months later, symptoms recur despite acid-suppressing therapy. He does not take nonsteroidal anti-inflammatory drugs. Stool analysis for *H. pylori* antigen is negative. Upper gastrointestinal (GI) endoscopy reveals prominent gastric folds together with the persistent ulceration in the duodenal bulb previously detected and the beginning of a new ulceration 4 cm proximal to the initial ulcer. Fasting gastrin levels are elevated, and basal acid secretion is 15 mEq/h. What is the best test to perform to make the diagnosis?

- A. No additional testing is necessary.
- B. Blood sampling for gastrin levels following a meal.
- C. Blood sampling for gastrin levels following secretin administration.
- D. Endoscopic ultrasonography of the pancreas.
- E. Genetic testing for mutations in the *MEN1* gene.

VII-20. Which of the following are side effects of therapies directed at peptic ulcer disease?

- A. Famotidine may have weak anti-androgenic side effects resulting in reversible gynecomastia and impotence, primarily in patients receiving high doses for prolonged periods of time (months to years).
- B. Long-term acid suppression, particularly with proton pump inhibitors, has been associated with a higher incidence of community-acquired pneumonia as well as community- and hospital-acquired *Clostridium difficile*-associated disease.
- C. Magnesium hydroxide can produce constipation and phosphate depletion.
- D. Rebound gastric acid hypersecretion has been described in *H. pylori*-negative individuals after discontinuation of H₂ blockers. It occurs even after relatively short-term usage (2 months) and may last for up to 2 months after the medication has been discontinued.
- E. The long-term use of aluminum hydroxide can lead to milk-alkali syndrome (hypercalcemia, hyperphosphatemia with possible renal calcinosis and progression to renal insufficiency).

VII-21. Which of the following statements regarding gastric acid secretion is true?

- A. Basal acid production occurs in a circadian pattern, with highest levels occurring during the morning and lowest levels during the night.
- B. Distention of the stomach wall directly activates acid secretion by parietal cells.
- C. Sight, smell, and taste of food are the components of the cephalic phase, which stimulates gastric secretion via the phrenic nerve.
- D. The gastric phase is activated once food enters the stomach. This component of secretion is driven by nutrients (amino acids and amines) that directly (via peptone and amino acid receptors) and indirectly (via stimulation of intramural gastrin releasing peptide neurons) stimulate the G cell to release gastrin, which in turn activates the parietal cell via direct and indirect mechanisms.

VII-22. A 23-year-old woman is evaluated by her primary care physician for diffuse, crampy abdominal pain. She reports that she has had abdominal pain for the last several years, but it is getting worse and is now associated with intermittent diarrhea without flatulence. This does not wake her at night. Stools do not float and are not hard to flush. She has not noted any worsening with specific foods, but she does have occasional rashes on her lower legs. She

has lost about 10 lb over the last year. She is otherwise healthy and takes no medications. Which of the following is the most appropriate recommendation at this point?

- A. Increased dietary fiber intake
- B. Measurement of antiendomysial antibody
- C. Measurement of 24-hour fecal fat
- D. Referral to gastroenterologist for endoscopy
- E. Trial of lactose-free diet

VII-23. A 54-year-old man is evaluated by a gastroenterologist for diarrhea that has been present for approximately 1 month. He reports stools that float and are difficult to flush down the toilet; these can occur at any time of day or night but seem worsened by fatty meals. In addition, he reports pain in many joints lasting days to weeks and not relieved by ibuprofen. His wife notes that the patient has had difficulty with memory for the last few months. He has lost 30 lb and reports intermittent low-grade fevers. He takes no medications and is otherwise healthy. Endoscopy is recommended. Which of the following is the most likely finding on small bowel biopsy?

- A. Dilated lymphatics
- B. Flat villi with crypt hyperplasia
- C. Mononuclear cell infiltrate in the lamina propria
- D. Normal small bowel biopsy
- E. Periodic acid-Schiff-positive macrophages containing small bacilli

VII-24. A 54-year-old man presents with 1 month of diarrhea. He states that he has 8 to 10 loose bowel movements a day. He has lost 8 lb during this time. Vital signs and physical examination are normal. Serum laboratory studies are normal. A 24-hour stool collection reveals 500 g of stool with a measured stool osmolality of 200 mOsm/L and a calculated stool osmolality of 210 mOsm/L. Based on these findings, which of the following is the most likely cause of this patient's diarrhea?

- A. Celiac sprue
- B. Chronic pancreatitis
- C. Lactase deficiency
- D. Vasoactive intestinal peptide tumor
- E. Whipple disease

VII-25. All of the following are direct functions of the intestinal epithelium **EXCEPT**:

- A. Barrier and immune defense
- B. Fluid and electrolyte absorption and secretion
- C. Production of several bioactive amines and peptides
- D. Secretion of bile acids
- E. Synthesis and secretion of apolipoproteins

VII-26. A 40-year-old male with multiple prior admissions for alcoholic pancreatitis presents for evaluation of several months of diarrhea. A 24-hour stool analysis confirms that he has steatorrhea. Which of the following mechanisms likely explains the elevation in stool fat content and resulting diarrhea?

- A. Abnormalities of intestinal lymphatics (e.g., intestinal lymphangiectasia)
- B. Decreased enterohepatic circulation of bile acids
- C. Decreased hydrolysis of triglycerides to free fatty acids, monoglycerides, and glycerol
- D. Impaired lipid absorption as a result of mucosal inflammation
- E. Increased dietary intake of medium chain triglycerides

VII-27. A 24-year-old woman is admitted to the hospital with a 1-year history of severe abdominal pain and chronic diarrhea that has been bloody for the past 2 months. She reports a 20-lb weight loss, frequent fevers, and night sweats. She denies vomiting. Her abdominal pain is crampy and primarily involves her right lower quadrant. She is otherwise healthy. Examination is concerning for an acute abdomen with rebound and guarding present. CT shows free air in the peritoneum. She is urgently taken to the operating room for surgical exploration, where she is found to have multiple strictures and a perforation of her bowel in the terminal ileum. The rectum was spared, and a fissure from the duodenum to the jejunum was found. The perforated area is resected and adhesions lysed. Which of the following findings on pathology of her resected area confirms her diagnosis?

- A. Crypt abscesses
- B. Flat villi
- C. Noncaseating granulomas throughout the bowel wall
- D. Special stain for *Clostridium difficile* toxin
- E. Transmural acute and chronic inflammation

VII-28. A 45-year-old man with ulcerative colitis has been treated for the last 5 years with infliximab with excellent resolution of his bowel symptoms and endoscopic evidence of normal colonic mucosa. He is otherwise healthy. He is evaluated by a dermatologist for a lesion that initially was a pustule over his right lower extremity but has since progressed in size with ulceration. The ulcer is moderately painful. He does not recall any trauma to the area. On examination, the ulcer measures 15 × 7 cm, and central necrosis is present. The edges of the ulcer are violaceous. No other lesions are identified. Which of the following is the most likely diagnosis?

- A. Erythema nodosum
- B. Metastatic Crohn disease
- C. Psoriasis
- D. Pyoderma gangrenosum
- E. Pyoderma vegetans

VII-29. Which of the following statements regarding the epidemiology of inflammatory bowel disease (IBD) is true?

- A. Children who receive frequent antibiotics in childhood are at lower risk of developing IBD.
- B. IBD appears with decreasing incidence in countries that are becoming more westernized.
- C. Patients who have had an appendectomy with confirmed appendicitis have a lower risk of developing ulcerative colitis.
- D. Peak incidence of ulcerative colitis and Crohn disease is in the 6th–7th decades of life.
- E. The highest incidence rates of ulcerative colitis and Crohn disease have been reported in Asia, the Middle East, and Northern Africa.

VII-30. A 36-year-old male has had Crohn disease for 5 years. His disease was initially managed with occasional tapers of oral prednisone, followed by use of oral budesonide. Over the last 6 months he has required two courses of oral prednisone and was hospitalized for worsening abdominal pain and fevers requiring IV antibiotics and corticosteroids. He has not had evidence of fistulizing disease. How should his Crohn disease be managed at this time?

- A. Begin IV cyclosporine A
- B. Provide intermittent infusions of infliximab each time he has a disease flare
- C. Start azathioprine orally at 2–3mg/kg with the potential addition of infliximab
- D. Start maintenance prednisone at 0.5mg/kg for 3 months
- E. Start monthly infusions of IV golimumab

VII-31. Which of the following patients requires no further testing before making the diagnosis of irritable bowel syndrome and initiating treatment?

- A. A 76-year-old woman with 6 months of intermittent crampy abdominal pain that is worse with stress and associated with bloating and diarrhea.
- B. A 25-year-old woman with 6 months of abdominal pain, bloating, and diarrhea that has worsened steadily and who now wakes from sleep at night to move her bowels.
- C. A 30-year-old man with 6 months of lower abdominal crampy pain relieved with bowel movements, usually loose. Symptoms are worse during the daytime at work and better on the weekend. Weight loss is not present.
- D. A 19-year-old female college student with 2 months of diarrhea and worsening abdominal pain with occasional blood in her stool.
- E. A 27-year-old woman with 6 months of intermittent abdominal pain, bloating, and diarrhea without associated weight loss. Crampy pain and diarrhea persist after a 48-hour fast.

VII-32. A 29-year-old woman comes to see you in the clinic because of abdominal discomfort. She feels abdominal discomfort on most days of the week, and the pain varies in location and intensity. She notes constipation as

well as diarrhea, but diarrhea predominates. Compared with 6 months ago, she has more bloating and flatulence than she has had before. She identifies eating and stress as aggravating factors, and her pain is relieved by defecation. You suspect irritable bowel syndrome. Laboratory data include white blood cell count 8000/ μ L, hematocrit 32%, platelets 210,000/ μ L, and erythrocyte sedimentation rate 44 mm/h. Stool studies show the presence of lactoferrin but no blood. Which of the following interventions is most appropriate at this time?

- A. Antidepressants
- B. Ciprofloxacin
- C. Colonoscopy
- D. Reassurance and patient counseling
- E. Stool bulking agents

VII-33. Which of the following statements regarding the pathophysiology of irritable bowel syndrome (IBS) is true?

- A. Abdominal distention, belching, and flatulence are most commonly due to increased amounts of gas in the gastrointestinal tract of patients with IBS.
- B. Compared with patients without IBS, IBS patients frequently exhibit blunted sensory responses to visceral stimulation.
- C. Gut dysbiosis acting in concert with genetic and environmental factors may alter intestinal permeability, increase antigen presentation resulting in mast cell activation.
- D. Almost all patients with IBS display persistent signs of low-grade mucosal inflammation with activated lymphocytes, mast cells, and enhanced expression of proinflammatory cytokines.
- E. The majority of IBS patients report a bout of gastroenteritis at the onset of their symptoms.

VII-34. A 34-year-old female with known irritable bowel syndrome has severe bouts of postprandial pain that have prevented her from being able to enjoy regular meals with her family. Which of the following therapies would be most likely to improve her discomfort?

- A. Activated charcoal taken immediately after eating a meal.
- B. An anticholinergic drug, such as dicyclomine, administered 30 minutes prior to eating.
- C. Citalopram administered once daily.
- D. Fiber supplementation with psyllium.
- E. Small doses of loperamide prior to meals.

VII-35. A 78-year-old woman is admitted to the hospital with fever, loss of appetite, and left lower quadrant pain. She is not constipated but has not moved her bowels recently. Laboratory examination is notable for an elevated white blood cell count. These symptoms began approximately 3 days ago and have steadily worsened. Which of the following statements regarding her likely condition is true?

- A. Air-fluid levels are commonly seen on plain abdominal films.
- B. Less than 25% of patients present with peritoneal signs.
- C. Lower gastrointestinal bleeding will likely be visualized on CT angiography.
- D. Thickened colonic wall is not required on CT for the diagnosis of her likely condition.
- E. Ultrasound of the pelvis is the best modality to visualize the likely pathologic process.

VII-36. A 67-year-old man is evaluated by the emergency department for blood in the toilet bowl after moving his bowels. Blood was also present on the toilet paper after wiping. He does report straining and recent constipation. He has a history of systemic hypertension and hyperlipidemia. Vital signs are normal, and he is not orthostatic. Anoscopy shows external hemorrhoids, hematocrit is normal, and bleeding does not recur during his 6-hour emergency department stay. Which of the following is the most appropriate management?

- A. Ciprofloxacin and metronidazole
- B. Cortisone suppositories and fiber supplementation
- C. Hemorrhoidal banding
- D. Operative hemorrhoidectomy
- E. Upper endoscopy

VII-37. An 88-year-old woman is brought to your clinic by her family because she has become increasingly socially withdrawn. The patient lives alone and has been reluctant to visit or be visited by her family. Family members, including seven children, also note a foul odor in her apartment and on her person. She has not had any weight loss. Alone in the examining room, she only complains of hemorrhoids. On mental status examination, she does have signs of depression. Which of the following interventions is most appropriate at this time?

- A. Head CT scan
- B. Treatment with an antidepressant medication
- C. Physical examination including genitourinary and rectal examination
- D. Screening for occult malignancy
- E. Serum thyroid-stimulating hormone

VII-38. A 59-year-old female develops acute diverticulitis and requires hospitalization for IV antibiotics. She has a CT scan that does not show evidence of abscess perforation, stricture, or fistula, and she improves in about a week. Which of the following should be undertaken to prevent further symptomatic disease?

- A. Annual colonoscopy
- B. Avoidance of nuts
- C. Daily mesalamine therapy
- D. Initiation of a high-fiber diet
- E. Subtotal colectomy to reduce the risk of perforation during a subsequent bout

VII-39. A 63-year-old man with poorly controlled type 2 diabetes presents to his primary care physician with 3 days of perianal pain, fever, and difficulty voiding. He has a history of uncomplicated stage 1 hemorrhoids, which have been treated in the past with cortisone suppositories and a high-fiber diet. On examination, he is febrile and there is a 2-cm fluctuant mass at the anal verge. Which of the following would be the most appropriate management?

- A. Excision and drainage of the mass
- B. Infrared anticoagulation
- C. Oral antibiotics and follow-up in 2 days
- D. Rubber band ligation
- E. Sclerotherapy

VII-40. A 63-year-old man with a history of diabetes and myocardial infarction was admitted to the medical intensive care unit 1 day ago with sepsis due to pneumococcal pneumonia with bacteremia. He was started on antibiotics immediately, but he initially required high doses of noradrenaline and fluids to stabilize his blood pressure. The noradrenaline was weaned off approximately 12 hours ago. Over the past 2 hours, he has had increasing abdominal pain, distension, and bloody stools. His physical examination is notable for blood pressure of 100/50, regular heart rate of 100 beats/min, respiratory rate of 22 breaths/min, and oxygen saturation of 93% on high-flow nasal oxygen. He has a diffusely tender abdomen with no audible bowel sounds. An abdominal radiograph shows multiple small bowel air-fluid levels. Which of the following is the most likely diagnosis?

- A. Arterial embolism
- B. *Clostridium difficile* colitis
- C. Inflammatory bowel disease
- D. Nonocclusive mesenteric ischemia
- E. Venous thrombosis

VII-41. A 79-year-old female with known coronary artery disease requiring previous percutaneous coronary intervention with stent placement presents to her primary care physician's office with 3 months of cramping abdominal pain after meals. She denies any change in the pain with defecation, and does not report increased flatulence or belching. She has had more frequent, loose stools but denies constipation. She has lost 15 lb during this time period. Which of the following would be the most appropriate next step in management?

- A. Noncontrast CT scan of the abdomen and pelvis
- B. Electrocardiogram to look for the presence of atrial fibrillation
- C. Initiation of a high fiber diet
- D. Initiation of an selective serotonin reuptake inhibitor medication for possible depression
- E. Mesenteric angiography

VII-42. A 74-year-old woman is 2 days post hip surgery for a fracture after a fall. Her only medication prior to admission was a calcium supplement, and she has no prior surgical history. Over the past 24 hours, she has had increasing

abdominal discomfort and distension. She received a dose of cefazolin prior to surgery but no other antibiotics. On physical examination, she is afebrile with a blood pressure of 140/80, heart rate of 110 beats/min, respiratory rate of 16 breaths/min, and oxygen saturation of 100% on 2 L of nasal oxygen. She has a distended tympanic abdomen with absent bowel sounds. There is no rebound tenderness. Her upright abdominal film is shown in Figure VII-42. Which of the following is the most likely diagnosis?



FIGURE VII-42 Reproduced with permission from Bongard FS et al: *Current Critical Care Diagnosis & Treatment*. New York: McGraw Hill, 1994.

- A. Acalculous cholecystitis
- B. Colonic pseudo-obstruction
- C. Perforated duodenal ulcer
- D. Small bowel obstruction
- E. Small bowel ileus

VII-43. A 52-year-old man presents to the emergency department with acute-onset abdominal pain that awoke him from sleep. He has a history of paroxysmal atrial fibrillation but has not had any recent episodes of atrial fibrillation on remote monitoring. He has no prior history of abdominal surgeries. His pain is 7/10 in severity, and comes in intense waves. He has vomited twice and each time his pain was improved for a few minutes afterward, but then it returned. He is afebrile, with a heart rate in the 120s that is confirmed to be sinus tachycardia by electrocardiogram. He is afebrile. His abdomen is tender and distended. He has high-pitched "musical" bowel sounds. CT of the abdomen is shown in Figure III-43. He has no evidence of gallstones or ductal dilatation on right upper quadrant ultrasound, and the pancreas appears normal on



FIGURE VII-43

CT (not shown). Labs are pending. Which of the following is the most likely cause of his symptoms?

- A. Acute mesenteric ischemia
- B. Acute cholecystitis
- C. Acute pancreatitis
- D. Small bowel obstruction
- E. Viral gastroenteritis

VII-44. A 32-year-old woman is evaluated in the emergency department for abdominal pain. She reports a vague loss of appetite for the last day and has had progressively severe abdominal pain, initially at her umbilicus, but now localized to her right lower quadrant. The pain is crampy. She has not moved her bowels or vomited. She reports that she is otherwise healthy and has had no sick contact. Examination is notable for a temperature of 38.2°C (100.7°F) and heart rate of 105 beats/min; otherwise, vital signs are normal. Her abdomen is tender in the right lower quadrant, and pelvic examination is normal. Urine pregnancy test is negative. Which of the following imaging modalities is most likely to confirm her diagnosis?

- A. CT of the abdomen without contrast
- B. Colonoscopy
- C. Pelvic ultrasound
- D. Plain film of the abdomen
- E. Ultrasound of the abdomen

VII-45. The patient in question VII-44 is diagnosed with acute appendicitis. Which of the following is the most important next step in management?

- A. Colonoscopy
- B. Empiric antibiotics and watchful waiting for 48 hours
- C. Exploratory laparotomy
- D. IV corticosteroids
- E. Laparoscopic appendectomy

VII-46. A 38-year-old man is seen in the urgent care center with several hours of severe abdominal pain. His symptoms began suddenly, but he reports several months of pain in the epigastrium after eating, with a resultant 10-lb weight loss. He takes no medications other than over-the-counter antacids and has no other medical problems or habits. On physical examination, temperature is 38.0°C (100.4°F), pulse is 130 beats/min, respiratory rate is 24 breaths/min, and blood pressure is 110/50. His abdomen has absent bowel sounds and is rigid with involuntary guarding diffusely. A plain film of the abdomen is obtained and shows free air under the diaphragm. Which of the following is most likely to be found in the operating room?

- A. Necrotic bowel
- B. Necrotic pancreas
- C. Perforated duodenal ulcer
- D. Perforated gallbladder
- E. Perforated gastric ulcer

VII-47. Which of the following statements regarding the signs and symptoms of a patient with acute appendicitis is true?

- A. Anorexia is an uncommon symptom.
- B. McBurney point describes pain in the mid-epigastrium.
- C. Patients with pelvic appendicitis commonly present with dysuria, urinary frequency, diarrhea, or tenesmus.
- D. Right lower quadrant tenderness is present in only 50% of patients.
- E. Rovsing sign is the most sensitive finding on physical examination.

VII-48. The greatest source of nutrients and calories in a typical individual's diet come from which source?

- A. Alcohol
- B. Carbohydrates
- C. Fat
- D. Protein

VII-49. Which of the following factors accounts for the largest amount of water loss per day?

- A. Evaporation and exhalation
- B. Fever
- C. Normal stool output (i.e., no diarrhea)
- D. Pregnancy
- E. Urine output in a person with a very low solute diet

VII-50. Mr. Ryan is a 56-year-old man who was admitted to the surgical service for care due to exposure injury and frostbite in his distal extremities. He has a long-standing history of alcoholism, drinking about 1 L of vodka on a daily basis. You are asked to consult due to concerns of bizarre behaviors exhibited by the patient. He is expressing a belief that his wounds were the result of burns that were inflicted on him by “torturers” in the government because he “knows too much” about government surveillance plans. The surgical team reports it is difficult to keep the patient in his bed, and he seems unsteady on his feet at times. He has been medicated throughout his stay for prevention of alcohol withdrawal via a symptom-triggered approach and last received lorazepam 2 mg orally about 2 hours ago. At that time, the patient was noted to be tremulous, tachycardic, and hypertensive. The delusional thoughts are not responsive to treatment of alcohol withdrawal symptoms. When you see the patient, he is sleeping quietly. Vital signs are blood pressure 110/82, heart rate 94 beats/min, respiratory rate 16 breaths/min, temperature 37.1°C (98.7°F), and SaO₂ is 97% on room air. He awakens easily and has a minimal resting tremor. On neurologic examination, he exhibits past-pointing, difficulty with rapid alternating movements, horizontal nystagmus, and decreased sensation to light touch and pinprick in the lower extremities below the mid-tibia. His gait is wide based and ataxic. He no longer expresses his prior delusional beliefs, but he is disoriented and thinks he is in jail. He states he was brought to “this gulag” so that the government could experiment on him. He has 5% dextrose in half-normal saline infusing at 100 mL/h and is also receiving nafcillin 2 g IV every 4 hours for cellulitis. Which of the following is the most likely cause of Mr. Ryan’s altered mental state and neurologic findings?

- A. Hypoglycemia
- B. Hyponatremia
- C. Niacin deficiency
- D. Thiamine deficiency
- E. Undertreated alcohol withdrawal

VII-51. A 48-year-old man is diagnosed with carcinoid syndrome after presenting with diarrhea, flushing, and hypotension. With appropriate treatment, he experiences an appropriate response biochemically, and his flushing and blood pressure are markedly improved. However, he continues to have some mild diarrhea and also has mouth soreness. He remains fatigued with a loss of appetite and irritability. On examination, you notice his tongue is bright red and somewhat enlarged. It is tender to touch. In addition, he has a pigmented and scaling rash that is most prominent around his neckline. Which of the following is the most likely vitamin or mineral deficiency in this patient?

- A. Copper
- B. Niacin
- C. Riboflavin
- D. Vitamin C
- E. Zinc

VII-52. A 51-year-old alcoholic man presents to the emergency department complaining of vomiting blood. On further evaluation including gastric lavage, you determine that he is not experiencing an upper gastrointestinal bleed, but he is having significant gingival bleeding. He is intoxicated and complains of fatigue. Reviewing his chart, you find that he had a hemarthrosis evacuated 6 months ago and has been lost to follow-up since then. He takes no medications. Laboratory data show platelets of 250,000 and international normalized ratio of 0.9. He has a diffuse hemorrhagic eruption on his legs that is centered around hair follicles. Which of the following is the recommended treatment for this patient’s underlying disorder?

- A. Folate
- B. Niacin
- C. Thiamine
- D. Vitamin C
- E. Vitamin K

VII-53. A 46-year-old male from Mexico was found to be purified protein derivative–positive on routine health screening prior to beginning a new job in a hospital. He was started on isoniazid for treatment of latent tuberculosis. Three months into his treatment course he develops numbness and tingling in his bilateral feet. On laboratory assessment his liver function tests are normal. He reports taking his isoniazid faithfully but does not remember being told to take any additional medications along with the isoniazid. Which of the following is the likely cause of his symptoms?

- A. Niacin deficiency
- B. Thiamine deficiency
- C. Vitamin A toxicity
- D. Vitamin B₆ deficiency
- E. Vitamin B₁₂ deficiency

VII-54. A 3-year-old who recently arrived from Indonesia comes in for a routine wellness check with his new adoptive parents. On physical examination he is noted to have white patches of keratinized epithelium appearing on the sclera. Which of the following would be the most appropriate treatment for this condition?

- A. Biotin 10 mg/day for 7 days
- B. Pyridoxine 50 mg/day for 14 days
- C. Vitamin A 200,000 IU orally on day 1, day 2, and day 15
- D. Vitamin B₁₂ 1000 mcg per day for 1 month
- E. Vitamin C 100 mg/day for 7 days

VII-55. A 21-year-old woman is admitted to the cardiac care unit after collapsing in her college dormitory. When emergency personnel arrived, she was found to be in a torsades de pointes arrhythmia and was pulseless. She received cardiopulmonary resuscitation, defibrillation, and magnesium en route to the hospital. On arrival, her initial potassium is 1.2 mEq/L. Her physical examination is remarkable for an excessively thin appearance with lanugo hair on arms and chest. Her body mass index is 14.6 kg/m².

Which of the following statements is true regarding this patient's nutritional state?

- A. Mortality in the disease is most commonly due to complications of malnutrition.
- B. Poor wound healing and frequent skin infections are common complications.
- C. Systemic inflammation is a predominant finding on laboratory examination.
- D. The serum albumin is typically less than 2.8 g/dL.
- E. Triceps skinfold <3 mm and mid-arm muscle circumference <15 cm are useful diagnostic criteria.

VII-56. You are caring for a 54-year-old woman in the intensive care unit who was admitted for treatment of severe sepsis and pneumonia. You would like to initiate enteral nutrition and plan to calculate basal energy expenditure for the patient. All of the following factors are used to determine the patient's caloric needs EXCEPT:

- A. Age
- B. Albumin
- C. Gender
- D. Height
- E. Weight

VII-57. Which of the following conditions would be best classified as starvation-associated malnutrition?

- A. Closed head injury
- B. Major depression
- C. Motor-vehicle accident
- D. Pancreatic cancer
- E. Sarcopenic obesity

VII-58. Which of the following statements regarding the methodologies for determining body composition and nutritional status is true?

- A. Albumin is useful in assessing nutritional status because of its long half-life.
- B. Anthropomorphic analyses such as measuring skinfolds and circumferences are reliable and reproducible.
- C. Bioelectrical impedance can be readily applied and interpreted in individuals of different races and ethnic groups.
- D. C-reactive protein is a specific marker for malnutrition.
- E. Dual energy x-ray absorptiometry scanning can be used to compare truncal and appendicular soft tissue.

VII-59. Which of the following is the most common symptom or sign of liver disease?

- A. Fatigue
- B. Itching
- C. Jaundice
- D. Nausea
- E. Right upper quadrant pain

VII-60. All of the following are CAGE questions, which should be a component of the medical history focusing on alcohol abuse and dependence, EXCEPT:

- A. Do you feel like you have a greater tolerance for alcohol than your friends?
- B. Have you ever felt you ought to cut down on your drinking?
- C. Have people annoyed you by criticizing your drinking?
- D. Have you ever felt guilty or bad about your drinking?
- E. Have you ever had a drink first thing in the morning to steady your nerves or get rid of a hangover?

VII-61. A 43-year-old female presents to her primary care physician with fatigue and malaise. On routine laboratory testing she is found to have an aspartate aminotransferase of 150 IU/L, and alanine aminotransferase of 165 IU/L, an alkaline phosphatase of 125 IU/L, a total bilirubin of 1.2 mg/dL, and an international normalized ratio of 0.9. Which of the following best describes the pattern of liver tests?

- A. Cholestatic
- B. Cirrhotic
- C. Hepatocellular
- D. Mixed
- E. Peritonitic

VII-62. All of the following are potential advantages of magnetic resonance cholangiopancreatography over endoscopic retrograde cholangiopancreatography EXCEPT:

- A. Faster image acquisition
- B. Higher sensitivity for identifying ampullary lesions
- C. No risk of ionizing radiation
- D. No risk of pancreatitis
- E. No need for contrast media

VII-63. Elevation in all of the following laboratory studies would be indicative of liver disease EXCEPT:

- A. 5'-Nucleotidase
- B. Aspartate aminotransferase
- C. Conjugated bilirubin
- D. Unconjugated bilirubin
- E. Urine bilirubin

VII-64. Which of the following statements regarding liver function tests is true?

- A. Alanine aminotransferase (ALT) is found in liver, cardiac muscle, skeletal muscle, and kidney.
- B. Elevation of aspartate aminotransferase (AST) and ALT to >1000 IU/L is typical of ischemic hepatitis.
- C. Elevation of AST is more specific for liver dysfunction than elevation of ALT.
- D. Increased AST and ALT with an AST:ALT ratio of >3 is typical of acute viral hepatitis.
- E. The magnitude of elevated AST and ALT has important prognostic significance in acute hepatitis.

VII-65. Which of the following is true about aspartate aminotransferase (AST) and alanine aminotransferase (ALT) in liver injury?

- A. A ratio of ALT:AST greater than 1 is suggestive of cirrhosis.
- B. ALT is usually equal to or greater than AST in most causes of acute liver injury.
- C. AST is often normal in cases of alcoholic cirrhosis.
- D. AST and ALT usually remain elevated for days after the passage of a gallstone.
- E. If the ALT:AST ratio is greater than 3, then alcoholic hepatitis should be suspected.

VII-66. Which of the following is true about biochemical liver testing?

- A. γ -Glutamyl transpeptidase elevation is specific for cholestatic liver injury.
- B. Increased IgM levels may be a marker of autoimmune hepatitis.
- C. Low factor VIII levels may indicate chronic liver disease.
- D. Patients over the age of 60 can have a mild elevation in their alkaline phosphatase levels up to 1.5 times the upper limit of normal.
- E. Serum albumin is a good marker of the severity of acute liver injury.

VII-67. A 26-year-old male resident is noticed by his attending physician to have yellow eyes after his 24-hour call period. When asked, the resident states he has no medical history and is a social drinker, but on occasion, he has thought he might have mild scleral icterus when he is stressed or has more than four to five alcoholic drinks. He never sought medical treatment because his eyes would return fully to normal within 2 days. He denies nausea, abdominal pain, dark urine, light-colored stools, pruritus, or weight loss. On examination, he has a body mass index of 20.1 kg/m², and his vital signs are normal. Scleral icterus is present. There are no stigmata of chronic liver disease. The patient's abdomen is soft and nontender. The liver span is 8 cm to percussion. The liver edge is smooth and palpable only with deep inspiration. The spleen is not palpable. Laboratory examinations are normal except for a total bilirubin of 3.0 mg/dL. Direct bilirubin is 0.2 mg/dL. Aspartate aminotransferase, alanine aminotransferase, and alkaline phosphatase are normal. Hematocrit, lactate dehydrogenase, and haptoglobin are normal. Which of the following is the most likely diagnosis?

- A. Autoimmune hemolytic anemia
- B. Crigler-Najjar syndrome type 1
- C. Choledocholithiasis
- D. Dubin-Johnson syndrome
- E. Gilbert syndrome

VII-68. Which of the following statements regarding the hyperbilirubinemia seen in patients with significant intravascular hemolysis is true?

- A. Bilirubin values <4 mg/dL imply concomitant gallbladder or biliary dysfunction.
- B. Bilirubin values >4 mg/dL (68 μ mol/L) imply concomitant liver dysfunction.
- C. It is typically composed of 50% conjugated and 50% unconjugated bilirubin.
- D. Prolonged hemolysis may result in the development of cirrhosis.

VII-69. A 26-year-old female who is 3-months pregnant presents for a routine prenatal visit and is found to be jaundiced. This is her first pregnancy and she reports no symptoms other than occasional early morning nausea that resolves after eating breakfast. She has no pruritus. Her physical examination is unremarkable with the exception of obvious jaundice. She has no hepatosplenomegaly and her fundus is not yet palpable. Labs are notable for a total bilirubin of 6 mg/dL, with a direct bilirubin of 5 mg/dL. Her aspartate aminotransferase, alanine aminotransferase, and alkaline phosphatase levels are all within the normal range. Which of the following is the most likely cause of her hyperbilirubinemia?

- A. Benign recurrent intrahepatic cholestasis
- B. Cholestasis of pregnancy
- C. Crigler-Najjar syndrome type 2
- D. Dubin-Johnson syndrome
- E. HELLP (hemolysis, elevated liver enzymes, and low platelet) syndrome

VII-70. A 34-year-old man presents to the physician complaining of yellow eyes. For the past week, he has felt ill, with decreased oral intake, low-grade fevers (~100°F), fatigue, nausea, and occasional vomiting. With the onset of jaundice, he has noticed pain in his right upper quadrant. He currently uses marijuana and ecstasy and has a prior history of injection drug use with cocaine. He has no other past medical history, but he was unable to donate blood for reasons that he cannot recall 4 years previously. His social history is remarkable for working as a veterinary assistant. On sexual history, he reports five male sexual partners over the past 6 months. He does not consistently use condoms. On physical examination, he appears ill and has obvious jaundice with scleral icterus. His liver is 15 cm to percussion, palpable 6 cm below the right costal margin. The edge is smooth and tender to palpation. The spleen is not enlarged. There are no stigmata of chronic liver disease. His aspartate aminotransferase is 1232 IU/L, alanine aminotransferase is 1560 IU/L, alkaline phosphatase is 394 IU/L, total bilirubin is 13.4 mg/dL, and direct bilirubin is 12.2 mg/dL. His international normalized ratio is 2.3, and activated partial thromboplastin time is 52 seconds. Hepatitis serologies are sent and reveal the following:

Hepatitis A IgM: negative

Hepatitis A IgG: negative

Hepatitis B core IgM: positive

Hepatitis B core IgG: negative

Hepatitis B surface antigen: positive

Hepatitis B surface antibody: negative

Hepatitis B e antigen: positive

Hepatitis B e antibody: negative

Hepatitis C antibody: positive

What is the cause of the patient's current clinical presentation?

- A. Acute hepatitis A infection
- B. Acute hepatitis B infection
- C. Acute hepatitis C infection
- D. Chronic hepatitis B infection
- E. Drug-induced hepatitis

VII-71. In the patient described in question VII-70, what would be the best approach to prevent development of chronic hepatitis?

- A. Administration of anti-hepatitis A virus IgG
- B. Administration of lamivudine
- C. Administration of pegylated interferon- α plus ribavirin
- D. Administration of prednisone beginning at a dose of 1 mg/kg daily
- E. Do nothing and observe as 99% of individuals with this disease recover

VII-72. A 16-year-old girl had visited your clinic 1 month ago with jaundice, vomiting, malaise, and anorexia. Two other family members were ill with similar symptoms. Based on viral serologies, including a positive anti-hepatitis A virus IgM, a diagnosis of hepatitis A was made. The patient was treated conservatively, and 1 week after first presenting, she appeared to have made a full recovery. She returns to your clinic today complaining of the same symptoms she had 1 month ago. She is jaundiced, and an initial panel of laboratory tests returns elevated transaminases. Which of the following offers the best explanation of what has occurred in this patient?

- A. Co-infection with hepatitis C
- B. Inappropriate treatment of initial infection
- C. Incorrect initial diagnosis; this patient likely has hepatitis B
- D. Reinfection with hepatitis A
- E. Relapse of hepatitis A

VII-73. An 18-year-old man presents to a rural clinic with nausea, vomiting, anorexia, abdominal discomfort, myalgias, and jaundice. He describes occasional alcohol use and is sexually active. He describes using heroin and cocaine "a few times in the past." He works as a short-order cook in a local restaurant. He has lost 15.5 kg (34 lb) since his last visit to the clinic and appears emaciated and ill. On examination, he is noted to have icteric sclerae and a palpable,

tender liver below the right costal margin. In regard to acute hepatitis, which of the following is true?

- A. A distinction between viral etiologies cannot be made using clinical criteria alone.
- B. Based on age and risk factors, he is likely to have a hepatitis B infection.
- C. He does not have hepatitis E virus, as this infects only pregnant women.
- D. This patient cannot have hepatitis C because his presentation is too acute.
- E. This patient does not have hepatitis A because his presentation is too fulminant.

VII-74. Which of the following best explains why neonatal infection with hepatitis B virus (HBV) does not result in significant hepatocellular inflammation early in life?

- A. Effective priming of HBV-specific T cells with HBV leads to the lack of an immune response.
- B. HBV does not easily cross the placenta.
- C. In utero exposure to HBV induces immune tolerance.
- D. Maternal antibodies against hepatitis core antigen result in immunity against infection.
- E. Natural killer cells are not particularly effective early in life.

VII-75. A 57-year-old female presents to your clinic for a routine health evaluation. She is complaining of intermittent abdominal pain so you decide to order serum chemistries. Her aspartate aminotransferase is mildly elevated at 75 IU/L, and her alanine aminotransferase is also mildly elevated at 66 IU/L. Her bilirubin and international normalized ratio are normal. She denies recent alcohol use and has never used injection drugs. She is sexually active and has had two recent male partners. She does not use barrier protection. She was born in China but has lived in the United States for 40 years. You decide to check hepatitis serologies and obtain the following results:

Hepatitis B surface antigen: positive
 Hepatitis B surface antibody: negative
 Hepatitis B core antibody: positive
 Hepatitis B e-antigen: negative
 Hepatitis B e-antibody: positive

Which of the following best describes her hepatitis B status?

- A. Acute hepatitis B
- B. Chronic Hepatitis B, high infectivity
- C. Immunization against hepatitis B
- D. Late acute or chronic hepatitis B, low infectivity
- E. Recovery from hepatitis B

VII-76. Which of the following statements is true about the prevention of viral hepatitis?

- A. Health care workers in endemic areas should receive the hepatitis D vaccine every 5 years.
- B. Patients with acute hepatitis A should be placed on contact isolation to prevent transmission to health care workers and other patients.
- C. Serum Ig has been shown to be effective in postexposure prophylaxis for hepatitis C.
- D. The hepatitis B vaccine is a live attenuated vaccine that should be avoided in immunocompromised patients.
- E. There is an effective vaccine against hepatitis E genotype 1 that also has effectiveness against other genotypes.

VII-77. A 32-year-old woman is admitted to the intensive care unit following an overdose of acetaminophen with co-ingestion of alcohol. She was known to be alert and interactive about 4 hours prior to her presentation when she had a fight with her boyfriend who then left the home. When he returned 6 hours later, he found an empty bottle of acetaminophen 500-mg capsules as well as an empty vodka bottle. The exact number of pills in the bottle is unknown, but the full bottle held as much as 50 capsules. The patient was unresponsive and had vomited, so her boyfriend called 911. On arrival to the emergency department, the patient is stuporous. Her vital signs are pulse 109 beats/min, respiratory rate 20 breaths/min, blood pressure 96/52, and oxygen saturation 95% on room air. Her examination shows mild nonspecific abdominal pain with palpation. The liver is not enlarged. Her initial laboratory values show a normal complete blood count, normal electrolytes, and kidney function. The aspartate aminotransferase is 68 IU/L, alanine aminotransferase is 46 IU/L, alkaline phosphatase is 110 IU/L, and total bilirubin is 1.2 mg/dL. Glucose and coagulation studies are normal. The serum alcohol level is 210 g/dL. The acetaminophen level is 350 µg/mL. What is the most appropriate next step in the treatment of this patient?

- A. Administration of activated charcoal or cholestyramine
- B. Administration of N-acetylcysteine 140 mg/kg followed by 70 mg/kg every 4 hours for a total of 15–20 doses
- C. Continued monitoring of liver function, glucose, and coagulation studies every 4 hours with administration of N-acetylcysteine if these begin to change
- D. Do nothing as normal liver function tests and coagulation studies are indicative of only a minor ingestion
- E. Initiate hemodialysis for toxin clearance

VII-78. A 31-year-old health care worker is found to have a newly positive tuberculin skin test 6 weeks after an exposure to a patient with active pulmonary tuberculosis. He is asymptomatic and has a normal chest radiograph. Which of the following statements regarding initiation of isoniazid (INH) prophylactic therapy is true?

- A. Acute hepatocellular injury due to INH is an idiosyncratic reaction that will manifest within the first 2 months of initiation of therapy.
- B. Controlled trials have demonstrated that monthly monitoring of aminotransferase levels reduces morbidity in U.S. health care workers receiving INH prophylaxis.
- C. Elevation of aminotransferase levels in the first 2 months of therapy is an indication to stop INH and switch to another drug.
- D. The patient has a 50–70% chance of transient elevation of his aminotransferase levels in the first 2 months of treatment.
- E. The frequency of acute hepatocellular injury due to INH is age dependent, increasing in patients >35 years of age.

VII-79. Which of the following statements is true about drug-induced liver injury (DILI)?

- A. Direct toxic hepatitis is usually clinically indistinguishable from viral hepatitis.
- B. Direct toxic hepatitis occurs with predictable regularity after exposure and is dose dependent.
- C. “Idiosyncratic” drug reactions are easily identified by their histologic pattern on liver biopsy.
- D. The incidence of DILI is higher in patients with chronic liver disease.
- E. The occurrence of jaundice in a phase 3 clinical trial is not predictive of severe hepatotoxicity in post-marketing surveillance.

VII-80. Which of the following statements regarding drug-induced liver injury is true?

- A. Amiodarone-induced liver injury resolves within days of stopping the drug.
- B. Phenytoin-induced liver injury is only rarely accompanied by systemic symptoms and is usually picked up on screening liver enzyme testing.
- C. Sodium valproate liver toxicity is more common in patients with mitochondrial deficiencies and can be ameliorated by the administration of IV carnitine.
- D. The incidence of amoxicillin-induced liver injury is decreased when administered in combination with clavulanate.
- E. When it occurs, nitrofurantoin liver toxicity happens soon after the initiation of the drug.

VII-81. A 38-year-old woman is evaluated for elevated transaminase levels that were identified during routine laboratory testing for life insurance. She is originally from Thailand and immigrated to the United States 10 years previously. She has been married to an American for the past 12 years, meeting him while he was living abroad for business. She previously worked in Thailand as a deputy tourism minister for the government but is not currently employed. She has no significant past medical history. She had one uncomplicated pregnancy at the age of 22. When queried about risk factors for liver disease, she denies alcohol intake or drug abuse. She has never

had a blood transfusion. She recalls an episode of jaundice that she did not seek evaluation for about 15 years ago as it resolved spontaneously. She currently feels well, and her husband wished to have her added to his life insurance policy. On physical examination there are no stigmata of chronic liver disease. Her laboratory studies reveal an aspartate aminotransferase of 346 IU/L, alanine aminotransferase of 412 IU/L, alkaline phosphatase of 98 IU/L, and total bilirubin 1.5 of mg/dL. Further workup includes the following viral studies: hepatitis A IgG positive, hepatitis B surface antigen positive, hepatitis B e antigen positive, anti-hepatitis B virus core IgG positive, and hepatitis C IgG negative. The hepatitis B virus DNA level is 4.8×10^4 IU/mL. Which of the following medications is indicated for this patient?

- A. Acyclovir
- B. Entecavir
- C. Ritonavir
- D. Simeprevir
- E. No treatment is necessary

VII-82. A 46-year-old man is known to have chronic hepatitis C virus (HCV) infection. He is a former IV drug user for more than 20 years who has been abstinent from drug use for 1 year. He was treated for tricuspid valve endocarditis 3 years previously. He does not know when he acquired HCV. His laboratory studies show a positive HCV IgG antibody with a viral load of greater than 1 million copies. The virus is genotype 2. His aspartate aminotransferase is 82 IU/L, and his alanine aminotransferase is 74 IU/L. He undergoes liver biopsy, which demonstrates a moderate degree of bridging fibrosis. Which of the following is the most predictive of the development of cirrhosis?

- A. Abnormal transaminases
- B. Bridging fibrosis on liver biopsy
- C. Genotype 2
- D. History of bacterial endocarditis
- E. History of IV drug use

VII-83. For the patient in described in question VII-82, which of the following statements is true about potential treatment options for his hepatitis C virus?

- A. A liver biopsy should be performed to determine the appropriate treatment regimen.
- B. Combination therapy with sofosbuvir and velpatasvir would be a reasonable first-line treatment based on his genotype.
- C. He should be treated with ribavirin and pegylated interferon as first-line agents.
- D. Monotherapy with boceprevir would avoid any potential interactions with the CYP3A4 pathway.
- E. Because his alanine aminotransferase and aspartate aminotransferase are not more than three times the

upper limit of normal he does not currently require treatment.

VII-84. A 34-year-old woman is evaluated for fatigue, malaise, arthralgias, and a 10-lb weight loss over the past 6–8 weeks. She has no past medical history. Since feeling poorly, she has taken approximately one or two tablets of acetaminophen 500 mg daily. On physical examination, her temperature is 37.9°F (100.2°F), respiratory rate is 18 breaths/min, blood pressure is 100/48, heart rate is 92 beats/min, and oxygen saturation is 96% on room air. She has scleral icterus. Her liver edge is palpable 3 cm below the right costal margin. It is smooth and tender. The spleen is not enlarged. She has mild synovitis in the small joints of her hands. Her aspartate aminotransferase is 542 IU/L, alanine aminotransferase is 657 IU/L, alkaline phosphatase is 102 IU/L, total bilirubin is 5.3 mg/dL, and direct bilirubin is 4.8 mg/dL. Which of the following tests would be LEAST likely to be positive in this diagnosis?

- A. Antinuclear antibodies in a homogeneous pattern
- B. Anti-liver/kidney microsomal antibodies
- C. Antimitochondrial antibodies
- D. Hypergammaglobulinemia
- E. Rheumatoid factor

VII-85. Which of the following statements is true about the mechanism of liver injury in patients with autoimmune hepatitis?

- A. Acute viral hepatitis has not been associated with the subsequent development of autoimmune hepatitis.
- B. Circulating autoantibodies in patients have been directly linked to hepatocyte injury.
- C. Immune complex deposition is an important mechanism of hepatocyte injury.
- D. Macrophages are the main effector cell of liver injury.
- E. Molecular mimicry by cross-reacting antigens that contain epitopes similar to liver antigens is postulated to activate cytotoxic T cells.

VII-86. All of the following statements regarding alcoholic liver disease are true EXCEPT:

- A. Fatty liver is present in >90% of daily and binge drinkers.
- B. Hepatitis C infection worsens the prognosis of alcoholic liver disease.
- C. Over 50% of alcoholics will develop alcoholic hepatitis.
- D. Quantity and duration of alcohol consumption are the most important risk factors for the development of alcoholic liver disease.
- E. The pathologic hallmarks of alcoholic liver disease are fatty liver, hepatitis, and cirrhosis.

VII-87. A 32-year-old woman is admitted to the hospital with fever, abdominal pain, and jaundice. She drinks approximately 6 beers daily and has recently increased her alcohol intake to more than 12 beers daily. She has no other substance abuse history and has no prior history of alcoholic liver disease or pancreatitis. She is not taking any medications. On physical examination, she appears ill and disheveled with a fruity odor to her breath. Her vital signs are heart rate 122 beats/min, blood pressure 95/56, respiratory rate 22 breaths/min, temperature 38.4°C (101.2°F), and oxygen saturation 98% on room air. She has scleral icterus, and spider angiomas are present on the trunk. The liver edge is palpable 10 cm below the right costal margin. The liver is smooth and tender to palpation. The spleen is not palpable. No ascites or lower extremity edema is present. Laboratory studies demonstrate an aspartate aminotransferase of 431 IU/L, alanine aminotransferase of 198 IU/L, bilirubin of 8.6 mg/dL, alkaline phosphatase of 201 IU/L, amylase of 88 U/L, and lipase of 50 U/L. Total protein is 6.2 g/dL, and albumin is 2.8 g/dL. The prothrombin time is 29 seconds (control, 13 seconds) with international normalized ratio of 2.2. What is the best approach to treatment of this patient?

- A. Administer IV fluids, thiamine, and folate and observe for improvement in laboratory tests and clinical condition.
- B. Administer IV fluids, thiamine, folate, and imipenem while awaiting blood culture results.
- C. Administer prednisone 40 mg daily for 4 weeks before beginning a taper.
- D. Consult surgery for management of acute cholecystitis.
- E. Perform an abdominal CT with IV contrast to assess for necrotizing pancreatitis.

VII-88. Which of the following is true about the pathophysiology of alcoholic liver disease?

- A. Alcoholic fatty liver disease is a benign condition and does not lead to lasting liver injury.
- B. Alcoholic fatty liver disease is characterized by ballooning degeneration, spotty necrosis, polymorphonuclear infiltrate, and fibrosis in the perivenular and perisinusoidal space of Disse.
- C. Fatty accumulation begins in the perivenular hepatocytes and can eventually involve the entire hepatic lobule.
- D. It is relatively easy to differentiate alcoholic from nonalcoholic fatty liver disease on liver biopsy.
- E. Once fatty accumulation occurs in the liver, it is not reversible.

VII-89. A 44-year-old man seeks evaluation for an abnormal finding on abdominal ultrasonography. He has a history of type 2 diabetes mellitus and is on insulin therapy. Last week, he was evaluated in the emergency department for mid-epigastric pain likely due to nonsteroidal anti-inflammatory drug therapy for muscle aches (he recently started exercising because his wife told him to lose weight). During the evaluation, an abdominal ultrasound showed

marked fatty infiltration of the liver. Laboratory studies show his transaminases are 2× normal with normal alkaline phosphatase, bilirubin, and prothrombin time. Other than insulin, he takes no medications, does not consume alcohol or illicit drugs, and has no family history of liver disease. On physical examination, he is obese (body mass index 32 kg/m²) with normal vital signs and no other abnormalities. You think he likely has nonalcoholic fatty liver disease (NAFLD). All of the following statements regarding his potential therapy are true EXCEPT:

- A. Bariatric surgery is safe in patients with NAFLD.
- B. Exercise may reduce hepatic steatosis.
- C. Statins may worsen inflammation in NAFLD.
- D. There are no therapies approved by the Food and Drug Administration for NAFLD.
- E. Vitamin E may reduce aminotransferase levels and hepatic steatosis.

VII-90. Which of the following statements about non-alcoholic steatohepatitis (NASH) is true?

- A. Insulin resistance is an important driver of lipid uptake, fat synthesis, and fat storage that leads to triglyceride accumulation in hepatocytes.
- B. Greater than 50% of patients with nonalcoholic fatty liver disease (NAFLD) also have some component of NASH.
- C. Serum biomarkers are useful in distinguishing NAFLD from NASH.
- D. The contribution of NASH to the development of cirrhosis is on the decline in the United States.
- E. Triglyceride accumulation within hepatocytes is directly cytotoxic.

VII-91. A 63-year-old man with cirrhosis and portal hypertension due to hemochromatosis presents with altered mental status. He has chronic ascites controlled with diet and spironolactone. He has a history of one esophageal bleed but none since starting propranolol. His family reports that over the last 2 days, he has become more confused, but he has had no melena or hematemesis. He is afebrile with normal vital signs, and physical examination is notable for ascites, asterixis, and being oriented only to person. His laboratory examination is notable for a hemoglobin of 10.1 g/dL (baseline 9.5), creatinine of 1.4 mg/dL (baseline 1.4), and blood urea nitrogen of 45 mg/dL (baseline 18). A paracentesis is performed that yields reveals clear fluid with 800 white blood cells/μL (40% neutrophils). Which of the following is the most indicated therapy?

- A. Ampicillin, ceftriaxone, vancomycin
- B. Cefotaxime
- C. Esophagogastroduodenoscopy with banding
- D. Hemodialysis
- E. Lactulose

VII-92. A 48-year-old woman presents complaining of fatigue and itching. She has been tired for the past 6 months and recently has developed itching diffusely. It is worse in the evening hours but is intermittent. She does

not note it to be worse following hot baths or showers. Her past medical history is significant only for hypothyroidism for which she takes levothyroxine 125 µg daily. On physical examination, she has mild jaundice and scleral icterus. The liver is enlarged to 15 cm on palpation and is palpable 5 cm below the right costal margin. Xanthomas are seen on both elbows. Hyperpigmentation is noticeable on the trunk and arms where the patient has excoriations. Laboratory studies demonstrate the following: white blood cells 8900/µL, hemoglobin 13.3 g/dL, hematocrit 41.6%, and platelets 160,000/µL. The creatinine is 1.2 mg/dL. The aspartate aminotransferase is 52 IU/L, alanine aminotransferase is 62 IU/L, alkaline phosphatase is 216 IU/L, total bilirubin is 3.2 mg/dL, and direct bilirubin is 2.9 mg/dL. The total protein is 8.2 g/dL, and albumin is 3.9 U/L. The thyroid-stimulating hormone is 4.5 U/mL. Antimitochondrial antibodies are positive. Perinuclear antineutrophil cytoplasmic antibodies (ANCA) and cytoplasmic ANCA are negative. What is the most likely cause of the patient's symptoms?

- A. Lymphoma
- B. Polycythemia vera
- C. Primary biliary cirrhosis
- D. Primary sclerosis cholangitis
- E. Uncontrolled hypothyroidism

VII-93. A 42-year-old man with cirrhosis related to hepatitis C and alcohol abuse has ascites requiring frequent large-volume paracentesis. All of the following therapies would be indicated for this patient EXCEPT:

- A. Fluid restriction to less than 2 L daily
- B. Furosemide 40 mg daily
- C. Sodium restriction to less than 2 g daily
- D. Spironolactone 100 mg daily
- E. Transjugular intrahepatic portosystemic shunt if medical therapy fails

VII-94. You are asked to consult on a 62-year-old white woman with pruritus for 4 months. She has noted progressive fatigue and a 5-lb weight loss. She has intermittent nausea but no vomiting and denies changes in her bowel habits. There is no history of prior alcohol use, blood transfusions, or illicit drug use. The patient is widowed and had two heterosexual partners in her lifetime. Her past medical history is significant only for hypothyroidism, for which she takes levothyroxine. Her family history is unremarkable. On examination, she is mildly icteric. She has spider angiomas on her torso. You palpate a nodular liver edge 2 cm below the right costal margin. The remainder of the examination is unremarkable. A right upper quadrant ultrasound confirms your suspicion of cirrhosis. You order a complete blood count and a comprehensive metabolic panel. What is the most appropriate next test?

- A. 24-Hour urine copper
- B. Antimitochondrial antibodies
- C. Endoscopic retrograde cholangiopancreatography
- D. Hepatitis B serologies
- E. Serum ferritin

VII-95. A 55-year-old man with cirrhosis thought secondary to nonalcoholic steatohepatitis presents with altered mental status. All of the following can precipitate hepatic encephalopathy in this type of patient EXCEPT:

- A. Dehydration
- B. Hyperkalemia
- C. Hypokalemia
- D. Medication nonadherence
- E. Spontaneous bacterial peritonitis

VII-96. A 64-year-old man with known cirrhosis is admitted to the intensive care unit with a large gastrointestinal bleed and altered mental status. He is confused and unable to provide any history. His initial hemoglobin is 6.9 g/dL, and his vital signs are notable for a heart rate of 115 beats/min and a blood pressure of 90/55. In addition to fluid resuscitation including transfusion of packed red blood cells, all of the following are appropriate therapy at this time EXCEPT:

- A. Endoscopic sclerotherapy
- B. Endoscopic variceal ligation
- C. Octreotide
- D. Propranolol
- E. Transjugular intrahepatic portosystemic shunt

VII-97. In the patient in question VII-96, all of the following could be used as prophylaxis for further variceal bleeding EXCEPT:

- A. Initiation of a nonselective beta blocker such as propranolol
- B. Initiation of daily subcutaneous octreotide injections
- C. Liver transplantation
- D. Repeated variceal ligation if the initial bleed was controlled with endoscopic variceal ligation
- E. Transjugular intrahepatic portosystemic shunt

VII-98. Which of the following statements is true about alcoholic cirrhosis?

- A. Alcoholic cirrhosis is characterized by predominantly large (>2 cm) nodules in the liver.
- B. If patients with alcoholic cirrhosis are able to stop drinking alcohol, their 5-year survival improves.
- C. It is the most common cause of cirrhosis in the United States.
- D. Parenterally administered tumor necrosis factor inhibitors have emerged as an important treatment option that can reduce mortality in patients with alcoholic cirrhosis.

VII-99. A 63-year-old female with primary biliary cirrhosis has refractory ascites that is managed with serial large volume paracentesis. She presents to the emergency department the night after an outpatient large volume tap with altered mental status. On arrival her laboratory studies are notable for a serum creatinine of 2.7 mg/dL (baseline 0.9 2 weeks ago) and a serum sodium of 127 mEq/L. Her urine output is low. You suspect possible hepatorenal syndrome. All of the following would be potential next steps in her evaluation and management EXCEPT:

- A. Administer IV albumin to restore effective circulating volume
- B. Look for alternative causes of renal failure
- C. Referral for liver transplantation if creatinine continues to worsen
- D. Start furosemide and spironolactone to increase her urine output
- E. Start midodrine and octreotide to try to reduce splanchnic vasodilation and improve renal perfusion

VII-100. Which of the following patients is the highest priority for liver transplantation?

- A. A 24-year-old woman with cirrhosis due to autoimmune hepatitis; she has been on the transplant list for 2 months and now has an elevated bilirubin, international normalized ratio (INR), and creatinine.
- B. A 38-year-old woman with chronic hepatitis C and normal bilirubin and INR.
- C. A 49-year-old man with alcoholic cirrhosis who has been on the transplant list for 6 months; he has had two esophageal variceal bleeds.
- D. A 59-year-old man with a history of hyperlipidemia who was admitted to the intensive care unit 2 days ago with fulminant hepatic failure due to mistakenly ingesting *Amanita* mushrooms from his lawn.
- E. A 64-year-old woman with primary hepatocellular carcinoma admitted to the hospital with acute renal failure.

VII-101. Which of the following statements regarding liver transplantation is true?

- A. ABO incompatibility is an absolute contraindication to transplant.
- B. Hepatocellular carcinoma is one of several conditions that receive disease-specific model for end-stage liver disease exceptions.
- C. HIV infection is a contraindication to transplant.
- D. Living donor transplants pose very little risk to the donor.
- E. Preformed cytotoxic human leukocyte antigen antibodies are a contraindication to transplantation.

VII-102. A 44-year-old woman is evaluated for complaints of abdominal pain. She describes the pain as a postprandial burning pain. It is worse with spicy or fatty foods and is relieved with antacids. She is diagnosed with a gastric ulcer and is treated appropriately for *Helicobacter pylori*. During the course of her evaluation for her abdominal

pain, the patient had a right upper quadrant ultrasound that demonstrated the presence of gallstones. Following treatment of *H. pylori*, her symptoms have resolved. She is requesting your opinion regarding whether treatment is required for the finding of gallstone disease. On review of the ultrasound report, there were numerous stones in the gallbladder, including in the neck of the gallbladder. The largest stone measures 2.8 cm. What is your advice to the patient regarding the risk of complications and the need for definitive treatment?

- A. Given the size and number of stones, prophylactic cholecystectomy is recommended.
- B. No treatment is necessary unless the patient develops symptoms of biliary colic frequent severe enough to interfere with the patient's life.
- C. The only reason to proceed with cholecystectomy is the development of gallstone pancreatitis or cholangitis.
- D. The risk of developing acute cholecystitis is about 5–10% per year.
- E. Ursodeoxycholic acid should be given at a dose of 10–15 mg/kg daily for a minimum of 6 months to dissolve the stones.

VII-103. A 62-year-old man has been hospitalized in intensive care for the past 3 weeks following an automobile accident resulting in multiple long bone fractures and acute respiratory distress syndrome. He has been slowly improving but remains on mechanical ventilation. He is now febrile and hypotensive requiring vasopressors. He is being treated empirically with cefepime and vancomycin. Multiple blood cultures are negative. He has no new infiltrates or increasing secretions on chest radiograph. His laboratory studies demonstrated a rise in his liver function tests, bilirubin, and alkaline phosphatase. Amylase and lipase are normal. A right upper quadrant ultrasound shows sludge in the gallbladder but no stones. The bile duct is not dilated. What is the next best step in the evaluation and treatment of this patient?

- A. Discontinue cefepime
- B. Initiate treatment with clindamycin
- C. Initiate treatment with metronidazole
- D. Perform hepatobiliary scintigraphy
- E. Refer for exploratory laparotomy

VII-104. A 41-year-old woman presents to your clinic with a week of jaundice. She notes pruritus, icterus, and dark urine. She denies fever, abdominal pain, or weight loss. The examination is unremarkable except for yellow discoloration of the skin. Total bilirubin is 6.0 mg/dL, and direct bilirubin is 5.1 mg/dL. Aspartate aminotransferase is 84 IU/L, alanine aminotransferase is 92 IU/L, and alkaline phosphatase is 662 IU/L. CT scan of the abdomen is unremarkable. Right upper quadrant ultrasound shows a normal gallbladder but does not visualize the common bile duct. What is the most appropriate next management step?

- A. Antibiotics and observation
- B. Endoscopic retrograde cholangiopancreatography
- C. Hepatitis serologies
- D. Hepatobiliary iminodiacetic acid scan
- E. Serologies for antimitochondrial antibodies

VII-105. All of the following are contributing factors to cholesterol stone formation EXCEPT:

- A. An excess of biliary cholesterol in relation to bile acids and phospholipids
- B. Increased biliary secretion of cholesterol
- C. Gallbladder hypomotility
- D. Nucleation of cholesterol monohydrate crystals
- E. Rapid weight gain through a high-fat diet

VII-106. A previously healthy 46-year-old female presents with 12 hours of right upper quadrant pain and nausea. The pain occasionally radiates to her right scapula. She has exquisite right upper quadrant tenderness on deep palpation. Laboratory studies show a mild elevation in her serum alkaline phosphatase. A right upper quadrant ultrasound shows stones in the bladder and a thickened gallbladder wall consistent with acute calculous cholecystitis. All of the following statements about cholecystectomy are true EXCEPT:

- A. Five to ten percent of patients who undergo elective cholecystectomy develop diarrhea.
- B. Atelectasis and subphrenic abscesses may occur after cholecystectomy.
- C. Cholecystectomy provides near total relief of symptoms in up to 75–90% of patients.
- D. The complication rate is higher in patients who undergo early cholecystectomy as opposed to delayed cholecystectomy (>6 weeks).
- E. Undetected common bile duct stones are left behind in 1–5% of patients.

VII-107. A 46-year-old woman presents with a 1-week history of right upper quadrant pain, intermittent fever, and nausea. Her initial labs are notable for a markedly elevated bilirubin, mild elevations in her transaminases, and a modest elevation in her alkaline phosphatase. Ultrasound shows no gallstones. Magnetic resonance cholangiopancreatography reveals a beaded appearance of her intrahepatic and extrahepatic bile ducts due to multiple discrete strictures. Which of the following lab tests would be most helpful in understanding the cause of her current disease?

- A. Hepatitis C antibody
- B. Procalcitonin
- C. Serum IgG-4 levels
- D. Serum triglycerides
- E. Urine histoplasma antigen

VII-108. All of the following disorders may be associated with a rise in serum amylase levels EXCEPT:

- A. Esophageal carcinoma
- B. Intestinal obstruction
- C. Mumps
- D. Pregnancy
- E. Renal Failure

VII-109. Which of the following statements about imaging of the pancreas is true?

- A. Abdominal CT has no role in the diagnosis of acute pancreatitis.
- B. Endoscopic ultrasound (EUS) and magnetic resonance cholangiopancreatography have largely replaced endoscopic retrograde cholangiopancreatography in the diagnostic evaluation of pancreatic disease.
- C. EUS is not helpful in the diagnosis of chronic pancreatitis.
- D. Plane radiographs of the abdomen are commonly used in the evaluation of acute and chronic pancreatitis.
- E. One advantage of pancreatic ultrasound is that intestinal bowel gas does not interfere with the ability to obtain images.

VII-110. Which of the following statements regarding pancreatic enzyme secretion is true?

- A. Acetylcholine is an important neurotransmitter that stimulates pancreatic enzyme secretion.
- B. All pancreatic enzymes have pH optima in the acidic range.
- C. All pancreatic enzymes are secreted in their active forms.
- D. Cholecystokinin receptors on human pancreatic acinar cells are important in pancreatic enzyme secretion.
- E. Enterokinase found in the duodenal mucosa activates vasoactive intestinal peptide which then cleaves zymogens into active proteases.

VII-111. A 45-year-old woman with a known history of cholelithiasis is admitted to the hospital with severe mid-epigastric pain, fever to 38.5°C (101.3°F), tachycardia to 110 beats/min, and a blood pressure of 100/50. Her examination shows a diffusely tender abdomen with guarding. Radiographs show an abdominal ileus with no free air. Labs are notable for a hemoglobin of 15 g/dL and elevations of amylase and lipase. Which of the following statements regarding this patient's likely diagnosis is true?

- A. Elevated lipase is more specific than elevated amylase for the diagnosis of acute pancreatitis.
- B. Hypercalcemia occurs in >75% of cases of acute pancreatitis.
- C. Magnitude of lipase elevation above normal is correlated with the severity of acute pancreatitis.
- D. Serum amylase levels will remain elevated for up to 30 days after the resolution of acute pancreatitis.
- E. The combination of elevated serum amylase and metabolic acidosis (pH <7.32) has a >90% positive predictive value for acute pancreatitis.

VII-112. A 27-year-old woman is admitted to the hospital with acute-onset severe right upper quadrant pain that radiates to the back. The pain is constant and not relieved with eating or bowel movements. Her labs show marked elevation in amylase and lipase, and acute pancreatitis is diagnosed. Which of the following is the best first test to demonstrate the etiology of her pancreatitis?

- A. Right upper quadrant ultrasound
- B. Serum alcohol level
- C. Serum triglyceride level
- D. Technetium hepatobiliary iminodiacetic acid scan
- E. Urine drug screen

VII-113. A 58-year-old female with known alcohol abuse to the hospital with abdominal pain after a drinking binge. Her symptoms have been present for 3 days, and she has persisted to drink heavily. She now has persistent vomiting and feels dizzy on standing. On examination, she has severe epigastric and right upper quadrant tenderness and decreased bowel sounds, and she appears uncomfortable. She has a blue-red discoloration of the flanks bilaterally but denies any recent trauma. What is the significance of this finding?

- A. Abdominal plain film is likely to show pancreatic calcification.
- B. CT of the abdomen is likely to show severe necrotizing pancreatitis with hemoperitoneum.
- C. Endoscopic ultrasound is likely to reveal a pancreatic mass.
- D. Right upper quadrant ultrasound will likely show acute cholecystitis.
- E. Transvaginal ultrasound will likely show a ruptured ectopic pregnancy.

VII-114. A 42-year-old male is admitted with severe epigastric pain radiating to the back. You suspect acute pancreatitis. All of the following are accepted diagnostic criteria for acute pancreatitis EXCEPT:

- A. Confirmatory findings of acute pancreatitis on cross-sectional abdominal imaging
- B. $\text{PaO}_2 < 60$
- C. Threefold or greater elevation in serum lipase and/or amylase
- D. Typical abdominal pain in the epigastrium that may radiate to the back

VII-115. A 45-year-old female with known gallstones presents to the hospital with 2 days of mid-epigastric tenderness radiating to her back, as well as nausea and vomiting. Her serum lipase is elevated to four times the upper limit of normal and abdominal ultrasound reveals gallstones and findings consistent with acute pancreatitis. Her initial hematocrit is 48 and her vital signs are notable for sinus

tachycardia with a pulse of 110 beats/min and a fever up to 38.1°C (100.6°F). After receiving 3 L of IV lactated ringers solution over the first 24 hours, her hematocrit drops to 38. Her heart rate and temperature have also improved. What is the significance of the drop in hematocrit?

- A. Current fluid management should be continued as her drop in hematocrit is evidence of effective fluid resuscitation.
- B. She has likely developed hemorrhage from severe necrotizing pancreatitis and should undergo urgent CT scanning of the abdomen.
- C. She should be evaluated for hemolysis as a cause of the drop in hematocrit.
- D. She should receive two units of packed red blood cells to prevent organ dysfunction from hypoperfusion related to evolving anemia.
- E. She should undergo endoscopic ultrasound to look for an evolving pancreatic pseudocyst.

VII-116. A 56-year-old male presents to the emergency department with 1–2 days of worsening abdominal pain, fever, tachycardia and shortness of breath. He denies any alcohol use and has no known history of gallstones. His initial labs are notable for a markedly elevated amylase and lipase. He is given 2 L of IV isotonic fluids and is started on empiric antibiotics. CT of the chest, abdomen, and pelvis reveal severe necrotizing pancreatitis, as well as bilateral alveolar infiltrates with basilar atelectasis. He requires intubation and mechanical ventilation for evolving respiratory failure. All cultures are negative. Which of the following best explains the etiology of his respiratory failure?

- A. He has developed acute respiratory distress syndrome (ARDS) from severe pancreatitis.
- B. He has developed heart failure from overly aggressive fluid resuscitation.
- C. He has developed pulmonary hemorrhage related to coagulopathy from sepsis.
- D. He likely has culture-negative pneumonia and developed ARDS from direct lung injury.
- E. He likely has sepsis from necrotizing pancreatitis and developed ARDS from the resulting bacteremia.

VII-117. Which of the following statements regarding enteral feeding in acute pancreatitis is true?

- A. A patient with persistent evidence of pancreatic necrosis on CT 2 weeks after acute presentation should be maintained on bowel rest.
- B. All patients with elevations of amylase and lipase and CT evidence of pancreatitis should be fasted until amylase and lipase normalize.
- C. Enteral feeding has been demonstrated to have fewer infectious complications than total parenteral

nutrition in the management of patients with acute pancreatitis.

- D. Patients requiring surgical removal of infected pancreatic pseudocysts should be treated with total parenteral nutrition.
- E. Total parenteral nutrition has been shown to maintain integrity of the intestinal tract in acute pancreatitis.

VII-118. A 25-year-old woman with cystic fibrosis is diagnosed with chronic pancreatitis. She is at risk for all of the following complications EXCEPT:

- A. Vitamin B₁₂ deficiency
- B. Vitamin A deficiency
- C. Pancreatic carcinoma
- D. Niacin deficiency
- E. Steatorrhea

ANSWERS

VII-1. **The answer is E.** (*Chap. 314*) The colon is primarily responsible for stool dehydration, decreasing volumes from 1000–1500 mL to 100–200 mL. The stomach triturates and mixes the food bolus with pepsin and acid. The stomach also secretes intrinsic factor, which is necessary for vitamin B₁₂ absorption. Gastric acid sterilizes the upper gut. Pancreatic juice contains enzymes for carbohydrate, protein, and fat digestion as well as bicarbonate to optimize the pH for enzyme activation.

VII-2. **The answer is C.** (*Chap. 314*) Pernicious anemia is associated with little or no gastric acid due to a lack of intrinsic factor. This is caused by autoantibodies that damage the gastric parietal cells. Gastrin G-cell hyperplasia results in increased gastrin levels, which leads to gastric acid hypersecretion. About 50% of G-cell hyperplasia is related to *Helicobacter pylori* infection and can be treated with eradication of the bacterium. Retained antrum syndrome is a rare postgastrectomy syndrome following a Billroth-II procedure in which the antrum has not been entirely removed. Some patients with duodenal ulcers have acid hypersecretion. Zollinger-Ellison syndrome is caused by a tumor (usually in the pancreas or duodenum) that secretes gastrin and therefore causes gastric acid hypersecretion.

VII-3. **The answer is D.** (*Chap. 314*) Mesenteric ischemia, biliary colic, and neoplasms most commonly cause noninflammatory visceral pain. Peptic ulcer, appendicitis, diverticulitis, inflammatory bowel disease, pancreatitis, cholecystitis, and infectious enterocolitis are all painful inflammatory diseases.

VII-4. **The answer is A.** (*Chap. 314*) Diarrhea from malabsorption usually improves with fasting, whereas secretory diarrhea persists without oral intake. Sudden awakening from sound sleep by pain suggests organic rather than functional disease. Symptoms from mechanical obstruction, ischemia, inflammatory bowel disease, and functional bowel disorders are worsened by meals. Ulcer pain occurs intermittently over weeks to months, whereas biliary colic has a sudden onset and lasts up to several hours. Ulcer symptoms may be relieved by eating or antacids.

VII-5. **The answer is B.** (*Chap. 315*) Since this patient is over 50 years old, even without a family history, screening for colorectal cancer is recommended. Colonoscopy is the gold standard for imaging the colonic mucosa. The cecum is reached in >95% of cases and the terminal ileum can often be examined. Colonoscopy has greater sensitivity than barium enema for colitis, polyps, and cancer. CT colonography rivals the accuracy of colonoscopy for detection of some polyps and cancer, although it is not as sensitive for the detection of flat lesions, such as serrated polyps. Flexible sigmoidoscopy is similar to colonoscopy, but it visualizes only the rectum and a variable portion of the left colon, typically to 60 cm from the anal verge.

VII-6. The answer is E. (*Chap. 315*) Patients undergoing percutaneous endoscopic gastrostomy tube placement should receive prophylactic antibiotics at the time as the procedure to prevent peristomal infection. Table VII-6 lists common endoscopic procedures and whether or not antibiotic prophylaxis is indicated.

TABLE VII-6 Antibiotic Prophylaxis for Endoscopic Procedures

Patient Condition	Procedure Contemplated	Goal of Prophylaxis	Periprocedural Antibiotic Prophylaxis
All cardiac conditions	Any endoscopic procedure	Prevention of infective endocarditis	Not indicated
Bile duct obstruction in the absence of cholangitis	ERCP with complete drainage	Prevention of cholangitis	Not recommended
Bile duct obstruction in absence of cholangitis	ERCP with anticipated incomplete drainage (e.g., sclerosing cholangitis, hilar strictures)	Prevention of cholangitis	Recommended; continue antibiotics after the procedure
Sterile pancreatic fluid collection (e.g., pseudocyst, necrosis), which communicates with pancreatic duct	ERCP	Prevention of cyst infection	Recommended; continue antibiotics after the procedure
Sterile pancreatic fluid collection	Transmural drainage	Prevention of cyst infection	Recommended
Solid lesion along upper GI tract	EUS-FNA	Prevention of local infection	Not recommended ^a
Solid lesion along lower GI tract	EUS-FNA	Prevention of local infection	Not recommended ^a
Cystic lesions along GI tract (including mediastinum and pancreas)	EUS-FNA	Prevention of cyst infection	Recommended
All patients	Percutaneous endoscopic feeding tube placement	Prevention of peristomal infection	Recommended ^b
Cirrhosis with acute GI bleeding	Required for all such patients, regardless of endoscopic procedures	Prevention of infectious complications and reduction of mortality	Recommended, on admission ^c
Continuous peritoneal dialysis	Lower GI tract endoscopy	Prevention of bacterial peritonitis	Recommended
Synthetic vascular graft and other nonvalvular cardiovascular devices	Any endoscopic procedure	Prevention of graft and device infection	Not recommended ^d
Prosthetic joints	Any endoscopic procedure	Prevention of septic arthritis	Not recommended ^e

^aLow rates of bacteremia and local infection.

^bCefazolin or an antibiotic with equivalent coverage of oral and skin flora.

^cRisk for bacterial infection associated with cirrhosis and GI bleeding is well established; ceftriaxone or a quinolone antibiotic recommended.

^dNo reported cases of infection associated with endoscopy.

^eVery low risk of infection.

Abbreviations: ERCP, endoscopic retrograde cholangiopancreatography; EUS-FNA, endoscopic ultrasound–fine-needle aspiration; GI, gastrointestinal.

Source: Adapted with permission from ASGE Standards of Practice Committee et al: Antibiotic prophylaxis for GI endoscopy. *Gastrointest Endosc* 81:81, 2015.

VII-7. The answer is E. (*Chap. 315*) Severe pain in and of itself is not an indication for urgent endoscopy in the setting of an upper GI bleed. The severity of the initial hemorrhage is the most important indication for urgent endoscopy, since a large initial bleed increases the likelihood of ongoing or recurrent bleeding. Patients with resting hypotension or orthostatic change in vital signs, repeated hematemesis, bloody nasogastric aspirate that does not clear with large volume lavage, or those requiring blood transfusions should be considered for urgent endoscopy. In addition, patients with cirrhosis, coagulopathy, respiratory or renal failure, and those over 70 years of age are more likely to have significant rebleeding and to benefit from prompt evaluation and treatment. Perforation of the esophagus, stomach, or duodenum may cause acute severe pain in conjunction with upper GI bleeding. Upright chest/abdomen radiography or CT scanning may be indicated.

VII-8. The answer is B. (*Chap. 315*) The endoscopic appearance of peptic ulcers provides useful prognostic information and guides the need for endoscopic therapy in patients with acute hemorrhage. A clean-based ulcer is associated with a low risk (3–5%) of rebleeding; patients with melena and a clean-based ulcer are often discharged home from the emergency room

or endoscopy suite if they are young, reliable, and otherwise healthy. Flat pigmented spots and adherent clots covering the ulcer base have a 10% and 20% risk of rebleeding, respectively. Endoscopic therapy may be considered for an ulcer with an adherent clot. When a fibrin plug is seen protruding from a vessel wall in the base of an ulcer (so-called sentinel clot or visible vessel), the risk of rebleeding from the ulcer is 40%. This finding generally leads to endoscopic therapy to decrease the rebleeding rate. When active spurting from an ulcer is seen, there is a 90% risk of ongoing bleeding without therapy.

- VII-9. The answer is A.** (*Chap. 315*) Endoscopy, also known as esophagogastroduodenoscopy, is the best test for evaluation of the proximal gastrointestinal tract. Because of high-quality images, disorders of color such as Barrett metaplasia and mucosal irregularities are easily demonstrated. Sensitivity of endoscopy is superior to that of barium radiography for mucosal lesions. Because the endoscope has an instrumentation channel, biopsy specimens are easily obtained, and dilation of strictures can also be performed. The sensitivity of radiography compared with endoscopy for detecting reflux esophagitis reportedly ranges from 22 to 95%, with higher grades of esophagitis (i.e., ulceration or stricture) exhibiting greater detection rates. Conversely, the sensitivity of barium radiography for detecting esophageal strictures is greater than that of endoscopy, especially when the study is done in conjunction with barium-soaked bread or a 13-mm barium tablet. Barium studies also provide an assessment of esophageal function and morphology that may be undetected on endoscopy. The major shortcoming of barium radiography is that it rarely obviates the need for endoscopy. Barium radiography does not require sedation, which in some populations at risk for conscious sedation is an important consideration.
- VII-10. The answer is E.** (*Chap. 316*) Intermittent solid food dysphagia is a classic symptom in Schatzki ring in which a distal esophageal ring occurs at the squamocolumnar mucosal junction. The origin of these rings is unknown, and smaller rings with a lumen of greater than 13 mm are common in the general population (up to 15%). When the lumen is less than 13 mm, dysphagia may occur. Schatzki rings typically occur in persons older than 40 years and often cause “steakhouse syndrome” from meat getting stuck at the ring. The rings are easily treated with dilation. Plummer-Vinson syndrome also includes esophageal rings, but typically, the rings occur in the proximal esophagus, are associated with iron-deficiency anemia, and occur in middle-aged women. Achalasia involves both solid and liquid dysphagia often with regurgitation. Adenocarcinoma often includes solid and liquid dysphagia at later stages. Most esophageal diverticula are asymptomatic.
- VII-11. The answer is A.** (*Chap. 316*) This patient has symptoms of esophagitis. In patients with HIV, various infections can cause esophagitis, including herpes simplex virus (HSV), cytomegalovirus (CMV), varicella-zoster virus, *Candida*, and HIV itself. The lack of thrush does not rule out *Candida* as a cause of esophagitis, and esophagogastroduodenoscopy (EGD) is necessary for diagnosis. CMV classically causes serpiginous ulcers in the distal esophagus that may coalesce to form large ulcers. Brushings alone are insufficient for diagnosis, and biopsies must be performed. Biopsies reveal intranuclear and intracytoplasmic inclusions with enlarged nuclei in large fibroblasts and endothelial cells. Given this patient’s notable swallowing symptoms, IV ganciclovir is the treatment of choice. Valganciclovir is an effective oral preparation. Foscarnet is useful in treating ganciclovir-resistant CMV. HSV manifests as vesicles and punched-out lesions in the esophagus with the characteristic findings on biopsy of ballooning degeneration with ground-glass changes in the nuclei. It can be treated with acyclovir or foscarnet in resistant cases. *Candida* esophagitis has the appearance of yellow nodular plaques with surrounding erythema. Treatment usually requires fluconazole therapy. Finally, HIV alone can cause esophagitis that can be quite resistant to therapy. On EGD, these ulcers appear deep and linear. Treatment with thalidomide or oral glucocorticoids is employed, and highly active antiretroviral therapy should be considered.
- VII-12. The answer is A.** (*Chap. 316*) The barium swallow image demonstrates achalasia with esophageal dilation narrowing at the gastroesophageal junction and an air-fluid level in the mid-esophagus. Achalasia is a rare disease caused by loss of ganglion cells within the

esophageal myenteric plexus with a population incidence of about 1:100,000; it usually presents between age 25 and 60. With long-standing disease, aganglionosis is noted. The disease involves both excitatory (cholinergic) and inhibitory (nitric oxide) ganglionic neurons. This leads to impaired deglutitive lower esophageal sphincter (LES) relaxation and absent peristalsis. Increasing evidence suggests that the ultimate cause of ganglion cell degeneration in achalasia is an autoimmune process attributable to a latent infection with human herpes simplex virus 1 combined with genetic susceptibility. Long-standing achalasia is characterized by progressive dilatation and sigmoid deformity of the esophagus with hypertrophy of the LES. Clinical manifestations may include dysphagia, regurgitation, chest pain, and weight loss. Most patients report solid and liquid food dysphagia. Regurgitation occurs when food, fluid, and secretions are retained in the dilated esophagus. Patients with advanced achalasia are at risk for bronchitis, pneumonia, or lung abscess from chronic regurgitation and aspiration. The differential diagnosis of achalasia includes diffuse esophageal spasm, Chagas disease, and pseudo-achalasia. Chagas disease is endemic in areas of central Brazil, Venezuela, and northern Argentina and spread by the bite of the reduviid (kissing) bug that transmits the protozoan *Trypanosoma cruzi*. The chronic phase of the disease develops years after infection and results from destruction of autonomic ganglion cells throughout the body, including the heart, gut, urinary tract, and respiratory tract. Tumor infiltration, most commonly seen with carcinoma in the gastric fundus or distal esophagus, can mimic idiopathic achalasia. The resulting pseudo-achalasia accounts for up to 5% of suspected cases and is more likely with advanced age, abrupt onset of symptoms (<1 year), and weight loss. Hence, endoscopy is a necessary part of the evaluation of achalasia. When the clinical suspicion for pseudo-achalasia is high and endoscopy nondiagnostic, CT scanning or endoscopic ultrasound may be of value. There is no known way of preventing or reversing achalasia. Therapy is directed at reducing LES pressure so that gravity and esophageal pressurization promote esophageal emptying. Peristalsis rarely, if ever, recovers. Botulinum toxin, injected into the LES under endoscopic guidance, inhibits acetylcholine release from nerve endings and improves dysphagia in about 66% of cases for at least 6 months. The only durable therapies for achalasia are pneumatic dilatation and Heller myotomy.

VII-13. The answer is B. (Chap. 316) These findings are characteristic of Barrett metaplasia, the most serious complication of gastroesophageal reflux disease. Barrett metaplasia has a strong association with the subsequent development of esophageal adenocarcinoma. The incidence of these lesions has increased, not decreased, in the era of potent acid suppression. Barrett metaplasia is endoscopically recognized by tongues of reddish mucosa extending proximally from the gastroesophageal junction or histopathologically identified by the finding of specialized columnar metaplasia. Barrett metaplasia can progress to adenocarcinoma through the intermediate stages of low- and high-grade dysplasia. Due to this risk, areas of Barrett metaplasia and especially any included areas of mucosal irregularity should be extensively biopsied. No high-level evidence confirms that aggressive antisecretory therapy or antireflux surgery causes regression of Barrett esophagus or prevents adenocarcinoma. Although the management of Barrett esophagus remains controversial, the finding of dysplasia in Barrett esophagus, particularly high-grade dysplasia, mandates further intervention. In addition to the high rate of progression to adenocarcinoma, there is also a high prevalence of unrecognized coexisting cancer with high-grade dysplasia. Nonetheless, treatment remains controversial. Esophagectomy, intensive endoscopic surveillance, and mucosal ablation have all been advocated. Currently, esophagectomy is the gold standard treatment for high-grade dysplasia in an otherwise healthy patient with minimal surgical risk. However, esophagectomy has a mortality ranging from 3 to 10%, along with substantial morbidity. As a result of these factors and the increasing evidence of the effectiveness of endoscopic therapy with purpose-built radiofrequency ablation devices, many now favor this therapy as a preferable management strategy.

VII-14. The answer is D. (Chap. 316) This person has eosinophilic esophagitis (EoE) as evidenced by his symptoms, esophagogastroduodenoscopy appearance, and pathologic findings. The biopsy shows infiltration of the esophageal squamous epithelium with eosinophils, with basal cell hyperplasia, and lamina propria fibrosis. EoE characteristically exhibits multiple esophageal rings, linear furrows, white punctate exudate, and

strictures on endoscopy. A trial of a proton pump inhibitor (PPI) leads to the resolution of eosinophilic esophagitis in about 30–50% of cases. If a PPI trial does not work, patients are often encouraged to start an elimination diet to try to avoid common food allergens that might be causative. Swallowed topical corticosteroids are effective but esophagitis often recurs on cessation of the topical steroid. Systemic corticosteroids are only used if all other treatment options fail. The role of newer anti-interleukin-5 therapies such as mepolizumab in treating eosinophilic esophagitis is not yet known.

VII-15. The answer is A. (*Chap. 316*) The symptoms and barium swallow are consistent with the diagnosis of diffuse esophageal spasm (DES), which is characterized by DES episodes of dysphagia and chest pain attributable to abnormal esophageal contractions with normal deglutitive LES relaxation. The barium swallow demonstrates the characteristic “corkscrew” esophagus results from spastic contraction of the helical muscle array in the esophageal wall. Manometry in patients with DES is characterized by contractions in the distal esophagus with short latency relative to the time of the pharyngeal contraction. However, DES may be confused with or actually overlap with some forms of achalasia. High-resolution manometry has led to the recognition of three distinct types of achalasia. All are characterized by impaired LES contraction and the absence of peristalsis. Classic achalasia has minimal pressurization of the esophageal body, whereas substantial fluid pressurization is observed in achalasia with esophageal compression, and spastic esophageal contractions are observed with spastic achalasia. DES may be particularly confused with spastic achalasia. Manometric studies of the esophagus performed to evaluate chest pain and/or dysphagia often report minor abnormalities (e.g., hypertensive or hypotensive peristalsis, hypertensive LES) that are insufficient to diagnose either achalasia or DES. These findings are of unclear significance.

VII-16. The answer is D. (*Chap. 316*) The side effects of proton pump inhibitor (PPI) therapy are generally minimal. Rare cases of interstitial nephritis and severe, reversible hypomagnesemia have been reported. Vitamin B₁₂ and iron absorption may be compromised and susceptibility to enteric infections, particularly *Clostridium difficile* colitis, increased with treatment. PPIs are more effective than H₂ receptor antagonists, but both are more effective than placebo. The effectiveness of PPIs is not substantially increased at higher doses. The most broadly applicable recommendation for gastroesophageal reflux disease (GERD) treatment is weight reduction. Even though the benefit with respect to reflux cannot be assured, the strong epidemiologic relationship between body mass index and GERD and the secondary health gains of weight reduction is beyond dispute.

VII-17. The answer is D. (*Chap. 317*) Noninvasive testing for *Helicobacter pylori* infection is recommended in patients with suggestive symptoms and no other indication for endoscopy (e.g., gastrointestinal bleeding, atypical symptoms). Several tests have good sensitivity and specificity (Table VII-17), including plasma serology for *H. pylori*, 14C- or 13C-urea

TABLE VII-17 Tests for Detection of *Helicobacter pylori*

Test	Sensitivity/Specificity (%)	Comments
Invasive (Endoscopy/Biopsy Required)		
Rapid urease	80–95/95–100	Simple, false negative with recent use of PPIs, antibiotics, or bismuth compounds
Histology	80–90/>95	Requires pathology processing and staining; provides histologic information
Culture	—/—	Time-consuming, expensive, dependent on experience; allows determination of antibiotic susceptibility
Noninvasive		
Serology	>80/>90	Inexpensive, convenient; not useful for early follow-up
Urea breath test	>90/>90	Simple, rapid; useful for early follow-up; false negatives with recent therapy (see rapid urease test); exposure to low-dose radiation with ¹⁴ C test
Stool antigen	>90/>90	Inexpensive, convenient

Abbreviation: PPIs, proton pump inhibitors.

breath test, and the fecal *H. pylori* antigen test. Sensitivity and specificity are >80% and >90%, respectively, for serology, whereas the sensitivity and specificity of the urea breath test and fecal antigen testing are >90% for both. Serology is not useful for early follow-up after therapy completion because antibody titers will take several weeks to months to fall. The urea breath test, which relies on the presence of urease secreted by *H. pylori* to digest the swallowed radioactive urea and liberate ^{14}C or ^{13}C as part of ammonia, is simple and rapid. It is useful for early follow-up because it requires living bacteria to secrete urease and produce a positive test. The limitations to the test include requirement for ingestion of radioactive materials, albeit low dose, and false-negative results with recent use of proton pump inhibitors, antibiotics, or bismuth compounds. Stool antigen testing is cheap and convenient but is not established for proof of eradication.

VII-18. The answer is A. (*Chap. 317*) Documented eradication of *Helicobacter pylori* in patients with peptic ulcer disease (PUD) is associated with a dramatic decrease in ulcer recurrence to <10–20% compared with 59% in gastric ulcer patients and 67% in duodenal ulcer patients when the organism is not eliminated. Eradication of the organism may lead to diminished recurrent ulcer bleeding. The effect of its eradication on ulcer perforation is unclear. Extensive effort has been made in determining who of the many individuals with *H. pylori* infection should be treated. The common conclusion arrived at by multiple consensus conferences around the world is that *H. pylori* should be eradicated in patients with documented PUD. This holds true independent of time of presentation (first episode or not), severity of symptoms, presence of confounding factors such as ingestion of nonsteroidal anti-inflammatory drugs, or whether the ulcer is in remission. Multiple drugs have been evaluated in the therapy of *H. pylori*. No single agent is effective in eradicating the organism. Combination therapy for 14 days provides the greatest efficacy, although regimens based on sequential administration of antibiotics also appear promising (Table VII-18). A shorter administration course (7–10 days), although attractive, has not proved as successful as the 14-day regimens. Suggested treatment regimens for *H. pylori* are outlined below. Choice of a particular regimen will be influenced by several factors, including efficacy, patient tolerance, existing antibiotic resistance, and cost of the drugs. The aim for initial eradication rates should be 85–90%. Dual therapy (proton pump inhibitor [PPI] plus amoxicillin, PPI plus clarithromycin, ranitidine bismuth citrate [Triatec] plus clarithromycin) is not recommended in view of studies demonstrating eradication rates of <80–85%. Addition of acid suppression assists in providing early symptom relief and enhances bacterial eradication. Triple therapy, although effective, has several drawbacks, including the potential for poor patient compliance and drug-induced side effects. Compliance is addressed by simplifying the regimens so that patients can take the medications twice a day. Simpler (dual therapy) and shorter regimens (7 and 10 days) are not as effective as triple therapy for 14 days. Two anti-*H. pylori* regimens are available in prepackaged formulation: Prevpac (lansoprazole, clarithromycin, and amoxicillin) and Helidac (bismuth subsalicylate, tetracycline, and metronidazole). The contents of the Prevpac are to be taken twice per day for 14 days, whereas Helidac constituents are taken four times per day with an antisecretory agent (PPI or H_2 blocker), also taken for at least 14 days. Clarithromycin-based triple therapy should be avoided in settings where *H. pylori* resistance to this agent exceeds 15–20%. Quadruple therapy should be reserved for patients with failure to eradicate *H. pylori* after an effective initial course.

VII-19. The answer is C. (*Chap. 317*) Fasting gastrin levels can be elevated in a variety of conditions, including atrophic gastritis with or without pernicious anemia, G-cell hyperplasia, and acid suppressive therapy (gastrin levels increase as a consequence of loss of negative feedback). The diagnostic concern in a patient with persistent ulcers following optimal therapy is Zollinger-Ellison syndrome (ZES). A fasting result is not sufficient to make a diagnosis because gastrin levels may be elevated in a variety of conditions. Elevated basal acid secretion also is consistent with ZES, but up to 12% of patients with peptic ulcer disease may have basal acid secretion as high as 15 mEq/h. Thus, additional testing is necessary. Gastrin levels may go up with a meal (>200%), but this test does not distinguish G-cell hyperfunction from ZES. The best test in this setting is the secretin stimulation test. An increase in gastrin levels >200 pg within 15 minutes of administering 2 $\mu\text{g}/\text{kg}$ of

TABLE VII-18 Recommended First-Line Therapies for *Helicobacter pylori* Infection

Regimen	Drugs (doses)	Dosing Frequency	Duration (days)	FDA Approval
Clarithromycin triple	PPI (standard or double dose) Clarithromycin (500 mg) Amoxicillin (1 g) or metronidazole (500 mg tid)	bid	14	Yes ^a
Bismuth quadruple	PPI (standard dose) Bismuth subcitrate (120–300 mg) or subsalicylate (300 mg) Tetracycline (500 mg) Metronidazole (250–500 mg)	bid qid qid qid (250) tid to qid (500)	10–14	No ^b
Concomitant	PPI (standard dose) Clarithromycin (500 mg) Amoxicillin (1 g) Nitroimidazole (500 mg) ^c	bid	10–14	No
Sequential	PPI (standard dose) PPI, clarithromycin (500 mg) + nitroimidazole (500 mg) ^c	bid bid	5–7 5–7	No
Hybrid	PPI (standard dose) + amoxicillin (1 g) PPI, amoxicillin, clarithromycin (500 mg), nitroimidazole (500 mg) ^c	bid bid	7 7	No
Levofloxacin triple	PPI (standard or double dose) + amoxicillin (1 g) Levofloxacin (500 mg) Amoxicillin (1 g)	bid qd bid	5–7	No
Levofloxacin sequential	PPI (standard or double dose) + amoxicillin (1 g) PPI, amoxicillin, levofloxacin (500 mg qd), nitroimidazole (500 mg) ^c	bid bid	5–7 5–7	No
LOAD	Levofloxacin (250 mg) PPI (double dose) Nitazoxanide (500 mg) Doxycycline (100 mg)	qd qd bid qd	7–10	No

^aSeveral PPI, clarithromycin, and amoxicillin combinations have achieved FDA approval. PPI, clarithromycin, and metronidazole is not an FDA-approved treatment regimen.

^bPPI, bismuth, tetracycline, and metronidazole combined with a PPI for 10 days is an FDA-approved treatment regimen.

^cMetronidazole or tinidazole.

Abbreviations: bid, twice daily; FDA, Food and Drug Administration; PPI, proton pump inhibitor; tid, three times daily; qd, once daily; qid, four times daily.

Source: Reproduced with permission from Chey WD et al: ACG Clinical Guideline: treatment of *Helicobacter pylori* Infection. *Am J Gastroenterol* 112:212, 2017.

secretin by IV bolus has a sensitivity and specificity of >90% for ZES. Endoscopic ultrasonography is useful in locating the gastrin-secreting tumor once the positive secretin test is obtained. Genetic testing for mutations in the gene that encodes the menin protein can detect the fraction of patients with gastrinomas that are a manifestation of multiple endocrine neoplasia type 1 (Wermer syndrome). Gastrinoma is the second most common tumor in this syndrome after parathyroid adenoma, but its peak incidence is generally in the third decade.

VII-20. The answer is B. (Chap. 317) Long-term acid suppression, especially with proton pump inhibitors (PPIs), has been associated with a higher incidence of community-acquired pneumonia as well as community- and hospital-acquired *Clostridium difficile*-associated disease. The impact of PPI-induced changes in the host microbiome is postulated to play a role in the increased risk of infection, but this theory needs to be confirmed. Cimetidine, not famotidine, may have weak antiandrogenic side effects resulting in reversible gynecomastia and impotence, primarily in patients receiving high doses for prolonged periods of time (months to years). The long-term use of calcium carbonate (converts to calcium chloride in the stomach) can lead to milk-alkali syndrome (hypercalcemia, hyperphosphatemia with possible renal calcinosis and progression to renal insufficiency). Rebound

gastric acid hypersecretion has been described in *Helicobacter pylori*-negative individuals after discontinuation of PPIs, not H₂ blockers. It occurs even after relatively short-term usage (2 months) and may last for up to 2 months after the PPI has been discontinued. The mechanism involves gastrin-induced hyperplasia and hypertrophy of histamine-secreting enterochromaffin-like cells. The clinical relevance of this observation is that individuals may have worsening symptoms of gastroesophageal reflux disease or dyspepsia on stopping the PPI. Gradual tapering of the PPI and switching to an H₂ receptor antagonist may prevent this from occurring. Aluminum hydroxide can produce constipation and phosphate depletion. Magnesium hydroxide can produce loose stools.

- VII-21. The answer is D.** (*Chap. 317*) Basal acid production occurs in a circadian pattern, with highest levels occurring during the night and lowest levels during the morning hours. Cholinergic input via the vagus nerve and histaminergic input from local gastric sources are the principal contributors to basal acid secretion. Stimulated gastric acid secretion occurs primarily in three phases based on the site where the signal originates (cephalic, gastric, and intestinal). Sight, smell, and taste of food are the components of the cephalic phase, which stimulates gastric secretion via the vagus nerve. The gastric phase is activated once food enters the stomach. This component of secretion is driven by nutrients (amino acids and amines) that directly (via peptone and amino acid receptors) and indirectly (via stimulation of intramural gastrin-releasing peptide neurons) stimulate the G cell to release gastrin, which in turn activates the parietal cell via direct and indirect mechanisms. Distention of the stomach wall also leads to gastrin release and acid production. The last phase of gastric acid secretion is initiated as food enters the intestine and is mediated by luminal distention and nutrient assimilation.
- VII-22. The answer is B.** (*Chap. 318*) The patient presents with nonspecific gastrointestinal symptoms, but the presence of weight loss does suggest malabsorption syndrome. Patients with lactose intolerance are usually able to relate symptoms to consumption of milk-based products and also report a strong history of crampy pain and flatulence. Therefore, a lactose-free diet is unlikely to be helpful. The patient does not have nocturnal diarrhea, which is commonly a feature of steatorrhea along with floating stools. In the absence of symptoms suggesting fat malabsorption, the first test should not be fecal fat measurement. Because the patient has weight loss, irritable bowel syndrome is less likely, and increased dietary fiber is unlikely to be useful. Finally, the patient's symptoms may be consistent with celiac disease. Antibodies to gliadin, endomysium, and tissue transglutaminase can be easily measured in peripheral blood. Antiendomysial antibody has 90–95% sensitivity and equal specificity, making it a reasonable first test in symptomatic individuals. The presence of the antibody is not diagnostic, however, and duodenal biopsy is recommended. Duodenal biopsy will show villous atrophy, absence or reduced height of villi, cuboidal appearance of surface epithelial cells, and increased lymphocytes and plasma cells in the lamina propria. These changes regress with complete removal of gluten from the diet.
- VII-23. The answer is E.** (*Chap. 318*) The patient presents with symptoms suggestive of Whipple disease, a chronic multisystem disease often including diarrhea/steatorrhea, migratory arthralgias, weight loss, and central nervous system (CNS) or cardiac problems. Generally the presentation is of insidious onset, and dementia is a late finding and a poor prognostic sign. The disease primarily occurs in middle-aged white males. The diagnosis requires small bowel biopsy and demonstration of periodic acid–Schiff-positive macrophages within the small bowel. Small bacilli are often present and suggest the diagnosis of Whipple disease. Similar macrophages may be found in other affected organs (e.g., the CNS). Dilated lymphatics are present in patients with intestinal lymphangiectasia. Mononuclear cell infiltrate in the lamina propria is often demonstrated in patients with tropical sprue, and fat villi with crypt hyperplasia is the hallmark of celiac disease.
- VII-24. The answer is D.** (*Chap. 318*) This patient has a stool osmolality gap (measured stool osmolality – calculated stool osmolality) of 10 mOsm/L, suggesting a secretory rather than an osmotic cause for diarrhea. Secretory causes of diarrhea include toxin-mediated diarrhea (cholera, enterotoxigenic *Escherichia coli*) and intestinal peptide-mediated

diarrhea in which the major pathophysiology is a luminal or circulating secretagogue. The distinction between secretory diarrhea and osmotic diarrhea aids in forming a differential diagnosis. Secretory diarrhea will not decrease substantially during a fast and has a low osmolality gap. Osmotic diarrhea will generally decrease during a fast and has a high (>50 mOsm/L) osmolality gap. Celiac sprue, chronic pancreatitis, lactase deficiency, and Whipple disease all cause osmotic diarrhea. A low stool osmolality (<290 mOsm/kg H_2O) reflects the addition of either dilute urine or water, indicating either collection of urine and stool together or so-called factitious diarrhea, a form of Munchausen syndrome.

- VII-25. The answer is D.** (*Chap. 318*) The intestine is exposed to a large number of potential antigens and enteric and invasive microorganisms, and it is extremely active at preventing the entry of almost all of these agents. The intestinal mucosa also synthesizes and secretes secretory IgA. The intestine absorbs ~ 7 – 8 L of fluid daily, a volume comprising dietary fluid intake (1 – 2 L/d) and salivary, gastric, pancreatic, biliary, and intestinal fluid (6 – 7 L/d). Several stimuli, especially bacteria and bacterial enterotoxins, induce fluid and electrolyte secretion that may lead to diarrhea. The intestinal mucosa is a major site for the production of proteins, including apolipoproteins. The intestine is one of the largest endocrine organs in the body and produces several amines (e.g., 5-hydroxytryptophan) and peptides that serve as paracrine and hormonal mediators of intestinal function. Bile acids are not present in the diet but are synthesized in the liver by a series of enzymatic steps that also represent cholesterol catabolism. Bile acids are either primary or secondary. Primary bile acids are synthesized in the liver from cholesterol, and secondary bile acids are synthesized from primary bile acids in the intestine by colonic bacterial enzymes.
- VII-26. The answer is C.** (*Chap. 318*) Impaired lipolysis can lead to steatorrhea and can occur in the presence of pancreatic insufficiency due to chronic pancreatitis in adults or cystic fibrosis in children and adolescents. Normal lipolysis can be maintained by $\sim 5\%$ of maximal pancreatic lipase secretion; thus, steatorrhea is a late manifestation of these disorders. Steatorrhea can also result from impaired movement of fatty acids across the unstirred aqueous fluid layer in two situations: (1) an increase in the relative thickness of the unstirred water layer that occurs in bacterial overgrowth syndromes secondary to functional stasis (e.g., scleroderma) and (2) a decrease in the duodenal concentration of conjugated bile acids below the critical micelle concentration, resulting in impaired micelle formation. Impaired lipid absorption as a result of mucosal inflammation (e.g., celiac disease) and/or intestinal resection can also lead to steatorrhea. Abnormalities of intestinal lymphatics (e.g., intestinal lymphangiectasia) may also be associated with steatorrhea as well as protein loss.
- VII-27. The answer is C.** (*Chap. 319*) Chronic, bloody diarrhea associated with weight loss and systemic symptoms in a young person is highly suggestive of inflammatory bowel disease. The patient's surgical findings suggest discontinuous lesions, which is typical of Crohn disease. Ulcerative colitis, in contrast, typically affects the rectum and proceeds caudally from there without normal mucosa until the area of inflammation terminates. The presence of strictures and fissures further supports the diagnosis of Crohn disease, as these are not features of ulcerative colitis. Microscopically, both ulcerative colitis and Crohn disease may have crypt abscess, and although Crohn disease is more often transmural, full-thickness disease may be present in ulcerative colitis. The hallmark of Crohn disease is granulomas that may be present throughout the bowel wall and involve the lymph nodes, mesentery, peritoneum, liver, and pancreas. Although pathognomonic for Crohn disease, granulomas are only found in about half of surgical resections. Flat villi are not always present in either disease and are more commonly found in isolation with celiac disease.
- VII-28. The answer is D.** (*Chap. 319*) There are a number of dermatologic manifestations of inflammatory bowel disease (IBD), and each type of IBD has a particular predilection for different dermatologic conditions. This patient has pyoderma gangrenosum. Pyoderma gangrenosum can occur in up to 12% of patients with ulcerative colitis and is characterized by a lesion that begins as a pustule and progresses concentrically to surrounding normal skin. The lesions ulcerate with violaceous, heaped margins and surrounding erythema.

They are typically found on the lower extremities. Often the lesions are difficult to treat and respond poorly to colectomy; similarly, pyoderma gangrenosum is not prevented by colectomy. Treatment commonly includes IV antibiotics, glucocorticoids, dapsone, infliximab, and other immunomodulatory agents. Erythema nodosum is more common in Crohn disease and attacks correlate with bowel symptoms. The lesions are typically multiple, red hot, tender nodules measuring 1–5 cm and are found on the lower legs and arms. Psoriasis is more common in ulcerative colitis. Finally, pyoderma vegetans is a rare disorder in intertriginous areas reported to be manifestation of IBD in the skin.

- VII-29. The answer is C.** (*Chap. 319*) There is a protective effect of previous appendectomy with confirmed appendicitis (risk reduction of 13–26%), particularly at a young age, on the development of ulcerative colitis (UC) across different geographical regions and populations. There is an association between antibiotic use and the development of childhood inflammatory bowel disease (IBD) with children who received one or more dispensations of antibiotics during the first year of life having a 2.9-fold increase in the risk of developing IBD during childhood. Peak incidence of UC and Crohn Disease (CD) is in the second to fourth decades, with 78% of CD studies and 51% of UC studies reporting the highest incidence among those aged 20–29 years old. A second modest rise in incidence occurs between the seventh and ninth decades of life. In countries that are becoming more westernized, including China, South Korea, India, Lebanon, Iran, Thailand, and countries in the French West Indies and North Africa, IBD appears to be emerging, emphasizing the importance of environmental factors in disease pathogenesis. The highest incidence rates of CD and UC have been reported in northern Europe, the United Kingdom, and North America.
- VII-30. The answer is C.** (*Chap. 319*) Azathioprine may be used as monotherapy in Crohn disease but is often used in combination with a biologic agent such as an anti-tumor necrosis factor therapy. Cyclosporine A is more commonly used in severe ulcerative colitis but given evidence of equal effectiveness, many providers prefer biologic therapies such as infliximab. Intermittent infusions of infliximab are discouraged since intermittent as opposed to regular (i.e., every 8 week) infusions increases the risk of antibody formation, which can reduce effectiveness and cause infusion-related symptoms. Maintenance prednisone should not be used in the management of Crohn disease given the potential for long-term side effects. Golimumab is approved by the Food and Drug Administration for the treatment of severe ulcerative colitis.
- VII-31. The answer is C.** (*Chap. 320*) Irritable bowel syndrome (IBS) is characterized by recurrence of lower abdominal pain with altered bowel habits over a period of time without progressive deterioration, onset of symptoms during periods of stress or emotional upset, absence of other systemic symptoms such as fever and weight loss, and small-volume stool without evidence of blood. Warning signs that the symptoms may be due to something other than IBS include presentation for the first time in old age, progressive course from the time of onset, persistent diarrhea after a 48-hour fast, and the presence of nocturnal diarrhea or steatorrheal stools. Each patient, except for patient C, has “warning” symptoms that should prompt further evaluation.
- VII-32. The answer is D.** (*Chap. 320*) Up to 80% of patients with irritable bowel syndrome (IBS) also have abnormal psychiatric features; however, no single psychiatric diagnosis predominates. The mechanism is not well understood but may involve altered pain thresholds. Although these patients are hypersensitive to colonic stimuli, this does not carry over to the peripheral nervous system. Functional brain imaging shows disparate activation in, for example, the mid-cingulate cortex, but brain anatomy does not discriminate IBS patients from those without IBS. An association between a history of sexual abuse and IBS has been reported. There is no reported association with sexually transmitted diseases. Patients with IBS do not have an increased risk of autoimmunity.
- VII-33. The answer is C.** (*Chap. 320*) Gut dysbiosis acting in concert with genetic and environmental factors may alter intestinal permeability and increase antigen presentation,

resulting in mast cell activation. Products of mast cell degranulation may alter neuronal and smooth muscle function causing irritable bowel syndrome (IBS) symptoms. The cytokines and chemokines generated from mucosal inflammation may cause symptoms such as fibromyalgia, chronic fatigue, and mood changes. IBS may be induced by gastrointestinal infection. About a third of IBS patients experienced an acute “gastroenteritis-like” illness at the onset of their chronic IBS symptomatology. This group of “postinfective” IBS occurs more commonly in females and affects younger rather than older patients. Patients with IBS frequently complain of abdominal distention and increased belching or flatulence, all of which they attribute to increased gas, but most patients who complain of increased gas generate no more than a normal amount of intestinal gas. Most IBS patients have impaired transit and tolerance of intestinal gas loads. As with studies of motor activity, IBS patients frequently exhibit exaggerated sensory responses to visceral stimulation. The frequency of perceptions of food intolerance is at least twofold more common than in the general population. Some (but not most) patients with IBS display persistent signs of low-grade mucosal inflammation with activated lymphocytes, mast cells, and enhanced expression of proinflammatory cytokines. Other studies also indicate that peripheral blood mononuclear cells from IBS patients show abnormal release of proinflammatory cytokines such as interleukin (IL)-6, IL-1 β , and tumor necrosis factor. These abnormalities may contribute to abnormal epithelial secretion and visceral hypersensitivity.

VII-34. The answer is B. (*Chap. 320*) Physiologic studies demonstrate that anticholinergic drugs inhibit the gastrocolic reflex; hence, postprandial pain is best managed by giving antispasmodics 30 minutes before meals so that effective blood levels are achieved shortly before the anticipated onset of pain. Fiber supplementation with psyllium has been shown to reduce perception of rectal distention, indicating that fiber may have a positive effect on visceral afferent function. This would not, however, be expected to help with postprandial pain. Avoiding flatulogenic foods, exercising, losing excess weight, and taking activated charcoal are safe but unproven remedies for excessive gas, so it is unclear if they would be effective in this case. The selective serotonin reuptake inhibitor (SSRI), citalopram, blunts perception of rectal distention and reduces the magnitude of the gastrocolic response in healthy volunteers. However, SSRIs have not been definitively shown to decrease pain in IBS patients.

VII-35. The answer is B. (*Chap. 321*) The patient presents with classic signs of diverticulitis with fever, abdominal pain that is usually left lower quadrant, anorexia or obstipation, and leukocytosis. This most commonly occurs in older individuals. Patients may present with an acute abdomen due to perforation, although this occurs in <25% of cases. Plain radiographs of the abdomen are seldom helpful but may show the presence of an air-fluid level in the left lower quadrant indicating a giant diverticulum with impending perforation. CT with oral contrast is the diagnostic modality of choice with the following findings: sigmoid diverticula, thickened colonic wall >4 mm, and inflammation within the pericolic space with or without the collection of contrast material or fluid. In 16% of patients, an abdominal abscess may be present. Symptoms of IBS may mimic those of diverticulitis. Therefore, suspected diverticulitis that does not meet CT criteria or is not associated with a leukocytosis or fever is not diverticular disease. Other conditions that can mimic diverticular disease include an ovarian cyst, endometriosis, acute appendicitis, and pelvic inflammatory disease. Although the benefit of colonoscopy in the evaluation of patients with diverticular disease has been called into question, its use is still considered important in the exclusion of colorectal cancer. The parallel epidemiology of colorectal cancer and diverticular disease provides enough concern for an endoscopic evaluation before operative management. Therefore, a colonoscopy should be performed approximately 6 weeks after an attack of diverticular disease. Although diverticular disease may result in hematochezia, this finding is generally not temporally linked to diverticulitis.

VII-36. The answer is B. (*Chap. 321*) Hemorrhoids can be internal or external; however, they are normally internal and may prolapse to the external position. Hemorrhoids are staged in the following manner: stage I, enlargement with bleeding; stage II, protrusion with spontaneous reduction; stage III, protrusion requiring manual reduction; and stage IV,

irreducible protrusion. Stage I, which this patient has, is treated with fiber supplementation, cortisone suppositories, and/or sclerotherapy. Stage II hemorrhoids are treated with fiber and cortisone suppositories. Stage III is offered the prior three therapies and banding or operative hemorrhoidectomy. Patients with stage IV hemorrhoids benefit from fiber and cortisone therapy as well as operative hemorrhoidectomy. Although substantial upper gastrointestinal bleeding may result in hematochezia, the absence of suggestive signs/symptoms and the consistent findings of hemorrhoids do not indicate the need for upper endoscopy in this case.

- VII-37. The answer is C.** (*Chap. 321*) This patient has symptoms (social isolation), signs (foul odor), and risk factors (multiparity) for procidentia (rectal prolapse) and fecal incontinence. Procidentia is far more common in women than men and is often associated with pelvic floor disorders. It is not uncommon for these patients to become socially withdrawn and suffer from depression because of the associated fecal incontinence. The foul odor is a result of poor perianal hygiene due to the prolapsed rectum. Although depression in the elderly is an important medical problem, it is too premature in the evaluation to initiate medical therapy for depression. Occult malignancy and thyroid abnormalities may cause fecal incontinence and depression, but a physical examination would be diagnostic and avoid costly tests. Often patients are concerned they have a rectal mass or carcinoma. Examination after an enema often makes the prolapse apparent. Medical therapy is limited to stool-bulking agents or fiber. Surgical correction is the mainstay of therapy.
- VII-38. The answer is D.** (*Chap. 321*) Patients with diverticular disease may benefit from a fiber-enriched diet that includes 30 g of fiber each day. The use of fiber increases colonic transit time, and, therefore, prevents increased intraluminal pressure leading to the development of diverticulosis. Diverticular disease is considered a functional bowel disorder associated with low-grade inflammation. However, the use of anti-inflammatory medications (mesalazine) in randomized clinical trials have not demonstrated any effect on recurrence rates over placebo alone. Probiotics containing *Lactobacillus acidophilus* and *Bifidobacterium* strains may be beneficial, but they have not been definitively shown to reduce the risk of symptomatic disease. The historical recommendation to avoid eating nuts is based on no more than anecdotal data. Colonoscopy should be considered 6 weeks after resolution of acute diverticulitis to rule out colon cancer. In the absence of complicated diverticular disease there is no indication for surgery following a single bout of diverticulitis. However, patients on immunosuppressive therapy, in chronic renal failure, or with a collagen-vascular disease have a fivefold greater risk of perforation during recurrent attacks. Surgical therapy is indicated in all low-surgical-risk patients with complicated diverticular disease.
- VII-39. The answer is A.** (*Chap. 321*) This patient has a perianal abscess. The development of a perianal abscess is more common in men than women by a ratio of 3:1. The peak incidence is in the third to fifth decade of life and it is more prevalent in immunocompromised patients such as those with diabetes, hematologic disorders, inflammatory bowel disease, and HIV. The other therapies listed are for management of noninfected hemorrhoids. While hemorrhoids can protrude from the anal canal, the presence of fever and a fluctuant mass are atypical for hemorrhoids.
- VII-40. The answer is D.** (*Chap. 322*) This patient with a history of atherosclerotic cardiovascular disease and diabetes is at high risk for nonocclusive mesenteric ischemia in the setting of sepsis, hypotension, and administered vasoconstrictors. Intestinal ischemia is further classified based on etiology, which dictates management (Table VII-40): (1) arterial occlusive mesenteric ischemia, (2) nonocclusive mesenteric ischemia, and (3) mesenteric venous thrombosis. Risk factors for arterial occlusive mesenteric ischemia are generally acute in onset and include atrial fibrillation, recent myocardial infarction, valvular heart disease, and recent cardiac or vascular catheterization, all of which result in embolic clot reaching the mesenteric circulation. Nonocclusive mesenteric ischemia, also known as “intestinal angina,” is generally more insidious and most often seen in the aging population affected by atherosclerotic disease. Nonocclusive mesenteric ischemia

TABLE VII-40 Overview of the Management of Acute Intestinal Ischemia

Condition	Key to Early Diagnosis	Treatment of Underlying Cause	Treatment of Specific Lesion	Treatment of Systemic Consequence
Arterio-occlusive mesenteric ischemia	CT angiography	Anticoagulation	Laparotomy	Ensure hydration
1. Arterial embolus	Early laparotomy	Cardioversion	Embolectomy	Give antibiotics
		Proximal thrombectomy	Vascular bypass	Reverse acidosis
			Assess viability and resect dead bowel	Optimize oxygen delivery
				Avoid vasoconstrictors
2. Arterial thrombosis	Duplex ultrasound	Anticoagulation	Endovascular approach: thrombolysis, angioplasty and stenting	Give antibiotics
	Angiography	Hydration	Endarterectomy/thrombectomy or vascular bypass	Reverse acidosis
			Assess viability and resect dead bowel	Optimize oxygen delivery
				Support cardiac output
				Avoid vasoconstrictors
Mesenteric venous thrombosis	Spiral CT	Anticoagulation	Anticoagulation ± laparotomy/thrombectomy/catheter-directed thrombolysis	Give antibiotics
Venous thrombosis	Angiography with venous phase	Massive hydration	Assess viability and resect dead bowel	Reverse acidosis
				Optimize oxygen delivery
				Support cardiac output
				Avoid vasoconstrictors
Nonocclusive mesenteric ischemia	Vasospasm: Angiography	Ensure hydration	Vasospasm	Ensure hydration
	Hypoperfusion: Spiral CT or colonoscopy	Support cardiac output	Intra-arterial vasodilators	Give antibiotics
		Avoid vasoconstrictors	Hypoperfusion	Reverse acidosis
			Delayed laparotomy	Optimize oxygen delivery
			Assess viability and resect dead bowel	Support cardiac output
				Avoid vasoconstrictors

Source: Modified with permission from Cameron JL: *Current Surgical Therapy*, 2nd ed. Toronto: BC Decker, 1986.

is also seen in patients receiving high-dose vasopressor infusions, patients with cardiogenic or septic shock, and patients with cocaine overdose. Nonocclusive mesenteric ischemia is the most prevalent gastrointestinal disease complicating cardiovascular surgery. Mesenteric venous thrombosis is less common and is associated with the presence of a hypercoagulable state including protein C or S deficiency, anti-thrombin III deficiency, polycythemia vera, and carcinoma. In the absence of atrial fibrillation, nonocclusive mesenteric ischemia is more likely in this patient. There is no reason to suspect inflammatory bowel disease as a cause of the bloody stools in this critically ill patient with known bacteremic sepsis, and the patient has only been on antibiotics for 1 day and is unlikely to have *Clostridium difficile* colitis at this point.

VII-41. The answer is E. (Chap. 322) Chronic intestinal ischemia presents with intestinal angina or postprandial abdominal pain associated with increased blood flow demand to the intestine following meals. Patients report abdominal cramping and pain following ingestion of a meal. Weight loss and chronic diarrhea may also be noted. Abdominal pain without weight loss is not chronic mesenteric angina. The gold standard for confirmation of mesenteric arterial occlusion is mesenteric angiography. Evaluation with mesenteric angiography allows for identification and possible intervention for the treatment of atherosclerosis within the vessel lumen and will also evaluate the patency of remaining mesenteric vessels. A noncontrast CT might show calcifications in mesenteric vessels but would not show the patency of those vessels. Atrial fibrillation increases the risk for acute mesenteric ischemia from thromboembolism but its presence or absence would not aid in the diagnosis of chronic ischemia. A high-fiber diet would potentially help if this patient's symptoms were from irritable bowel syndrome, but her age, known atherosclerotic disease, and weight loss would make that a less likely diagnosis. Although depression can lead to chronic pain and weight loss, the more life-threatening diagnosis of chronic intestinal ischemia needs to be considered first.

VII-42. The answer is B. (Chap. 323) The radiograph shows a massively dilated colon extending to the rectum. This radiograph is consistent with colonic pseudo-obstruction or Ogilvie syndrome. Ogilvie syndrome may be seen in elderly patients after nonabdominal surgery or in patients with underlying autonomic dysfunction. The presence of gas in the colon makes small bowel obstruction unlikely. There is no extraintestinal air that would

be suggestive of small or large bowel perforation. Small bowel ileus is characterized by multiple small bowel air-fluid levels on radiograph. The differential for extensive colonic dilation includes toxic megacolon due to *Clostridium difficile* infection. In this case, that is less likely given the recent surgery and lack of antibiotic treatment.

VII-43. The answer is D. (Chap. 323) This patient likely has a small bowel obstruction as evidenced by his symptoms, examination findings, and presence of air-fluid levels and dilated small bowel loops on the CT scan. While the most common cause of small bowel obstruction is postsurgical adhesions, a number of other conditions can lead to acute small bowel obstruction (Table VII-43). While he does have a history atrial fibrillation in the past, the presentation is not consistent with acute mesenteric ischemia and he is currently in sinus rhythm. He has no clear signs or symptoms of acute cholecystitis and the normal right upper quadrant ultrasound is reassuring that he does not have gallstone pancreatitis with a resulting ileus.

TABLE VII-43 Acute Small-Intestinal and Colonic Obstruction Incidences

Cause	Incidence
Postoperative adhesions	>50%
Neoplasms	~20%
Hernias (especially ventral or internal types, where the risk of strangulation is increased)	~10%
Inflammatory bowel disease, other inflammation (obstruction may resolve if acute inflammation and edema subside)	~5%
Intussusception, volvulus, other miscellaneous diseases	<15%

VII-44. The answer is A. (Chap. 324) The patient presents with typical findings for acute appendicitis with anorexia, progressing to vague periumbilical pain, followed by localization to the right lower quadrant. Low-grade fever and leukocytosis are frequently present. Although acute appendicitis is primarily a clinical diagnosis, imaging modalities are frequently employed because the symptoms are not always classic. Plain radiographs are rarely helpful except when an opaque fecalith is found in the right lower quadrant (<5% of cases). Ultrasound may demonstrate an enlarged appendix with a thick wall but is most useful to rule out ovarian pathology, tubo-ovarian abscess, or ectopic pregnancy. The effectiveness of ultrasonography as a tool to diagnose appendicitis is highly operator dependent. Even in very skilled hands, the appendix may not be visualized. Its overall sensitivity is 0.86, with a specificity of 0.81. Nonenhanced and contrast-enhanced CT are superior to ultrasound or plain radiograph in the diagnosis of acute appendicitis with a sensitivity of 0.94 and specificity of 0.95. Findings often include a thickened appendix with periappendiceal stranding and often the presence of a fecalith (Figure VII-44). Free air is uncommon, even in the case of a perforated appendix. Nonvisualization of the



FIGURE VII-44

appendix on CT is associated with surgical findings of a normal appendix 98% of the time. Colonoscopy has no role in the diagnosis of acute appendicitis.

VII-45. The answer is E. (*Chap. 324*) This patient likely has acute appendicitis. Laparoscopic appendectomy is preferred over an open procedure because of less postoperative pain, shorter lengths of stay, faster return to normal activity, and likely fewer superficial wound complications. There may be a higher risk of intra-abdominal abscess formation with the laparoscopic approach. Although some bouts of appendicitis might resolve on their own or with antibiotic therapy, it is difficult to predict which patients will develop a perforation or abscess, so surgical therapy is most commonly offered in a patient with no contraindications to surgical therapy. IV corticosteroids would not be helpful and might increase the risk of infection. Colonoscopy has no role in the management of acute appendicitis.

VII-46. The answer is C. (*Chap. 324*) The patient presents with several months of epigastric abdominal pain that is worse after eating. His symptoms are highly suggestive of peptic ulcer disease, with the worsening pain after eating suggesting a duodenal ulcer. The current presentation with an acute abdomen and free air under the diaphragm points to a diagnosis of perforated viscus. Perforated gallbladder is less likely in light of the duration of symptoms and the absence of the significant systemic symptoms that often accompany this condition. Because the patient is relatively young with no risk factors for mesenteric ischemia, necrotic bowel from an infarction is highly unlikely. Pancreatitis can have a similar presentation, but the pancreas cannot perforate and liberate free air. This patient likely has peritonitis caused by the presence of gastric contents in the peritoneal space.

VII-47. The answer is C. (*Chap. 324*) Arriving at the diagnosis of appendicitis can be challenging when the appendix is not located in the right lower quadrant, in women of childbearing age, and in the very young or elderly. Patients with pelvic appendicitis commonly present with dysuria, urinary frequency, diarrhea, or tenesmus. For this reason a pelvic and rectal examination should be performed in patients with suspected appendicitis. Anorexia is such a common symptom in appendicitis that its absence should prompt a search for an alternative diagnosis. Classically, maximal tenderness is identified in the right lower quadrant at or near McBurney point, which is located approximately one-third of the way along a line originating at the anterior iliac spine and running to the umbilicus. Right lower quadrant tenderness is present in over 90% of patients with appendicitis, while Rovsing sign, which is pain in the right lower quadrant during left lower quadrant palpation, is only present in 5% of patients with acute appendicitis (Table VII-47).

TABLE VII-47 Relative Frequency of Some Presenting Signs

Signs	Frequency (%)
Abdominal tenderness	>95
Right lower quadrant tenderness	>90
Rebound tenderness	30–70
Rectal tenderness	30–40
Cervical motion tenderness	30
Rigidity	~10
Psoas sign	3–5
Obturator sign	5–10
Rovsing sign	5
Palpable mass	<5

VII-48. The answer is B. (*Chap. 325*) Carbohydrates, proteins, and fats are the primary sources of nutrients in the diet. The largest source of calories and nutrients in the diet is carbohydrates. At least 45–55% of a person's total calories are derived from carbohydrates. The second biggest source of calories in the diet is fats. Although it is recommended that no more than 30% of one's calories come from fats, a typical American diet consists of about 34% fat. Proteins contribute 10–15% of daily caloric intake. Finally, although alcohol can contribute calories to one's diet, it does not contribute any nutrients.

- VII-49. The answer is A.** (*Chap. 325*) Evaporation and exhalation can account for 500–1000 mL of water loss per day. Normal stool output includes about 50–100 mL of water per day. Pregnancy increases daily water needs by about 30 mL/d. Urine output would be expected to be lower in a person with a lower solute load. Fever can increase water needs by up to 200 mL/d.
- VII-50. The answer is D.** (*Chap. 326*) This patient is manifesting symptoms of thiamine deficiency. The most common causes of thiamine deficiency in developed countries are alcoholism and chronic illnesses such as cancer. Alcohol interferes with the absorption of thiamine as well as the synthesis of thiamine pyrophosphate. In addition, alcohol increases the excretion of thiamine in the urine. Moreover, most chronic alcoholics have low dietary intake of thiamine. Chronic thiamine deficiency manifests as beriberi, which is classically described as “wet” or “dry,” depending on whether significant heart failure symptoms are present. However, there is often overlap between the syndromes of “wet” and “dry” beriberi. Both forms of the disease often exhibit paresthesias and neuropathic pain. Wet beriberi symptoms include those of high-output heart failure with tachycardia, cardiomegaly, and peripheral edema. Dry beriberi symptoms more commonly are those of both a motor and sensory neuropathy, predominantly affecting the legs. Alcoholic patients also demonstrate central nervous system effects. Acutely, this can consist of horizontal nystagmus, ophthalmoplegia, cerebellar ataxia, and mental impairment. This constellation of findings is called Wernicke encephalopathy. When there is additional loss of memory with confabulatory psychosis, as this individual is exhibiting, the syndrome is called Wernicke-Korsakoff syndrome. Rehydration with glucose-containing solutions without thiamine repletion can precipitate acute worsening of thiamine deficiency with lactic acidosis and coma and should be studiously avoided. In this case, the IV fluids with glucose should be held until thiamine repletion has begun. Thiamine should be repleted at a dose of 200 mg IV three times daily until there is no further improvement in acute symptoms, and then long-term oral thiamine at a dose of 10 mg daily should be continued until recovery is complete. In cases of Wernicke-Korsakoff syndrome, long-term memory loss is common.
- VII-51. The answer is B.** (*Chap. 326*) Niacin (vitamin B₃) has high bioavailability from beans, milk, meat, and eggs. While the bioavailability from grains is lower, most flour is enriched with “free” niacin; thus, deficiency of niacin is rare in western diets. Niacin deficiency can be found in individuals with corn-based diets in some parts of China, Africa, and India; individuals with alcoholism; and individuals with genetic defects limiting absorption of tryptophan. In addition, individuals with carcinoid syndrome are at increased risk of niacin deficiency because of increased conversion of tryptophan to serotonin. Clinically, the syndrome of niacin deficiency is known as pellagra. Early symptoms of niacin deficiency are loss of appetite, generalized weakness, abdominal pain, and vomiting. Glossitis is characteristic of pellagra with a beefy red tongue. Pellagra also has many dermatologic manifestations including a characteristic skin rash appearing on sun-exposed areas. The rash is typically scaling and erythematous. Often, the rash forms a ring around the neck, known as the Casal necklace. The four d’s of niacin deficiency—diarrhea, dermatitis, dementia, and death—are seen only in the most severe cases.
- VII-52. The answer is D.** (*Chap. 326*) This patient presents with gingival bleeding and has the classic perifollicular hemorrhagic rash of scurvy (vitamin C deficiency). In the United States, scurvy is primarily a disease of alcoholics and the elderly who consume <10 mg/d of vitamin C. Other individuals who are at risk of the disease are the poor and those who consume macrobiotic diets that are high in grains and seafood but avoid citrus fruits. In addition to nonspecific symptoms of fatigue, these patients also have impaired ability to form mature connective tissue and can bleed into various sites, including the joints, skin, and gingiva. A normal international normalized ratio excludes symptomatic vitamin K deficiency. Thiamine, niacin, and folate deficiencies are also seen in patients with alcoholism. Thiamine deficiency may cause peripheral neuropathy (beriberi), high-output heart failure, ataxia, and memory impairment. Folate deficiency causes macrocytic anemia and thrombocytopenia. Niacin deficiency causes pellagra, which is characterized

by glossitis and a pigmented, scaling rash that may be particularly noticeable in sun-exposed areas.

- VII-53. The answer is D.** (*Chap. 326*) This patient likely developed vitamin B₆ deficiency as a result of isoniazid treatment in the absence of taking pyridoxine. Vitamin B₆ refers to a family of compounds that includes pyridoxine, pyridoxal, pyridoxamine, and their 5'-phosphate derivatives. 5'-Pyridoxal phosphate (PLP) is a cofactor for >100 enzymes involved in amino acid metabolism. Vitamin B₆ also is involved in heme and neurotransmitter synthesis and in the metabolism of glycogen, lipids, steroids, sphingoid bases, and several vitamins, including the conversion of tryptophan to niacin. Certain medications, such as isoniazid, L-dopa, penicillamine, and cycloserine, interact with PLP due to a reaction with carbonyl groups. In the case of isoniazid, this results in a functional deficiency of pyridoxine that can lead to peripheral neuropathy. Pyridoxine should be given to all patients who are taking isoniazid to avoid this complication.
- VII-54. The answer is C.** (*Chap. 326*) This child likely has vitamin A deficiency. Vitamin A deficiency worldwide is present in 190 million preschool-age children, among whom >5 million have an ocular manifestation of deficiency termed xerophthalmia. This condition includes milder stages of night blindness and conjunctival xerosis (dryness) with Bitot spots (white patches of keratinized epithelium appearing on the sclera) that may affect 1–5% of children in deficient populations as well as rare, potentially blinding corneal ulceration and necrosis. Keratomalacia (softening of the cornea) leads to corneal scarring that blinds an estimated quarter of a million children each year and is associated with fatality rates of 4–25%. Vitamin A deficiency can compromise barrier, innate, and acquired immune defenses to infection. Supplementation in areas of endemic deficiency can markedly reduce mortality. Maternal vitamin A deficiency is a common cause of night blindness and can also increase the risk of maternal death from infection.
- VII-55. The answer is E.** (*Chap. 327*) The patient in this scenario has evidence of chronic starvation-related malnutrition, most likely related to anorexia nervosa. Starvation-related malnutrition that occurs without evidence of systemic inflammation is also known as marasmus and develops over months or years due to prolonged decreases in energy and protein intake. The patient exhibits a starved appearance with a low body mass index (<18.5 kg/m²). The diagnosis is based on diminished skinfold thickness, which reflects loss of fat stores and reduced arm circumference and demonstrates muscle wasting. In addition, temporal and interosseous muscle wasting is also commonly seen. Routine laboratory testing is not remarkably abnormal. Albumin may be low, but it is typically not less than 2.8 g/dL. However, despite a morbid appearance, immunocompetence and wound healing are preserved. Because the process is a chronic and fairly well-adapted process, treatment should be planned in concert with a dietitian to slowly allow a return to normal body weight. Overly aggressive nutritional support can lead to life-threatening metabolic imbalances. When mortality occurs in anorexia nervosa, it is most commonly related to complications of the disease and rarely due to malnutrition itself. Indeed, suicide is a more common cause of death in the disease than malnutrition. In certain processes such as cancer or chronic obstructive pulmonary disease, a similar loss of fat and protein can occur in the setting of systemic inflammation resulting in a wasted appearance. This process is known as cachexia. The diagnostic criteria are similar with regard to skinfold thickness and arm circumference. However, due to the concomitant systemic inflammation, individuals with cachexia are more likely to have lower albumin levels and can be immunocompromised.
- VII-56. The answer is B.** (*Chap. 327*) A patient's basal energy expenditure (BEE) can be calculated by the Harris-Benedict equation. The factors that are used for determining BEE are age, gender, height, and weight. The BEE for hospitalized patients is then adjusted by a factor of 1.1–1.4 depending on the severity of illness, with the highest values used for patients admitted with marked stress such as trauma or severe sepsis. The BEE serves as an estimate only. If it is important to have an exact calculation of energy expenditure,

indirect calorimetry can be performed. Protein needs can also be calculated more definitively by the use of urine urea nitrogen as an estimate of protein catabolism.

- VII-57. The answer is B.** (*Chap. 327*) Starvation-associated malnutrition is chronic starvation without inflammation (anorexia nervosa or major depression with lack of interest in eating). Chronic disease-associated malnutrition occurs when inflammation is chronic and of mild to moderate degree (e.g., organ failure, pancreatic cancer, or sarcopenic obesity). Acute disease or injury-associated malnutrition occurs when inflammation is acute and of severe degree (e.g., major infection, burns, trauma, or closed head injury).
- VII-58. The answer is E.** (*Chap. 327*) Dual energy x-ray absorptiometry (DEXA) is often used to measure bone density but it can also be for soft tissue measurements with appropriate software. DEXA can compare truncal and appendicular soft tissue components but does have the disadvantage of a modest x-ray exposure. Anthropomorphic analyses such as measuring skinfolds and circumferences require training for reliability. The typical coefficient of variation is $\geq 10\%$. Bioelectrical impedance is based on differential resistance of body tissues. The equipment is easily portable and provides good measurement of body water. This method requires population specific validation of regression equations. Albumin, with a serum half-life of 14–20 days, lacks sensitivity and specificity for malnutrition. It is a potent risk indicator for morbidity and mortality. It is a proxy measure for underlying injury, disease, or inflammation. In addition to malnutrition, a low albumin can be caused by liver disease, nephrotic syndrome, and protein-wasting enteropathy. C-reactive protein is a positive acute phase reactant. It is generally elevated if an active inflammatory process is manifest but it is not specific for malnutrition.
- VII-59. The answer is A.** (*Chap. 329*) The most common and most characteristic symptom of liver disease is fatigue. Unfortunately, it is also very nonspecific, with little specific diagnostic utility. The fatigue in liver disease seems to improve in the morning and worsen throughout the day, but it can be intermittent. Jaundice is the hallmark of liver disease and is much more specific. Jaundice, however, is typically a sign of more advanced disease. Itching is also typically a symptom of more advanced disease and is more common in cholestatic causes of liver disease. Nausea often occurs in severe disease and can be accompanied by vomiting. Right upper quadrant pain is a less common symptom and indicates stretching of the liver capsule.
- VII-60. The answer is A.** (*Chap. 329*) In assessing alcohol intake, the history should also focus on whether alcohol abuse or dependence is present. Alcoholism is usually defined by the behavioral patterns and consequences of alcohol intake, not by the amount. Abuse is defined by a repetitive pattern of drinking alcohol that has adverse effects on social, family, occupational, or health status. Dependence is defined by alcohol-seeking behavior, despite its adverse effects. Many alcoholics demonstrate both dependence and abuse, and dependence is considered the more serious and advanced form of alcoholism. A clinically helpful approach to diagnosis of alcohol dependence and abuse is the use of the CAGE questionnaire, which is recommended for all medical history taking. One “yes” response should raise suspicion of an alcohol use problem, and more than one “yes” response is a strong indication of abuse of dependence.
- VII-61. The answer is C.** (*Chap. 329*) While there are many causes of liver disease, these disorders generally present clinically in a few distinct patterns and are usually classified as hepatocellular, cholestatic (obstructive), or mixed (Table VII-61). In hepatocellular diseases (such as viral hepatitis and alcoholic liver disease), features of liver injury, inflammation, and necrosis predominate. In cholestatic diseases, such as gallstones or malignant obstruction, primary biliary cholangitis (previously referred to as primary biliary cirrhosis), and some drug-induced liver diseases, features of inhibition of bile flow predominate. In a mixed pattern, features of both hepatocellular and cholestatic injury are present (such as in cholestatic forms of viral hepatitis and many drug-induced liver diseases).

TABLE VII-61 Liver Diseases

Inherited hyperbilirubinemia	Liver involvement in systemic diseases
Gilbert syndrome	Sarcoidosis
Crigler-Najjar syndrome, types 1 and 2	Amyloidosis
Dubin-Johnson syndrome	Glycogen storage diseases
Rotor syndrome	Celiac disease
Viral hepatitis	Tuberculosis
Hepatitis A	<i>Mycobacterium avium-intracellulare</i> infection
Hepatitis B	Cholestatic syndromes
Hepatitis C	Benign postoperative cholestasis
Hepatitis D	Jaundice of sepsis
Hepatitis E	Total parenteral-nutrition-induced jaundice
Others (Epstein-Barr virus [mononucleosis]	Cholestasis of pregnancy
herpesvirus, cytomegalovirus, adenovirus	Cholangitis and cholecystitis
hepatitis)	Extrahepatic biliary obstruction (stone, stricture,
Cryptogenic hepatitis	cancer)
Immune and autoimmune liver diseases	Biliary atresia
Primary biliary cholangitis	Caroli disease
Autoimmune hepatitis	Cryptosporidiosis
Sclerosing cholangitis	Drug-induced liver disease
Overlap syndromes	Hepatocellular patterns (isoniazid,
Graft-versus-host disease	acetaminophen)
Allograft rejection	Cholestatic patterns (methyltestosterone)
Genetic liver diseases	Mixed patterns (sulfonamides, phenytoin)
Alpha-1 antitrypsin deficiency	Micro- and macrovesicular steatosis
Hemochromatosis	(methotrexate, fialuridine)
Wilson disease	Vascular injury
Benign recurrent intrahepatic cholestasis	Sinusoidal obstruction syndrome
Progressive familial intrahepatic cholestasis,	Budd-Chiari syndrome
types I–III	Ischemic hepatitis
Others (galactosemia, tyrosinemia, cystic	Passive congestion
fibrosis, Niemann-Pick-disease, Gaucher disease)	Portal vein thrombosis
Alcoholic liver disease	Nodular regenerative hyperplasia
Acute fatty liver	Mass lesions
Acute alcoholic hepatitis	Hepatocellular carcinoma
Laënnec cirrhosis	Cholangiocarcinoma
Nonalcoholic fatty liver	Adenoma
Steatosis	Focal nodular hyperplasia
Steatohepatitis	Metastatic tumors
Acute fatty liver of pregnancy	Abscess
	Cysts
	Hemangioma

VII-62. The answer is B. (Chap. 329) Magnetic resonance cholangiopancreatography (MRCP) and endoscopic retrograde cholangiopancreatography (ERCP) are the procedures of choice for visualization of the biliary tree. MRCP offers several advantages over ERCP: there is no need for contrast media or ionizing radiation, images can be acquired faster, the procedure is less operator dependent, and it carries no risk of pancreatitis. MRCP is superior to ultrasound and CT for detecting choledocholithiasis but is less specific. MRCP is useful in the diagnosis of bile duct obstruction and congenital biliary abnormalities, but ERCP is more valuable in evaluating ampullary lesions and primary sclerosing cholangitis. ERCP permits biopsy, direct visualization of the ampulla and common bile duct, and intraductal ultrasonography and brushings for cytological evaluation of malignancy. It also provides several therapeutic options in patients with obstructive jaundice, such as sphincterotomy, stone extraction, and placement of nasobiliary catheters and biliary stents.

VII-63. The answer is D. (Chap. 330) It is important to understand the patterns of laboratory abnormalities that indicate liver disease is present. One way to consider laboratory evaluation of liver disease is to consider three general categories of tests: those based on excretory function of the liver, those of biosynthetic activity of the liver, and those of coagulation factors. The most common tests of liver function fall under the category of

tests based on the detoxification and excretory function of the liver. These include serum bilirubin, ammonia and enzyme levels, and urine bilirubin. Bilirubin can exist as a conjugated and an unconjugated form. The unconjugated form is often referred to as the indirect portion. Isolated elevation in the unconjugated form of bilirubin is typically not related to liver disease but is most commonly seen in hemolysis and a number of benign genetic conditions such as Gilbert syndrome. In contrast, conjugated (or direct) hyperbilirubinemia almost always indicates disease of the liver or biliary tract. Conjugated bilirubin is water soluble and excreted in the urine, but unconjugated bilirubin is not; rather, it binds to albumin in the blood. Therefore, bilirubinuria implies liver disease as well. Among the serum enzymes, it is useful to consider enzymes as those that are associated with hepatocellular injury versus those that reflect cholestasis. Alanine and aspartate aminotransferases are the primary enzymes that indicate hepatocyte injury. Alkaline phosphatase is the most common enzyme elevated in cholestasis, but bone disease also causes increased alkaline phosphatase. In some cases, one needs additional information to determine whether the alkaline phosphatase is liver or bone in origin. Other tests that would be elevated in cholestatic liver disease are 5'-nucleotidase and γ -glutamyl transferase. The primary test of hepatic synthetic function is measurement of serum albumin. Coagulation factors can be directly measured, but impaired production of coagulation factors in liver disease is primarily inferred from elevations in prothrombin time.

VII-64. The answer is B. (*Chap. 330*) The aminotransferases are sensitive indicators of liver cell injury and are most helpful in recognizing acute hepatocellular diseases such as hepatitis. They include aspartate aminotransferase (AST) and alanine aminotransferase (ALT). AST is found in the liver, cardiac muscle, skeletal muscle, kidneys, brain, pancreas, lungs, leukocytes, and erythrocytes in decreasing order of concentration. ALT is found primarily in the liver and is therefore a more specific indicator of liver injury. The aminotransferases are normally present in the serum in low concentrations. These enzymes are released into the blood in greater amounts when there is damage to the liver cell membrane, resulting in increased permeability. Liver cell necrosis is not required for the release of the aminotransferases, and there is a poor correlation between the degree of liver cell damage and the level of the aminotransferases. Thus, the absolute elevation of the aminotransferases is of no prognostic significance in acute hepatocellular disorders. Any type of liver cell injury can cause modest elevations in the serum aminotransferases. Levels of up to 300 IU/L are nonspecific and may be found in any type of liver disorder. Minimal ALT elevations in asymptomatic blood donors rarely indicate severe liver disease; studies have shown that fatty liver disease is the most likely explanation. Striking elevations—i.e., aminotransferases >1000 IU/L—occur almost exclusively in disorders associated with extensive hepatocellular injury such as viral hepatitis, ischemic liver injury (prolonged hypotension or acute heart failure), or toxin- or drug-induced liver injury. The pattern of the aminotransferase elevation can be helpful diagnostically. In most acute hepatocellular disorders, the ALT is higher than or equal to the AST. Whereas the AST:ALT ratio is typically <1 in patients with chronic viral hepatitis and nonalcoholic fatty liver disease, a number of groups have noted that as cirrhosis develops, this ratio rises to >1. An AST:ALT ratio >2:1 is suggestive, whereas a ratio >3:1 is highly suggestive, of alcoholic liver disease. The AST in alcoholic liver disease is rarely >300 IU/L, and the ALT is often normal. The aminotransferases are usually not greatly elevated in obstructive jaundice.

VII-65. The answer is B. (*Chap. 330*) The pattern of the aminotransferase elevation can be helpful diagnostically. In most acute hepatocellular disorders, the alanine aminotransferase (ALT) is higher than or equal to the aspartate aminotransferase (AST). Whereas the AST:ALT ratio is typically <1 in patients with chronic viral hepatitis and nonalcoholic fatty liver disease, a number of groups have noted that as cirrhosis develops, this ratio rises to >1. An AST:ALT ratio >2:1 is suggestive, whereas a ratio >3:1 is highly suggestive, of alcoholic liver disease. The AST in alcoholic liver disease is rarely >300 IU/L, and the ALT is often normal. A low level of ALT in the serum is due to an alcohol-induced deficiency of pyridoxal phosphate. The aminotransferases are usually not greatly elevated in obstructive jaundice. One notable exception occurs during the acute phase of biliary

obstruction caused by the passage of a gallstone into the common bile duct. In this setting, the aminotransferases can briefly be in the 1000–2000 IU/L range. However, aminotransferase levels decrease quickly, and the biochemical tests rapidly evolve into those typical of cholestasis.

VII-66. The answer is D. (*Chap. 330*) Elevation of liver-derived alkaline phosphatase is not totally specific for cholestasis, and a less than threefold elevation can be seen in almost any type of liver disease. Patients over age 60 can have a mildly elevated alkaline phosphatase (1–1.5 times normal), whereas individuals with blood types O and B can have an elevation of the serum alkaline phosphatase after eating a fatty meal due to the influx of intestinal alkaline phosphatase into the blood. It is also elevated in children and adolescents undergoing rapid bone growth because of bone alkaline phosphatase, and late in normal pregnancies due to the influx of placental alkaline phosphatase. γ -Glutamyl transpeptidase is located in the endoplasmic reticulum and in bile duct epithelial cells and is less specific for cholestasis than are elevations of alkaline phosphatase or 5'-nucleotidase. Diffuse polyclonal increases in IgG levels are common in autoimmune hepatitis; increases >100% should alert the clinician to this possibility. Increases in the IgM levels are common in primary biliary cirrhosis, whereas increases in the IgA levels occur in alcoholic liver disease. With the exception of factor VIII, which is produced by vascular endothelial cells, the blood clotting factors are made exclusively in hepatocytes. Serum albumin has a long half-life: 18–20 days, with ~4% degraded per day. Because of this slow turnover, the serum albumin is not a good indicator of acute or mild hepatic dysfunction.

VII-67. The answer is E. (*Chap. 331*) This patient is presenting with an asymptomatic and mild elevation in unconjugated hyperbilirubinemia that has occurred during a time of increased stress, fatigue, and likely decreased caloric intake. This presentation is characteristic of Gilbert syndrome, an inherited disorder of bilirubin conjugation. In Gilbert syndrome, there is a mutation of the *UGT1A1* gene that encodes bilirubin UDP-glucuronosyltransferase, which leads to a reduction in activity on the enzyme to 10 to 35% of normal. This enzyme is of critical importance in the conjugation of bilirubin. Most of the time, there is no apparent jaundice because the reduced ability to conjugate bilirubin is not reduced to a degree that leads to an elevation of bilirubin. However, during times of stress, fatigue, alcohol use, decreased caloric intake, or intercurrent illness, the enzyme can become overwhelmed, leading to a mild hyperbilirubinemia. Typical bilirubin levels are less than 4.0 mg/dL unless the individual is ill or fasting. The level of hyperbilirubinemia may be enough to cause scleral icterus and jaundice. Diagnosis usually occurs during young adulthood, and episodes are self-limited and benign. If a liver biopsy were to be performed (not indicated in this case), hepatic histology would be normal. No treatment is necessary because there are no long-term consequences of Gilbert syndrome, and patient reassurance is recommended. Other inherited disorders of bilirubin conjugation are Crigler-Najjar syndromes types 1 and 2. Crigler-Najjar syndrome type 1 is a congenital disease characterized by more dramatic elevations in bilirubin as high as 20–45 mg/dL that is first diagnosed in the neonatal period and is present throughout life. This rare disorder was once fatal in early childhood due to the development of kernicterus. However, with phototherapy, individuals are now able to survive into adulthood, although neurologic deficits are common. Crigler-Najjar syndrome type 2 is similar to type 1, but the elevations in bilirubin are not as great. Kernicterus is rare. This is because there is some residual function of the bilirubin UDP-glucuronosyltransferase enzyme (<10%), which is totally absent in type 1 disease. Hemolysis is another frequent cause of elevated unconjugated bilirubin. Hemolysis can be caused by many factors including medications, autoimmune disorders, and inherited disorders, among others. However, the normal hematocrit, lactate dehydrogenase, and haptoglobin eliminate hemolysis as a possibility in this case. Dubin-Johnson syndrome is another congenital hyperbilirubinemia. However, it is a predominantly conjugated hyperbilirubinemia caused by a defect in biliary excretion from hepatocytes. Obstructive choledocholithiasis is characterized by right upper quadrant pain that is often exacerbated by fatty meals. The absence of symptoms or elevation in other liver function tests, especially alkaline phosphatase, also makes this diagnosis unlikely.

VII-68. The answer is B. (*Chap. 331*) Increased destruction of erythrocytes leads to increased bilirubin turnover and unconjugated hyperbilirubinemia; the hyperbilirubinemia is usually modest in the presence of normal liver function. Hemolysis alone cannot result in a sustained hyperbilirubinemia of more than 4 mg/dL (~68 μ mol). Higher values imply concomitant hepatic dysfunction. When hemolysis is the only abnormality in an otherwise healthy individual, the result is a purely unconjugated hyperbilirubinemia, with the direct-reacting fraction as measured in a typical clinical laboratory being $\leq 15\%$ of the total serum bilirubin. In the presence of systemic disease, which may include a degree of hepatic dysfunction, hemolysis may produce a component of conjugated hyperbilirubinemia in addition to an elevated unconjugated bilirubin concentration. Prolonged hemolysis may lead to the precipitation of bilirubin salts within the gallbladder or biliary tree, resulting in the formation of gallstones in which bilirubin, rather than cholesterol, is the major component. Such pigment stones may lead to acute or chronic cholecystitis, biliary obstruction, or any other biliary tract consequence of calculous disease.

VII-69. The answer is D. (*Chap. 331*) This patient likely has Dubin-Johnson Syndrome (DJS). This benign, relatively rare disorder is characterized by low-grade, predominantly conjugated hyperbilirubinemia (Table VII-69). Total bilirubin concentrations are typically between 34 and 85 μ mol/L (2 and 5 mg/dL) but on occasion can be in the normal range or as high as 340–430 μ mol/L (20–25 mg/dL) and can fluctuate widely in any given patient. The degree of hyperbilirubinemia may be increased by intercurrent illness, oral contraceptive use, and pregnancy. Because the hyperbilirubinemia is due to a predominant rise in conjugated bilirubin, bilirubinuria is characteristically present. Aside from elevated serum bilirubin levels, other routine laboratory tests are normal. Physical examination is usually normal except for jaundice, although an occasional patient may have hepatosplenomegaly. Patients with DJS are usually asymptomatic, although some may have vague constitutional symptoms. In women, the condition may be subclinical until the patient becomes pregnant or receives oral contraceptives, at which time chemical hyperbilirubinemia becomes frank jaundice. Even in these situations, other routine liver function tests, including serum alkaline phosphatase and transaminase activities, are normal. Benign recurrent intrahepatic cholestasis would be expected to have biochemical abnormalities other than isolated hyperbilirubinemia and is often accompanied by intense pruritus. Cholestasis of pregnancy usually occurs in the late second trimester or early third trimester and is most often accompanied by severe pruritus that precedes biochemical abnormalities by several weeks. Crigler-Najjar syndrome type 2 is characterized by marked unconjugated hyperbilirubinemia. HELLP syndrome is characterized

TABLE VII-69 Principal Differential Characteristics of Inheritable Disorders of Bile Canalicular Function

	DJS	Rotor	PFIC1	BRIC1	PFIC2	BRIC2	PFIC3
Gene	<i>ABCCA</i>	<i>SLCO1B1/SLCO1B3</i>	<i>ATP8B1</i>	<i>ATP8B1</i>	<i>ABCB11</i>	<i>ABCB11</i>	<i>ABCB4</i>
Protein	MRP2	OATP1B1/1B3	FIC1	FIC1	BSEP	BSEP	MDR3
Cholestasis	No	No	Yes	Episodic	Yes	Episodic	Yes
Serum GGT	Normal	Normal	Normal	Normal	Normal	Normal	$\uparrow\uparrow$
Serum bile acids	Normal	Normal	$\uparrow\uparrow$	$\uparrow\uparrow$ during episodes	$\uparrow\uparrow$	$\uparrow\uparrow$ during episodes	$\uparrow\uparrow$
Clinical features	Mild conjugated hyperbilirubinemia; otherwise normal liver function; dark pigment in liver; characteristic pattern of urinary coproporphyrins	Mild conjugated hyperbilirubinemia; otherwise normal liver function; liver without abnormal pigmentation	Severe cholestasis beginning in childhood	Recurrent episodes of cholestasis beginning at any age	Severe cholestasis beginning in childhood	Recurrent episodes of cholestasis beginning at any age	Severe cholestasis beginning in childhood; decreased phospholipids in bile

Abbreviations: BRIC, benign recurrent intrahepatic cholestasis; BSEP, bile salt excretory protein; DJS, Dubin-Johnson syndrome; GGT, γ -glutamyltransferase; MRP2, multidrug resistance-associated protein 2; OATP1A/1B, organic anion transport proteins 1B1 and 1B3; PFIC, progressive familial intrahepatic cholestasis; $\uparrow\uparrow$, increased.

by hemolysis, elevated liver enzymes, and low platelets, and would be expected to have an unconjugated hyperbilirubinemia.

VII-70. The answer is B. (*Chap. 332*) This patient presents with acute hepatitis, which has numerous etiologies. These include viruses, toxins/drugs, autoimmune diseases, metabolic diseases, alcohol, ischemia, pregnancy, and other infectious etiologies including rickettsial diseases and leptospirosis. In this clinical scenario, the patient does have risk factors for hepatitis A, B, and C infection, including being a man who has sex with men and a prior history of injection drug use. All acute viral hepatitis presents with a similar clinical pattern, although incubation periods vary after exposure. The most common initial symptoms are fatigue, anorexia, nausea, vomiting, myalgias, and headache. These symptoms precede the onset of jaundice by about 1–2 weeks. Once jaundice develops, the prodromal symptoms regress. On physical examination, there is usually obvious icterus with an enlarged and tender liver. Splenomegaly can occur. Aspartate aminotransferase (AST) and alanine aminotransferase (ALT) are elevated with peak levels that are quite variable between 400 and 4000 IU/L, and alkaline phosphatase levels are increased to a much lesser degree. Hyperbilirubinemia (levels 5–20 mg/dL) occurs with primarily increased levels of conjugated bilirubin. Thus, it is important to recognize the patterns of antibody production in the viral hepatitis. Hepatitis A is an RNA virus that presents with acute hepatitis and is transmitted by the fecal-oral route. In the acute state, the IgM would be elevated, which is not seen in this scenario. Hepatitis B virus is a DNA virus with three common antigens that are tested serologically to determine the time course of the illness. These antigens are the surface antigen, the core antigen, and the e antigen, which is a nucleocapsid protein produced from the same gene as the core antigen but immunologically distinct. Several distinct patterns can be observed. In acute hepatitis B, the core IgM, surface antigen, and e antigen are all positive, which is what is seen in this case. At this point, the patient is highly infectious, with viral shedding in body fluids, including saliva. In a late acute infection, core IgG may be positive at the same time as surface and e antigen positivity. In chronic hepatitis B, this same pattern of serologies is seen. If a patient has a prior infection without development of chronic hepatitis, the core IgG and surface antibody is positive. However, when immunity is obtained via vaccination, only the surface antibody (SAb) is positive, the e antigen and surface antigen will be negative since the patient was never infected. The variety of positive antigen-antibody results that can result are outlined in Table VII-70. Acute hepatitis C often is detectable with contemporary immunoassays early during the disease when the aminotransferases are positive. Thus, a positive hepatitis C virus antibody could indicate acute hepatitis C in this individual. However, given his clinical history of prior injection drug use and inability to donate blood, this likely indicates chronic hepatitis C infection. In some instances, ecstasy has been reported to cause drug-induced hepatitis, but given the viral serologies in this patient, this would be unlikely.

TABLE VII-70 Simplified Diagnostic Approach in Patients Presenting with Acute Hepatitis

Serologic Tests of Patient's Serum				Diagnostic Interpretation
HBsAg	IgM Anti-HAV	IgM Anti-HBc	Anti-HCV	
+	–	+	–	Acute hepatitis B
+	–	–	–	Chronic hepatitis B
+	+	–	–	Acute hepatitis A superimposed on chronic hepatitis B
+	+	+	–	Acute hepatitis A and B
–	+	–	–	Acute hepatitis A
–	+	+	–	Acute hepatitis A and B (HBsAg below detection threshold)
–	–	+	–	Acute hepatitis B (HBsAg below detection threshold)
–	–	–	+	Acute hepatitis C

Note: See text for abbreviations.

- VII-71. The answer is E.** (*Chap. 332*) No treatment is recommended for acute hepatitis B in most individuals because 99% of infected individuals recover without assistance. Therefore, it would not be expected that an individual would derive any particular benefit from treatment. In severe acute hepatitis B, nucleoside analogues, including lamivudine, have been used successfully, although there are no clinical trial data to support such an approach. Hepatitis A is an acute and self-limited illness that does not progress to chronic liver disease. Thus, no treatment is required. Anti-hepatitis A virus Ig can be given prophylactically following a known exposure to prevent development of disease, but it is not helpful in established disease. There is no role for oral or IV corticosteroids in the treatment of acute viral hepatitis of any etiology. It has demonstrated no clinical benefit and may increase the risk of developing chronic disease.
- VII-72. The answer is E.** (*Chap. 332*) Hepatitis A virus (HAV) is an acute, self-limited virus that is acquired almost exclusively via the fecal-oral route. It is classically a disease of poor hygiene and overcrowding. Outbreaks have been traced to contaminated water, milk, frozen raspberries and strawberries, green onions, and shellfish. Infection occurs mostly in children and young adults. It almost invariably resolves spontaneously and results in lifelong immunity. Fulminant disease occurs in $\leq 0.1\%$ of cases, and there is no chronic form (in contrast to hepatitis B and C). Diagnosis is made by demonstrating a positive IgM antibody to HAV, as described in this case. An IgG antibody to HAV indicates immunity, obtained by a previous infection or vaccination. A small proportion of patients will experience relapsing hepatitis weeks to months after a full recovery from HAV infection. This too is self-limited. There is no approved antiviral therapy for HAV. An inactivated vaccine has decreased the incidence of the disease, and it is recommended for all U.S. children, for high-risk adults, and for travelers to endemic areas. Passive immunization with Ig is also available, and it is effective in preventing clinical disease before exposure or during the early incubation period.
- VII-73. The answer is A.** (*Chap. 332*) A clear distinction between viral etiologies of acute hepatitis cannot be made on clinical or epidemiologic features alone. This patient is at risk of many forms of hepatitis due to his lifestyle. Given his occupation in food services, from a public health perspective, it is important to make an accurate diagnosis. Serologies must be obtained to make a diagnosis. Although hepatitis C virus typically does not present as an acute hepatitis, this is not absolute. Hepatitis E virus infects men and women equally and resembles hepatitis A virus in clinical presentation. This patient should be questioned regarding IV drug use, and in addition to hepatitis serologies, an HIV test should be performed.
- VII-74. The answer is C.** (*Chap. 332*) In HBeAg-expressing transgenic mice, in utero exposure to HBeAg, which is sufficiently small to traverse the placenta, induces T-cell tolerance to both nucleocapsid proteins. This, in turn, may explain why, when infection occurs so early in life, immunologic clearance does not occur, and protracted, lifelong infection ensues. An alternative explanation proposed to explain why robust liver injury does not accompany neonatal HBV infection but predisposes to chronic infection is defective priming of HBV-specific T cells during in utero exposure to HBV. Hepatitis B must cross the placenta for this immune tolerance to occur. Antibodies to hepatitis B core Ag do not confer immunity to infection. Antibodies to hepatitis B Surface Antigen (anti-HBs) confers immunity to infection. Natural killer cells are an important part of the innate immune response in neonates, and likely play an important role in hepatocellular injury in hepatitis B infection.
- VII-75. The answer is B.** (*Chap. 332*) This patient likely has chronic hepatitis B, as evidenced by the positive HBsAg and the presence of Anti-HBc IgG antibodies. The absence of the e-antigen likely means that she is in a low infectivity state and does not have high levels of viral replication. Table VII-75 shows the common serologic patterns for various hepatitis B disease states.

TABLE VII-75 Commonly Encountered Serologic Patterns of Hepatitis B Infection

HBsAg	Anti-HBs	Anti-HBc	HBeAg	Anti-HBe	Interpretation
+	–	IgM	+	–	Acute hepatitis B, high infectivity ^a
+	–	IgG	+	–	Chronic hepatitis B, high infectivity
+	–	IgG	–	+	1. Late acute or chronic hepatitis B, low infectivity 2. HBeAg-negative (“precore-mutant”) hepatitis B (chronic or, rarely, acute)
+	+	+	+/-	+/-	1. HBsAg of one subtype and heterotypic anti-HBs (common) 2. Process of seroconversion from HBsAg to anti-HBs (rare)
–	–	IgM	+/-	+/-	1. Acute hepatitis B ^a 2. Anti-HBc “window”
–	–	IgG	–	+/-	1. Low-level hepatitis B carrier 2. Hepatitis B in remote past
–	+	IgG	–	+/-	Recovery from hepatitis B
–	+	–	–	–	1. Immunization with HBsAg (after vaccination) 2. Hepatitis B in the remote past (?) 3. False positive

^aIgM anti-HBc may reappear during acute reactivation of chronic hepatitis B.

Note: See text for abbreviations.

VII-76. The answer is E. (*Chap. 332*) There is an effective vaccine against hepatitis E genotype 1 that also has effectiveness against other genotypes. It can be considered in endemic areas. There is no effective vaccine against hepatitis D. Because hepatitis D occurs in the setting of hepatitis B, vaccination against hepatitis B is effective in preventing hepatitis D. It is no longer standard practice to use strict contact isolation precautions for hepatitis A patients since person-to-person transmission is not very common and universal precautions should be sufficient. Serum Ig is not effective in preventing hepatitis C infection after a high-risk exposure. The hepatitis B vaccine contains inactive viral proteins and can be safely administered to immunocompromised patients.

VII-77. The answer is B. (*Chap. 333*) Acetaminophen overdose is the most common cause of acute liver failure and the most common cause of drug-induced liver failure that leads to transplantation. Acetaminophen is metabolized in the liver through two pathways. The primary pathway is a phase II reaction that produces nontoxic sulfate and glucuronide metabolites. The minor pathway occurs through a phase I reaction leading to production of N-acetyl-p-benzoquinone imine (NAPQI). This metabolite is directly toxic to liver cells and can lead to hepatocyte necrosis. With therapeutic use of acetaminophen, glutathione in the liver rapidly converts NAPQI to a nontoxic metabolite that is excreted in the urine. However, glutathione stores can become depleted in the setting of a large acute ingestion, chronic alcoholism, or chronic ingestion of increased acetaminophen. In addition, alcohol upregulates the first enzyme in the metabolic pathway, causing NAPQI to accumulate more quickly in alcoholics. Given the known hepatotoxicity of acetaminophen, the U.S. Food and Drug Administration has recommended a maximum daily dose of no more than 3.25 g, with lower doses in individuals with chronic alcohol use. Acute ingestions of 10–15 g of acetaminophen are sufficient to cause clinical evidence of liver injury, and doses higher than 25 g can lead to fatal hepatic necrosis. The course of illness with acute acetaminophen ingestion follows a predictable pattern. Nausea, vomiting, abdominal pain, and shock occur within 4–12 hours after ingestion. Liver enzymes and synthetic function are normal at this time. Within 24–48 hours, these symptoms subside and are followed by evidence of hepatic injury. Maximal levels of aminotransferases can reach more than 10,000 IU/L and may not occur until 4–6 days after ingestion. These patients must be followed carefully for fulminant hepatic failure with serious complications including encephalopathy, cerebral edema, marked coagulopathy,

renal failure, metabolic acidosis, electrolyte abnormalities, and refractory shock. Levels of acetaminophen are predictive of the development of hepatotoxicity. The first level should be measured no sooner than 4 hours after a known ingestion. Levels should be plotted on a nomogram that relates levels to the time after ingestion. If, at 4 hours, the acetaminophen level is greater than 300 $\mu\text{g/mL}$, significant hepatotoxicity is likely. In the setting of overdose, it may be difficult to know the exact quantity and timing of an ingestion. For the patient presenting in the clinical scenario in this question, her level of $>300 \mu\text{g/mL}$ is quite concerning for a large ingestion, and treatment should be initiated immediately. The primary treatment for acetaminophen overdose is N-acetylcysteine, which acts to replenish glutathione levels in the liver and also provides a reservoir of sulfhydryl groups to bind to the toxic metabolites. The typical regimen of N-acetylcysteine is 140 mg/kg given as a loading dose, followed by 70 mg/kg every 4 hours for a total of 15–20 doses. This drug can also be given by continuous infusion. Activated charcoal or cholestyramine should only be given if the patient presents within 30 minutes after ingestion. Hemodialysis will not accelerate clearance of acetaminophen and will not protect the liver. Most patients with fulminant hepatic failure develop acute renal failure, often requiring hemodialysis. If a patient survives an acetaminophen overdose, there is usually no chronic liver injury.

- VII-78. The answer is E.** (*Chap. 333*) Isoniazid (INH) remains central to most antituberculous prophylactic and therapeutic regimens, despite its long-standing recognition as a hepatotoxin. In 10% of patients treated with INH, elevated serum aminotransferase levels develop during the first few weeks of therapy; however, these elevations in most cases are self-limited, mild (values for alanine aminotransferase $<200 \text{ IU/L}$), and resolve despite continued drug use. This adaptive response allows continuation of the agent if symptoms and progressive enzyme elevations do not follow the initial elevations. Acute hepatocellular drug-induced liver injury secondary to INH is evident with a variable latency period up to 6 months and is more frequent in alcoholics and patients taking certain other medications, such as barbiturates, rifampin, and pyrazinamide. If the clinical threshold of encephalopathy is reached, severe hepatic injury is likely to be fatal or to require liver transplantation. Liver biopsy reveals morphologic changes similar to those of viral hepatitis or bridging hepatic necrosis. Substantial liver injury appears to be age related, increasing substantially after age 35; the highest frequency is in patients over age 50. Even for patients >50 years of age monitored carefully during therapy, hepatotoxicity occurs in only $\sim 2\%$, well below the risk estimate derived from earlier experiences. Many public health programs that require INH prophylaxis for a positive tuberculin skin test or QuantiFERON test include monthly monitoring of aminotransferase levels, although this practice has been called into question. Even more effective in limiting serious outcomes may be encouraging patients to be alert for symptoms such as nausea, fatigue, or jaundice, because most fatalities occur in the setting of continued INH use despite clinically apparent illness.
- VII-79. The answer is B.** (*Chap. 333*) Direct toxic hepatitis occurs with predictable regularity after exposure and is dose dependent. The latent period between exposure and liver injury is usually short (often several hours), although clinical manifestations may be delayed for 24–48 hours. Agents producing toxic hepatitis are generally systemic poisons or are converted in the liver to toxic metabolites. The direct hepatotoxins result in morphologic abnormalities that are reasonably characteristic and reproducible for each toxin. Idiosyncratic reactions can produce a variety of liver injury patterns and can be similar to acute viral hepatitis. The occurrence of such reactions is infrequent, unpredictable, and not dose dependent. The incidence of drug-induced liver injury (DILI) is not generally higher in patients with chronic liver injury although the consequences may be more severe. The occurrence of jaundice in phase 3 clinical trials predicts severe hepatotoxicity in a ratio of 10:1 (10 cases of jaundice for every case of severe hepatotoxicity—also called Hy's law after Hyman Zimmerman, a pioneer in the field of DILI).
- VII-80. The answer is C.** (*Chap. 333*) Valproate hepatotoxicity is more common in persons with mitochondrial enzyme deficiencies and may be ameliorated by IV administration of

carnitine, which valproate therapy can deplete. Amiodarone liver toxicity can persist for a prolonged period after discontinuation of the drug because of the drug's extremely long half-life. In many patients, hepatic injury from phenytoin is associated with striking fever, lymphadenopathy, rash (Stevens-Johnson syndrome or exfoliative dermatitis), leukocytosis, and eosinophilia, suggesting an immunologically mediated hypersensitivity mechanism. Amoxicillin liver injury is potentiated in the setting of clavulanate for reasons that are unknown. Nitrofurantoin may have a longer latency period to onset of liver injury, in part because of its intermittent and recurrent use.

- VII-81. The answer is B.** (*Chap. 334*) The patient in this scenario has evidence of chronic active hepatitis B virus (HBV) infection. The presence of hepatitis B e antigen (HBeAg) is indicative of ongoing viral replication, and individuals with HBeAg positivity typically have high levels of HBV DNA on testing. The spectrum of clinical infection in chronic hepatitis B is quite variable, and often, individuals are asymptomatic with elevated liver enzymes identified on testing for other reasons. Thus, the decision to treat chronic HBV infection should not be based on clinical features. Most experts recommend treatment of HBeAg-positive chronic HBV infection with HBV DNA levels $>2 \times 10^4$ IU/mL if the alanine aminotransferase is elevated greater than twice the upper limit of normal (Table VII-81). To date, the following drugs have been approved for the treatment of chronic HBV: injectable interferon (IFN)- α ; pegylated IFN (long-acting IFN bound to polyethylene glycol [PEG IFN]); and the oral agents lamivudine, adefovir dipivoxil, entecavir, telbivudine, and tenofovir. PEG IFN, entecavir, or tenofovir is recommended as first-line therapy. Simeprevir is a protease inhibitor that is approved for the treatment of hepatitis C virus infection. Acyclovir is used to treat herpes simplex viral infections, and ritonavir is a protease inhibitor used to treat HIV infection. The patient's husband should also be screened for HBV given the continued viremia.
- VII-82. The answer is B.** (*Chap. 334*) Chronic hepatitis develops in about 85% of all individuals affected with hepatitis C virus (HCV), and 20–25% of these individuals will progress to cirrhosis over about 20 years. Among those infected with HCV, about one-third of individuals will have normal or near-normal levels of aminotransferases, although liver biopsy demonstrates active hepatitis in as many as one-half of patients. Moreover, about 25% of individuals with normal aminotransferase levels at one point in time will develop elevations in these enzymes later and can develop progressive liver disease. Thus, normal aminotransferase levels at a single point in time do not definitively rule out the possibility that cirrhosis can develop. Progression to end-stage liver disease in individuals with chronic HCV hepatitis is more likely in older individuals and in patients with longer duration of infection, advanced histologic stage and grade, genotype 1 infection, more complex quasi-species diversity, concomitant other liver disease, HIV infection, and obesity. Among these factors, the best prognostic indicator for the development of progressive liver disease is liver histology. Specifically, patients who have moderate to severe inflammation or necrosis, including septal or bridging fibrosis, have the greatest risk of developing cirrhosis over the course of 10–20 years.
- VII-83. The answer is B.** (*Chap. 334*) Sofosbuvir and velpatasvir in combination would be the most appropriate first-line agents based on his genotype and current recommendations (Table VII-83). A liver biopsy can provide important prognostic information but with the advent of fibrosis prediction scores and newer imaging modalities, a biopsy is not necessary for most patients prior to initiation of treatment. Ribavirin may be indicated in some combination regimens but its use with pegylated interferon as first-line therapy is no longer common. Both telaprevir and boceprevir are eliminated by and inhibit CYP3A4 so these agents cannot be administered with other medications that induce CYP3A4 or are dependent on CYP3A4 for elimination. Patients with chronic hepatitis C who have detectable HCV RNA in serum, whether or not aminotransferase levels are increased, and chronic hepatitis of any grade and stage are candidates for antiviral therapy with DAA agents. The only exception would be patients with short life expectancies, for whom treating hepatitis C would have no influence on longevity.

TABLE VII-81 Comparison of Pegylated Interferon, Lamivudine, Adefovir, Entecavir, Telbivudine, and Tenofovir Therapy for Chronic Hepatitis B^a

Feature	PEG IFN ^b	Lamivudine	Adefovir	Entecavir	Telbivudine	Tenofovir
Route of administration	Subcutaneous injection	Oral	Oral	Oral	Oral	Oral
Duration of therapy ^c	48–52 weeks	≥52 weeks	≥48 weeks	≥48 weeks	≥52 weeks	≥48 weeks
Tolerability	Poorly tolerated	Well tolerated	Well tolerated; creatinine monitoring recommended	Well tolerated	Well tolerated	Well tolerated; creatinine monitoring recommended
HBeAg seroconversion						
1 yr Rx	18–20%	16–21%	12%	21%	22%	21%
>1 yr Rx	NA	Up to 50% @ 5 yrs	43% @ 3 yrs ^d	31% @ 2 yrs 44% @ 6 yrs	30% @ 2 yrs	40% @ 5 yrs
Log ₁₀ HBV DNA reduction (mean copies/mL)						
HBeAg-reactive	4.5	5.5	Median 3.5–5	6.9	6.4	6.2
HBeAg-negative	4.1	4.4–4.7	Median 3.5–3.9	5.0	5.2	4.6
HBV DNA PCR negative (<300–400 copies/mL; <1000 copies/mL for adefovir) at end of yr 1						
HBeAg-reactive	10–25%	36–44%	13–21%	67% (91% @ 4 yrs)	60%	76%
HBeAg-negative	63%	60–73%	48–77%	90%	88%	93%
ALT normalization at end of yr 1						
HBeAg-reactive	39%	41–75%	48–61%	68%	77%	68%
HBeAg-negative	34–38%	62–79%	48–77%	78%	74%	76%
HBsAg loss yr 1	3–4%	≤1%	0%	2%	<1%	3%
>yr 1	12% 5 yr after 1 yr of Rx	No data	5% at yr 5	6% at yr 6	No data	8% at yr 5
Histologic improvement (≥2 point reduction in HAI) at yr 1						
HBeAg-reactive	38% 6 months after	49–62%	53–68%	72%	65%	74%
HBeAg-negative	48% 6 months after	61–66%	64%	70%	67%	72%
Viral resistance	None	15–30% @ 1 yr 70% @ 5 yrs	None @ 1 yr 29% @ 5 yrs	≤1% @ 1 yr ^e 1.2% @ 6 yrs ^e	Up to 5% @ yr 1 Up to 22% @ yr 2	0% @ yr 1 0% through yr 8
Pregnancy category	C	C ^f	C	C	B	B
Cost (US\$) for 1 yr	~\$18,000	~\$2500	~\$6500	~\$8700 ^g	~\$6000	~\$6000

^aGenerally, these comparisons are based on data on each drug tested individually versus placebo in registration clinical trials; because, with rare exception, these comparisons are not based on head-to-head testing of these drugs, relative advantages and disadvantages should be interpreted cautiously.

^bAlthough standard interferon-α administered daily or three times a week is approved as therapy for chronic hepatitis B, it has been supplanted by PEG IFN, which is administered once a week and is more effective. Standard interferon has no advantages over PEG IFN.

^cDuration of therapy in clinical efficacy trials; use in clinical practice may vary.

^dBecause of a computer-generated randomization error that resulted in misallocation of drug versus placebo during the second year of clinical trial treatment, the frequency of HBeAg seroconversion beyond the first year is an estimate (Kaplan-Meier analysis) based on the small subset in whom adefovir was administered correctly.

^eSeven percent during a year of therapy (43% at year 4) in lamivudine-resistant patients.

^fDespite its Class C designation, lamivudine has an extensive pregnancy safety record in women with HIV/AIDS.

^gApproximately \$17,400 for lamivudine-refractory patients.

Abbreviations: ALT, alanine aminotransferase; HAI, histologic activity index; HBeAg, hepatitis B e antigen; HBsAg, hepatitis B surface antigen; HBV, hepatitis B virus; NA, not applicable; PEG IFN, pegylated interferon; PCR, polymerase chain reaction; Rx, therapy.

TABLE VII-83 Indications and Recommendations for Antiviral Therapy of Chronic Hepatitis C^a

Standard Indications for Therapy All patients with chronic HCV infection (detectable HCV RNA, with or without elevated ALT) except for those with short life expectancies because of comorbid conditions. Any stage of fibrosis; highest priority for advanced fibrosis (METAVIR stage 3)/cirrhosis (METAVIR stage 4) (pretreatment biopsy is no longer embraced and has been supplanted by noninvasive measures of fibrosis, e.g., imaging to determine liver elasticity) Responsiveness in groups previously refractory to IFN-based therapy (HIV-HCV co-infection, renal insufficiency, African American and Latino ethnicity, <i>IL28B</i> non-C haplotype, obesity, insulin resistance, hepatic decompensation, etc.) is not diminished to contemporary direct-acting oral combination regimens. Retreatment Recommended Relapsers, partial responders, or nonresponders after a previous course of IFN-based therapy or prior direct-acting antiviral therapy (see genotype-specific recommendations below). Antiviral Therapy Not Recommended Pregnancy: No clinical studies of direct-acting antivirals during pregnancy are available. Ribavirin is contraindicated during pregnancy; therefore, any regimen including ribavirin should not be used. Sofosbuvir; sofosbuvir + ledipasvir; and paritaprevir/ritonavir + ombitasvir + dasabuvir are classified as pregnancy category B, but the other direct-acting antivirals do not have a pregnancy classification. Therefore, these therapies are not indicated routinely in pregnancy and should be used, with caution, only if the benefit of treatment outweighs the potential for fetal risk. Therapeutic Regimens (based on AASLD-IDSA recommendations, www.hcvguidelines.org)^b The European Association for the Study of the Liver (EASL) issued recommendations in 2016; divergences from AASLD-IDSA recommendations are summarized as a footnote below.^c Treatment-Naïve or Relapsed After Prior PEG IFN/Ribavirin Therapy Genotype 1a ledipasvir + sofosbuvir 12 weeks (consider 8 weeks for noncirrhotic patients with HCV RNA $<6 \times 10^6$ IU/mL) paritaprevir/ritonavir + ombitasvir + dasabuvir + RBV 12 weeks (no cirrhosis) or 24 weeks (cirrhosis) sofosbuvir + simeprevir 12 weeks (no cirrhosis) or $\pm$ RBV 24 weeks (cirrhosis) daclatasvir + sofosbuvir 12 weeks (no cirrhosis) or $\pm$ RBV 24 weeks (cirrhosis) grazoprevir + elbasvir 12 weeks (no cirrhosis or cirrhosis sans ELB NS5A RASs) or + RBV $\times$ 16 weeks (ELB NS5A RASs) sofosbuvir + velpatasvir 12 weeks Genotype 1b ledipasvir + sofosbuvir 12 weeks (consider 8 weeks for noncirrhotic patients with HCV RNA $<6 \times 10^6$ IU/mL) paritaprevir/ritonavir + ombitasvir + dasabuvir 12 weeks sofosbuvir + simeprevir 12 weeks (no cirrhosis) or $\pm$ RBV 24 weeks (cirrhosis) daclatasvir + sofosbuvir 12 weeks (no cirrhosis) or $\pm$ RBV 24 weeks (cirrhosis) grazoprevir + elbasvir 12 weeks sofosbuvir + velpatasvir 12 weeks Genotype 2 sofosbuvir + velpatasvir 12 weeks daclatasvir + sofosbuvir (no cirrhosis) 12 weeks or 16–24 weeks (cirrhosis)	Failed Prior PEG IFN/Ribavirin Therapy, No Cirrhosis Genotype 1a ledipasvir + sofosbuvir 12 weeks paritaprevir/ritonavir + ombitasvir + dasabuvir + RBV 12 weeks sofosbuvir + simeprevir 12 weeks daclatasvir + sofosbuvir 12 weeks grazoprevir + elbasvir 12 weeks (without ELB NS5A RASs) or + RBV $\times$ 16 weeks (ELB NS5A RASs) sofosbuvir + velpatasvir 12 weeks Genotype 1b ledipasvir + sofosbuvir 12 weeks paritaprevir/ritonavir + ombitasvir + dasabuvir 12 weeks sofosbuvir + simeprevir 12 weeks daclatasvir + sofosbuvir 12 weeks grazoprevir + elbasvir 12 weeks sofosbuvir + velpatasvir 12 weeks Genotype 2 sofosbuvir + velpatasvir 12 weeks daclatasvir + sofosbuvir 12 weeks Genotype 3 sofosbuvir + velpatasvir 12 weeks daclatasvir + sofosbuvir 12 weeks Genotype 4 sofosbuvir + velpatasvir 12 weeks ledipasvir + sofosbuvir 12 weeks paritaprevir/r + ombitasvir + RBV 12 weeks (no dasabuvir) grazoprevir + elbasvir 12 weeks (prior relapse) or + RBV 16 weeks (prior nonresponse) Genotypes 5 and 6 sofosbuvir + velpatasvir 12 weeks ledipasvir + sofosbuvir 12 weeks Failed Prior PEG IFN/Ribavirin Therapy, Compensated Cirrhosis Genotype 1a ledipasvir + sofosbuvir + RBV 12 weeks ledipasvir + sofosbuvir 24 weeks sofosbuvir + velpatasvir 12 weeks grazoprevir + elbasvir 12 weeks (without ELB NS5A RASs) or + RBV $\times$ 16 weeks (ELB NS5A RASs) paritaprevir/ritonavir + ombitasvir + dasabuvir + RBV 24 weeks sofosbuvir + simeprevir $\pm$ RBV 24 weeks (no Q80K variant) daclatasvir + sofosbuvir $\pm$ RBV 24 weeks Genotype 1b ledipasvir + sofosbuvir + RBV 12 weeks ledipasvir + sofosbuvir 24 weeks sofosbuvir + velpatasvir 12 weeks grazoprevir + elbasvir 12 weeks paritaprevir/ritonavir + ombitasvir + dasabuvir 12 weeks sofosbuvir + simeprevir $\pm$ RBV 24 weeks daclatasvir + sofosbuvir $\pm$ RBV 24 weeks Genotype 2 sofosbuvir + velpatasvir 12 weeks sofosbuvir + daclatasvir 16 or 24 weeks Genotype 3 sofosbuvir + velpatasvir 12 weeks daclatasvir + sofosbuvir + RBV 24 weeks Genotype 4 sofosbuvir + velpatasvir 12 weeks ledipasvir + sofosbuvir + RBV 12 weeks
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(continued)

TABLE VII-83 Indications and Recommendations for Antiviral Therapy of Chronic Hepatitis C^a (Continued)

Genotype 3	Features Associated With Reduced Responsiveness To Direct-Acting Antiviral Combination Therapy
sofosbuvir + velpatasvir 12 weeks	Genotype and subtype (genotype 1a less responsive than genotype 1b for several drugs)
daclatasvir + sofosbuvir 12 weeks (no cirrhosis) or ± RBV 24 weeks (cirrhosis)	Treatment experience
Genotype 4	Advanced fibrosis (bridging fibrosis, cirrhosis)
sofosbuvir + velpatasvir 12 weeks	Reduced adherence
ledipasvir + sofosbuvir 12 weeks	
paritaprevir/r + ombitasvir + RBV 12 weeks (no dasabuvir)	
grazoprevir + elbasvir 12 weeks	
Genotypes 5 and 6	
sofosbuvir + velpatasvir 12 weeks	
ledipasvir + sofosbuvir 12 weeks	
paritaprevir/ritonavir + ombitasvir + RBV 12 weeks (no dasabuvir)	
grazoprevir + elbasvir 12 weeks (prior relapse) or + RBV 16 weeks (prior nonresponse)	
ledipasvir + sofosbuvir 24 weeks	
Genotypes 5 and 6	
sofosbuvir + velpatasvir 12 weeks	
ledipasvir + sofosbuvir 12 weeks	

^aFor rapidly evolving new recommendations continue to be issued; for up-to-date treatment recommendations, please see www.hcvguidelines.org.

^bClass I recommendations in **bold** font, all others are class II recommendations.

^cThe following EASL recommendations differ from those of AASLD-IDSA. (Please note that, although mentioned in EASL recommendations, testing for baseline RASs is not recommended routinely, but, if reliable resistance testing available, results can be used to guide therapy.):

Genotype 1

For genotype 1, simeprevir + sofosbuvir is not recommended.

For genotype 1a, treatment-experienced patients (IFN-based regimen failures) treated with sofosbuvir + ledipasvir should have weight-based ribavirin added. If reliable testing for RASs is available, ribavirin is needed only if baseline RASs are present, and, in such patients, if ribavirin is contraindicated, sofosbuvir + ledipasvir should be extended to 24 weeks.

For genotype 1b, in treatment-naïve, noncirrhotic patients receiving paritaprevir/ritonavir + ombitasvir + dasabuvir a treatment duration of 8 weeks can be considered.

For genotype 1a, in patients treatment with grazoprevir + elbasvir, EASL recommends testing for ELB RASs even in noncirrhotics. If resistance testing is not done, the level of baseline HCV RNA should determine whether ribavirin is added and the duration of therapy. If HCV RNA >800,000 IU/mL, add ribavirin and treat for 16 weeks; if HCV RNA ≤800,000 IU/mL, ribavirin is not added, and treatment for 12 weeks suffices. If baseline testing for RASs is available, patients with HCV RNA >800,000 IU/mL and detectable RASs should be treated with ribavirin for 16 weeks. Treatment without ribavirin and for 12 weeks suffices if HCV RNA ≤800,000 IU/mL even with detectable RASs or even if HCV RNA >800,000 IU/mL with undetectable RASs.

For genotype 1a, in treatment-experienced patients (IFN-based regimen failures) treated with daclatasvir + sofosbuvir, follow the same recommendations described above for ledipasvir + sofosbuvir regarding the addition of ribavirin.

Genotype 2

EASL recommendations are the same as those of AASLD-IDSA.

Genotype 3

For treatment-experienced patients (IFN-based regimen failures) treated with sofosbuvir + velpatasvir or sofosbuvir + daclatasvir, if testing for baseline RASs is not available, add weight-based ribavirin. If resistance testing is available, ribavirin is needed only if baseline RASs are present, and, in such patients, if ribavirin is contraindicated, treatment should be extended to 24 weeks.

Genotype 4

Treatment-experienced patients (IFN-based regimen failures) treated with sofosbuvir + ledipasvir should have weight-based ribavirin added, and, in such patients, if ribavirin is contraindicated, treatment should be extended to 24 weeks.

In treatment-experienced patients (IFN-based regimen failures) treated with grazoprevir + elbasvir, if HCV RNA >800,000 IU/mL, weight-based ribavirin should be added, and treatment should be extended to 16 weeks.

EASL recommends two additional treatment options for genotype 4 (noncirrhotic or cirrhotic) that are not included in AASLD-IDSA guidelines: sofosbuvir + daclatasvir and sofosbuvir + simeprevir. For both these options, treatment-naïve patients should be treated for 12 weeks without ribavirin; treatment-experienced (IFN-based regimen failures) patients should be treated with ribavirin for 12 weeks or, if ribavirin is contraindicated, without ribavirin for 24 weeks.

Genotypes 5 and 6

Treatment-experienced patients (IFN-based regimen failures) treated with sofosbuvir + ledipasvir should have weight-based ribavirin added, and, in such patients, if ribavirin is contraindicated, treatment should be extended to 24 weeks.

EASL recommends an additional treatment option for genotypes 5 and 6 (noncirrhotic or cirrhotic) that is not included in AASLD-IDSA guidelines: sofosbuvir + daclatasvir. Treatment-naïve patients should be treated for 12 weeks without ribavirin; treatment-experienced (IFN-based regimen failures) patients should be treated with ribavirin for 12 weeks or, if ribavirin is contraindicated, without ribavirin for 24 weeks.

Drug doses: sofosbuvir 400 mg; ledipasvir 90 mg; paritaprevir 150 mg; ritonavir 100 mg; ombitasvir 25 mg; dasabuvir 250 mg; ribavirin, weight-based: 1000 mg (<75 kg) to 1200 mg (≥75 kg); simeprevir 150 mg; daclatasvir 60 mg; elbasvir 50 mg; grazoprevir 100 mg; velpatasvir 100 mg. **Abbreviations:** AASLD, American Association for the Study of Liver Diseases; ALT, alanine aminotransferase; EASL, European Association for the Study of the Liver; ELB NS5A RASs, elbasvir NS5A resistance-associated substitutions; HCV, hepatitis C virus; IFN, interferon; IDSA, Infectious Diseases Society of America; IU, international units (1 IU/mL is equivalent to ~2.5 copies/mL); PEG IFN, pegylated interferon; RASs, resistance-associated substitutions; RBV, ribavirin.

- VII-84. The answer is C.** (*Chap. 334*) Three types of autoimmune hepatitis have been identified based on clinical and laboratory characteristics. Type I autoimmune hepatitis is typically a disorder seen in young women. The clinical characteristics can be variable from those of chronic hepatitis to fulminant hepatic failure, and many of the features are difficult to distinguish from other causes of chronic hepatitis. In some individuals, extrahepatic manifestations, including fatigue, malaise, weight loss, anorexia, and arthralgias, can be quite prominent. Liver enzymes are elevated but may not correlate with the clinical severity of disease. In more severe cases, elevations in serum bilirubin between 3 and 10 mg/dL can be seen. Hypoalbuminemia occurs in advanced disease, and hypergammaglobulinemia (>2.5 g/dL) is common. The circulating antibody profile in autoimmune hepatitis depends to some extent on the type of hepatitis. Antinuclear antibodies are positive in a homogeneous staining pattern almost invariably in the disease, and rheumatoid factor is also common. Perinuclear antineutrophilic cytoplasmic antibody (p-ANCA) may be positive, but in an atypical fashion. Anti-smooth muscle antibodies and anti-liver/kidney microsomal antibodies are frequently seen, but these are nonspecific because other causes of chronic hepatitis can lead to positivity of these enzymes. Because of the lack of a specific autoimmune profile, the diagnostic criteria for autoimmune hepatitis incorporate a variety of clinical and laboratory features. Specific features that argue against this diagnosis include prominent alkaline phosphatase elevation; presence of mitochondrial antibodies; markers of viral hepatitis; history of hepatotoxic drugs or excess alcohol intake; histologic evidence of bile duct injury; or atypical biopsy features including excess hepatic iron, fatty infiltration, and viral inclusions. Antimitochondrial antibodies are typically seen in primary biliary cirrhosis, not autoimmune hepatitis.
- VII-85. The answer is E.** (*Chap. 334*) Cellular immune mechanisms appear to be important in the pathogenesis of autoimmune hepatitis. In vitro studies have suggested that in patients with this disorder, CD4⁺ T lymphocytes are capable of becoming sensitized to hepatocyte membrane proteins and of destroying liver cells. Molecular mimicry by cross-reacting antigens that contain epitopes similar to liver antigens is postulated to activate these T cells, which infiltrate, and result in injury to the liver. Abnormalities of immunoregulatory control over cytotoxic lymphocytes (impaired regulatory CD4⁺CD25⁺ T-cell influences) may play a role as well. Circulating autoantibodies are prevalent in patients with autoimmune hepatitis and may aid in diagnosis. However, they have not been firmly linked to direct hepatocyte injury. Immune complex deposition is important in the extrahepatic manifestations of autoimmune hepatitis such as arthralgias, arthritis, cutaneous vasculitis, and glomerulonephritis. Immune complex deposition leads to complement activation, inflammation, and tissue injury.
- VII-86. The answer is C.** (*Chap. 335*) The pathology of alcoholic liver disease consists of three major lesions, with the progressive injury rarely existing in a pure form: (1) fatty liver, (2) alcoholic hepatitis, and (3) cirrhosis. Fatty liver is present in $>90\%$ of daily as well as binge drinkers. A much smaller percentage of heavy drinkers will progress to alcoholic hepatitis, thought to be a precursor to cirrhosis. The prognosis of severe alcoholic liver disease is dismal; the mortality of patients with alcoholic hepatitis concurrent with cirrhosis is nearly 60% at 4 years. Although alcohol is considered a direct hepatotoxin, only between 10 and 20% of alcoholics will develop alcoholic hepatitis. The explanation for this apparent paradox is unclear but involves the complex interaction of facilitating factors, such as drinking patterns, diet, obesity, and gender. There are currently no diagnostic tools that can predict individual susceptibility to alcoholic liver disease. Quantity and duration of alcohol intake are the most important risk factors involved in the development of alcoholic liver disease (Table VII-86). The roles of beverage type(s) (i.e., wine, beer, or spirits) and pattern of drinking (daily vs binge drinking) are less clear. Chronic infection with hepatitis C virus (HCV) is an important comorbidity in the progression of alcoholic liver disease to cirrhosis. Even moderate alcohol intake of 20–50 g/d increases the risk of cirrhosis and hepatocellular cancer in HCV-infected individuals. Patients with both alcoholic liver injury and HCV infection develop decompensated liver disease at a younger age and have poorer overall survival.

TABLE VII-86 Risk Factors for Alcoholic Liver Disease

Risk Factor	Comment
Quantity	In men, 40–80 g/d of ethanol produces fatty liver; 160 g/d for 10–20 years causes hepatitis or cirrhosis Only 15% of alcoholics develop alcoholic liver disease
Gender	Women exhibit increased susceptibility to alcoholic liver disease at amounts >20 g/d; two drinks per day is probably safe
Hepatitis C	Hepatitis C virus infection concurrent with alcoholic liver disease is associated with younger age for severity, more advanced histology, and decreased survival
Genetics	Patatin-like phospholipase domain-containing protein 3 has been associated with alcoholic cirrhosis
Fatty liver	Alcohol injury does not require malnutrition, but obesity and nonalcoholic fatty liver are risk factors Patients should receive vigorous attention to nutritional support

VII-87. The answer is C. (*Chap. 335*) This patient presents with severe acute alcoholic hepatitis. In its earliest form, alcoholic liver disease is marked by fatty infiltration of the liver. In more acute alcoholic hepatitis, there is hepatocyte injury with balloon degeneration and necrosis. Many cases of alcoholic hepatitis are asymptomatic. However, as in this case, the severe manifestations can include fever, jaundice, spider nevi, and abdominal pain that can mimic an acute abdomen in its severity. On laboratory examination, the aspartate aminotransferase is typically elevated more than the alanine aminotransferase, although the total transaminase levels are rarely greater than 400 IU/L. Hyperbilirubinemia can be quite marked with lesser elevation in alkaline phosphatase. Hypoalbuminemia and coagulopathy are poor prognostic indicators. A discriminate function (DF) can be calculated as follows: $(4.6 \times \text{the prolongation of prothrombin time above control}) + \text{serum bilirubin}$. A DF >32 is associated with poor prognosis and is an indication for treatment of acute alcoholic hepatitis. The model for end-stage liver disease score can also be used for prognostication in acute alcoholic hepatitis, with a score greater than 21 being an indication for treatment as well. This patient has a DF of >80, indicating very severe disease and a poor prognosis. Complete abstinence from alcohol is imperative. Treatment with prednisone 40 mg daily (or prednisolone 32 mg daily) for 4 weeks should be initiated. Following the initial period, a taper should be achieved over a period of 4 weeks. The role of tumor necrosis factor (TNF)- α expression and receptor activity in alcoholic liver injury has led to an examination of TNF inhibition as an alternative to glucocorticoids for severe alcoholic hepatitis. The nonspecific TNF inhibitor pentoxifylline (400 mg three times daily for 4 weeks) demonstrated improved survival in the therapy of severe alcoholic hepatitis, primarily due to a decrease in hepatorenal syndrome. Monoclonal antibodies that neutralize serum TNF- α should not be used in alcoholic hepatitis because of studies reporting increased deaths secondary to infection and renal failure. Liver transplantation is an accepted indication for treatment in selected and motivated patients with end-stage cirrhosis. Outcomes are equal or superior to other indications for transplantation.

VII-88. The answer is C. (*Chap. 335*) Alcoholic fatty liver was once thought to be relatively benign, but in the presence of features such as giant mitochondria, perivenular fibrosis, and macrovesicular fat it may be associated with progressive liver injury. Fatty accumulation begins in the perivenular hepatocytes and can eventually involve the entire hepatic lobule. Ballooning degeneration, spotty necrosis, polymorphonuclear infiltrate, and fibrosis in the perivenular and perisinusoidal space of Disse are hallmarks of alcoholic hepatitis. The changes of both alcoholic fatty liver disease and alcoholic hepatitis are reversible after the cessation of drinking. It can be hard to differentiate alcoholic from nonalcoholic fatty liver disease. A careful drinking history is essential to identifying alcohol as a risk factor.

VII-89. The answer is C. (*Chap. 336*) At present, there are no Food and Drug Administration-approved therapies for the treatment of nonalcoholic fatty liver disease (NAFLD). Thus, the current approach to NAFLD management focuses on treatment to improve the risk

factors for nonalcoholic steatohepatitis (NASH; i.e., obesity, insulin resistance, metabolic syndrome, dyslipidemia). Based on our understanding of the natural history of NAFLD, only patients with NASH or those with features of hepatic fibrosis on liver biopsy are considered currently for targeted pharmacologic therapies. Lifestyle changes and dietary modification are the foundation for NAFLD treatment. Many studies indicate that lifestyle modification can improve serum aminotransferases and hepatic steatosis, with loss of at least 3–5% of body weight improving steatosis, but greater weight loss (up to 10%) necessary to improve steatohepatitis. The benefits of different dietary macronutrient content (e.g., low-carbohydrate vs low-fat diets, saturated vs unsaturated fat diets) and different intensities of calorie restriction appear to be comparable. In adults with NAFLD, exercise regimens that improve fitness may be sufficient to reduce hepatic steatosis, but their impact on other aspects of liver histology remains unknown. Antioxidants have also been evaluated for the treatment of NAFLD because oxidant stress is thought to contribute to the pathogenesis of NASH. Vitamin E, an inexpensive yet potent antioxidant, has been examined in several small pediatric and adult studies with varying results. In all of those studies, vitamin E was well tolerated, and most showed modest improvements in aminotransferase levels, radiographic features of hepatic steatosis, and/or histologic features of NASH. Statins are an important class of agents to treat dyslipidemia and decrease cardiovascular risk. There is no evidence to suggest that statins cause liver failure in patients with any chronic liver disease, including NAFLD. The incidence of liver enzyme elevations in NAFLD patients taking statins is also no different than that of healthy controls or patients with other chronic liver diseases. Moreover, several studies have suggested that statins may improve aminotransferases and histology in patients with NASH. However, there is continued reluctance to use statins in patients with NAFLD. The lack of evidence that statins harm the liver in NAFLD patients, combined with the increased risk for cardiovascular morbidity and mortality in NAFLD patients, warrants the use of statins to treat dyslipidemia in patients with NAFLD/NASH. Although interest in bariatric surgery as a treatment for NAFLD exists, a recently published *Cochrane Review* concluded that lack of randomized clinical trials or adequate clinical studies prevents definitive assessment of benefits and harms of bariatric surgery as a treatment for NASH. Most studies of bariatric surgery have shown that it is generally safe in individuals with well-compensated chronic liver disease and improves hepatic steatosis and necroinflammation (i.e., features of NAFLD/NASH); however, effects on hepatic fibrosis have been variable.

VII-90. The answer is A. (*Chap. 336*) In some studies up to 25% of Americans have hepatic steatosis on imaging. About 25% of patients with nonalcoholic fatty liver disease (NAFLD) will go on to develop nonalcoholic steatohepatitis (NASH), which can lead to cirrhosis. By some estimates, NAFLD will be the most common indication for liver transplant in the next decade. Insulin resistance, in large part secondary to obesity, is a big driver of NAFLD and NASH, as insulin stimulates lipid uptake, fat synthesis, and fat storage leading to triglyceride accumulation in hepatocytes. The triglycerides themselves are not hepatotoxic. However, their precursors (e.g., fatty acids and diacylglycerols) and metabolic by-products (e.g., reactive oxygen species) may damage hepatocytes, leading to hepatocyte lipotoxicity. Lipotoxicity also triggers the generation of other factors (e.g., inflammatory cytokines, hormonal mediators) that deregulate systems that normally maintain hepatocyte viability. The net result is increased hepatocyte death. There are no serum biomarkers that can reliably make the diagnosis of NASH. In NAFLD the levels of serum aminotransferases (aspartate aminotransferase and alanine aminotransferase) do not reliably reflect the severity of liver cell injury, extent of liver cell death, or related liver inflammation and fibrosis. Thus, they are imperfect for determining which individuals with NAFLD have NASH.

VII-91. The answer is B. (*Chap. 337*) Spontaneous bacterial peritonitis (SBP) is a common and severe complication of ascites characterized by spontaneous infection of the ascitic fluid without an intra-abdominal source. In patients with cirrhosis and ascites severe enough for hospitalization, SBP can occur in up to 30% of individuals and can have a 25% in-hospital mortality rate. Bacterial translocation is the presumed mechanism for development of SBP, with gut flora traversing the intestine into mesenteric lymph nodes, leading

to bacteremia and seeding of the ascitic fluid. The most common organisms are *Escherichia coli* and other gut bacteria; however, gram-positive bacteria, including *Streptococcus viridans*, *Staphylococcus aureus*, and *Enterococcus* spp., can also be found. If more than two organisms are identified, secondary bacterial peritonitis due to a perforated viscus should be considered. The diagnosis of SBP is made when the fluid sample has an absolute neutrophil count $>250/\mu\text{L}$. In this case, the patient has an absolute neutrophil count of $320/\mu\text{L}$ (800×0.4), and, therefore, has SBP. Patients with ascites may present with fever, altered mental status, elevated white blood cell count, and abdominal pain or discomfort, or they may present without any of these features. Therefore, it is necessary to have a high degree of clinical suspicion, and peritoneal taps are important for making the diagnosis. Treatment is with a second-generation cephalosporin, with cefotaxime as the most commonly used antibiotic. In patients with variceal hemorrhage, the frequency of SBP is significantly increased, and prophylaxis against SBP is recommended when a patient presents with upper gastrointestinal bleeding. Furthermore, in patients who have had an episode(s) of SBP and recovered, once-weekly administration of antibiotics is used as prophylaxis for recurrent SBP. There is no indication for hemodialysis with the normal serum creatine or esophagogastroduodenoscopy with no history of bleeding and a stable hemoglobin. Blood urea nitrogen (BUN) may increase as a result of the infection. Similarly, although the BUN is elevated and the patient has altered mental status, lactulose would not treat the primary disorder causing the altered mental status. Given the likely diagnosis of SBP, empiric therapy for meningitis is not warranted at this time.

- VII-92. The answer is C.** (*Chap. 337*) The clinical presentation is consistent with a cholestatic picture, which can present with painless jaundice and pruritus. The pruritus can be prominent and is present in 50% of individuals at the time of diagnosis. The pruritus is typically intermittent and worse in the evening. There is no other prominent association, such as following hot baths or showers, which occurs in polycythemia vera. Other causes of pruritus outside of cholestasis include lymphoma and uncontrolled hypo- or hyperthyroidism. However, the laboratory studies in this patient clearly represent cholestasis with an elevation in alkaline phosphatase and bilirubin. The clinical characteristics are more commonly seen in primary biliary cirrhosis compared with primary sclerosis cholangitis as the patient is a middle-aged female with positive antimitochondrial antibodies. In contrast, primary sclerosing cholangitis is associated with positive perinuclear antineutrophil cytoplasmic antibodies in 65% of patients, and 50% of individuals with primary sclerosing cholangitis have a history of ulcerative colitis.
- VII-93. The answer is A.** (*Chap. 337*) The cornerstone of the management of ascites is sodium restriction to less than 2 g daily. A common misconception is to institute a fluid restriction as well. However, this is neither effective nor necessary. With a sodium restriction to 2 g daily, most mild ascites can be managed quite well. If sodium restriction alone fails to correct ascites, then initiation of diuretics is required. Spironolactone at a dose of 100–200 mg daily is the initial diuretic used for ascites and can be titrated as high as 400–600 mg daily if tolerated. Loop diuretics can be added to spironolactone. The typical agent is furosemide beginning at 40–80 mg daily with maximum doses of about 120–160 mg daily. Care must be taken to avoid renal dysfunction with loop diuretics, and higher doses may not be tolerated. If ascites is refractory to these treatments, transjugular intrahepatic portosystemic shunt (TIPS) can be considered. This procedure creates a direct portocaval shunt by introducing an expandable metal stent from the hepatic veins through the substance of the liver into the portal veins. Thus, TIPS decreases portal pressure, which in turn should decrease ascites and the risk of variceal bleeding. However, hepatic encephalopathy typically worsens following TIPS.
- VII-94. The answer is B.** (*Chap. 337*) The presence of cirrhosis in an elderly woman with no prior risk factors for viral or alcoholic cirrhosis should raise the possibility of primary biliary cirrhosis (PBC). PBC is characterized by chronic inflammation and fibrous obliteration of intrahepatic ductules. The cause is unknown, but autoimmunity is assumed because there is an association with other autoimmune disorders, such as autoimmune thyroiditis, CREST (calcinosis, Raynaud's phenomenon, esophageal dysmotility, sclerodactyly,

telangiectasia) syndrome, and sicca syndrome. The vast majority of patients with symptomatic disease are women. The antimitochondrial antibody (AMA) test is positive in over 90% of patients with PBC and only rarely is positive in other conditions. This makes it the most useful initial test in the diagnosis of PBC. Because there are false-positive results, if AMA is positive, a liver biopsy is performed to confirm the diagnosis. The 24-hour urine copper collection is useful in the diagnosis of Wilson disease. Hepatic failure from Wilson disease typically occurs before age 50 years. Hemochromatosis may result in cirrhosis. It is associated with lethargy, fatigue, loss of libido, discoloration of the skin, arthralgias, diabetes, and cardiomyopathy. Ferritin levels are usually increased, and the most suggestive laboratory abnormality is an elevated transferrin saturation percentage. Although hemochromatosis is a possible diagnosis in this case, PBC is more likely in light of the clinical scenario. Although chronic hepatitis B and hepatitis C are certainly in the differential diagnosis and must be ruled out, they are unlikely because of the patient's history and lack of risk factors.

VII-95. The answer is B. (*Chap. 337*) Infection is probably the most common, and the most concerning cause of worsening hepatic encephalopathy in a patient with known liver disease. Dehydration and electrolyte abnormalities are common causes of hepatic encephalopathy. Dehydration can lead to a contraction alkalosis that causes more ammonium ions (NH_4^+) to become NH_3 , which can more easily cross the blood-brain barrier to provoke encephalopathy. Hypokalemia can have the same effect. Hyperkalemia would be associated with the extracellular shifting of hydrogen ions and have the opposite effect, and therefore is less likely. Hyponatremia can directly cause altered mental status but can also be associated with dehydration, both of which can worsen encephalopathy. Medication nonadherence is a common cause of worsening hepatic encephalopathy.

VII-96. The answer is D. (*Chap. 337*) The approach to patients once they have had a variceal bleed is first to treat the acute bleed, which can be life-threatening, and then to prevent further bleeding. The medical management of acute variceal hemorrhage includes the use of vasoconstricting agents, usually somatostatin or octreotide. Balloon tamponade (Sengstaken-Blakemore tube or Minnesota tube) can be used in patients who cannot receive endoscopic therapy immediately or who need stabilization prior to endoscopic therapy. Endoscopic intervention is used as first-line treatment to control bleeding acutely. Some endoscopists will use variceal injection therapy (sclerotherapy) as initial therapy, particularly when bleeding is vigorous. Variceal band ligation is used to control acute bleeding in over 90% of cases and should be repeated until obliteration of all varices is accomplished. When esophageal varices extend into the proximal stomach, band ligation is less successful. In these situations, when bleeding continues from gastric varices, consideration for transjugular intrahepatic portosystemic shunt (TIPS) should be made. This offers an alternative to surgery for acute decompression of portal hypertension. Encephalopathy can occur in as many as 20% of patients after TIPS and is particularly problematic in elderly patients and in patients with pre-existing encephalopathy. TIPS should be reserved for individuals who fail endoscopic or medical management or who are poor surgical risks. TIPS can sometimes be used as a bridge to transplantation. Beta blockers, such as propranolol and nadolol, have been shown to decrease the risk of recurrent variceal bleeding and reduce mortality from a subsequent bleed but should not be used in the setting of an acutely bleeding patient.

VII-97. The answer is B. (*Chap. 337*) Prevention of further or subsequent variceal bleeding can be accomplished with repeated variceal band ligation until varices are obliterated. Non-selective beta blockers, such as propranolol and nadolol, have been shown to decrease the risk of recurrent variceal bleeding and reduce mortality from a subsequent bleed. Their efficacy is lessened once the esophageal varices have been banded, but they can be useful to treat portal hypertensive gastropathy that is not directly treated with variceal ligation. A TIPS would reduce portal pressure and lead to a reduction in recurrent variceal bleeding. Transplant would likewise restore normal portal pressure and lead to resolution of existing varices. A continuous infusion of octreotide may promote splanchnic vasoconstriction and can be used to manage acute variceal bleeding. Subcutaneous octreotide is not used as a prophylactic treatment.

- VII-98. The answer is B.** (*Chap. 337*) Cessation of drinking markedly improves survival in patients with alcoholic cirrhosis and there is evidence that fibrosis can improve after cessation. Alcoholic cirrhosis is usually “micronodular” (<3 mm) but with the cessation of drinking larger nodules may form. Alcohol is the second most common cause of cirrhosis in the United States, behind hepatitis C. Pentoxifylline is an oral agent that reduces production of tumor necrosis factor (TNF)- α and has been used in the treatment of acute alcoholic hepatitis. Recent studies have used parenterally administered inhibitors of TNF- α such as infliximab or etanercept for the treatment of alcoholic hepatitis. Early results have shown no adverse events; however, there was no clear-cut improvement in survival for acute hepatitis or cirrhosis.
- VII-99. The answer is D.** (*Chap. 337*) Hepatorenal syndrome (HRS) is caused by a combination of increased renovascular constriction and reduced effective circulatory volume. The use of diuretics would potentially worsen renal perfusion and could precipitate worsening renal failure. A number of treatments have been tried in the setting of HRS with limited efficacy including IV albumin along with midodrine and octreotide. Most patients will recover renal function if they undergo liver transplantation. It is always advisable to consider a wide range of potential causes for renal failure in patients with chronic liver disease.
- VII-100. The answer is D.** (*Chap. 338*) Patients with fulminant hepatic failure are highest priority for cadaveric liver transplantation. Currently in the United States, all donor livers are distributed through a nationwide organ-sharing network (United Network for Organ Sharing) designed to allocate available organs based on regional considerations and recipient acuity. Recipients who have the highest disease severity generally have the highest priority, but allocation strategies that balance highest urgency against best outcomes continue to evolve to distribute cadaver organs most effectively. Allocation is based on the model for end-stage liver disease (MELD) score, which is based on a mathematical model that includes bilirubin, creatinine, and international normalized ratio. Neither waiting time (except as a tiebreaker between two potential recipients with the same MELD scores) nor posttransplantation outcome is taken into account. Use of the MELD score has been shown to reduce waiting list mortality, to reduce waiting time prior to transplantation, and to be the best predictor of pretransplantation mortality. The highest priority (status 1) for liver transplantation continues to be reserved for patients with fulminant hepatic failure or primary graft nonfunction.
- VII-101. The answer is B.** (*Chap. 338*) Several conditions receive a disease-specific model for end-stage liver disease exception, which allows patients to receive a liver when they might not be as clinically ill. Such conditions include hepatocellular carcinoma, portopulmonary hypertension, hepatopulmonary syndrome, familial amyloid polyneuropathy, primary hyperoxaluria (necessitating liver-kidney transplantation), cystic fibrosis liver disease, and highly selected cases of hilar cholangiocarcinoma. ABO incompatibility and preformed cytotoxic human leukocyte antigen antibodies are not contraindications to transplant. HIV patients have successfully undergone liver transplantation. While living donor transplants have increased the number of potential donors, the procedure comes with several important risks to the donor including biliary complications in ~5%; postoperative complications such as wound infection, small-bowel obstruction, and incisional hernias in 9–19%, and, even death in 0.2–0.4%.
- VII-102. The answer is B.** (*Chap. 339*) In the National Health and Nutrition Examination Survey, the prevalence of gallstone disease in the United States was 7.9% in men and 16.6% in women. Although the disease is quite prevalent, not all patients with gallstone disease require cholecystectomy. It is estimated that 1–2% of patients with asymptomatic gallstone disease will develop complications that will require surgery yearly. Therefore, it is important to know which patients with asymptomatic gallstones require referral for surgery. The first factor to consider is whether the patient has symptoms that are caused by gallstones and are frequent enough and severe enough to necessitate

surgery. Commonly called biliary colic, the classic symptoms of gallstone disease are right upper quadrant pain and fullness that begins suddenly and can last as long as 5 hours. Nausea and vomiting can accompany the episode. Vague symptoms of epigastric fullness, dyspepsia, and bloating following meals should not be considered biliary colic. A second factor that would be considered in recommending a patient for cholecystectomy is whether the patient has a prior history of complications of gallstone disease such as pancreatitis or acute cholecystitis. A final factor that would lead to the recommendation for cholecystectomy is the presence of anatomical factors that would increase the likelihood of complications such as a porcelain gallbladder or congenital abnormalities of the biliary tract. Individuals with very large stones (>3 cm) would also need to be considered carefully for cholecystectomy. Ursodeoxycholic acid can be used in some instances to dissolve gallstones. It acts to decrease the cholesterol saturation of bile and also allows dispersion of cholesterol from stones by producing a lamellar crystalline phase. However, it is only effective in individuals with radiolucent stones measuring less than 10 mm.

VII-103. The answer is D. (*Chap. 339*) A practitioner needs to have a high index of suspicion for acalculous cholecystitis in critically ill patients who develop decompensation during the course of treatment for the underlying disease and have no other apparent source of infection. Some predisposing conditions for the development of acalculous cholecystitis include serious trauma or burns, postpartum following prolonged labor, prolonged parenteral hyperalimentation, and during the postoperative period following orthopedic and other major surgical procedures. The clinical manifestations of acalculous cholecystitis are identical to calculous disease, but the disease is more difficult to diagnose. Ultrasonography and CT scanning typically only show biliary sludge, but they may demonstrate large and tense gallbladders. Hepatobiliary scintigraphy often shows delayed or absent gallbladder emptying. Successful management relies on accurate and early diagnosis. In critically ill patients, a percutaneous cholecystostomy may be the safest immediate procedure to decompress an infected gallbladder. Once the patient is stabilized, early elective cholecystectomy should be considered. Metronidazole to provide anaerobic coverage should be added, but this would not elucidate or adequately treat the underlying condition.

VII-104. The answer is B. (*Chaps. 45 and 339*) The clinical presentation is consistent with a cholestatic picture. Painless jaundice always requires an extensive workup, as many of the underlying pathologies are ominous and early detection and intervention often offers the only hope for a good outcome. In this case, the gallbladder showed no evidence of stones, and the patient shows no evidence of clinical cholecystitis, and so a hepatobiliary iminodiacetic acid scan is not indicated. Similarly, antibiotics are not necessary at this point. The cholestatic picture without significant elevation of the transaminases on the liver function tests makes acute hepatitis unlikely. Antimitochondrial antibodies are elevated in cases of primary biliary cirrhosis (PBC), which may present in a similar fashion. However, PBC is far more common in women than in men, and the average age of onset is the fifth or sixth decade. The lack of an obvious lesion on CT scan does not rule out a source of the cholestasis in the biliary tree. Malignant causes, such as cholangiocarcinoma and tumor of the ampulla of Vater, and non-malignant causes, such as sclerosing cholangitis and Caroli disease, may be detected only by direct visualization with endoscopic retrograde cholangiopancreatography (ERCP). ERCP is useful both diagnostically and therapeutically as stenting procedures may be done to alleviate the obstruction.

VII-105. The answer is E. (*Chap. 339*) Approximately 10–20% of persons with rapid weight reduction achieved through very-low-calorie dieting develop gallstones. An excess of biliary cholesterol in relation to bile acids and phospholipids promotes the formation of unstable, cholesterol-rich vesicles in which cholesterol crystals can form. The most important factor in cholesterol stone formation is increased biliary secretion of cholesterol. This may occur in association with obesity, the metabolic syndrome, high-caloric and cholesterol-rich diets, or drugs (e.g., clofibrate) and may result from increased activity of

hydroxymethylglutaryl-coenzyme A reductase, the rate-limiting enzyme of hepatic cholesterol synthesis, and increased hepatic uptake of cholesterol from blood. Gallbladder hypomotility is an important risk factor for stone formation. If the gallbladder empties all supersaturated or crystal containing bile completely, stones are not able to grow. Nucleation of cholesterol monohydrate crystals is greatly accelerated in human lithogenic bile and is an important factor in stone formation.

- VII-106. The answer is D.** (*Chap. 339*) Cholecystectomy is a very successful operation that provides total or near-total relief of preoperative symptoms in 75–90% of patients. The optimal timing of surgical intervention in patients with acute cholecystitis depends on stabilization of the patient. The clear trend is toward earlier surgery, and this is due in part to requirements for shorter hospital stays. Urgent (emergency) cholecystectomy or cholecystostomy is probably appropriate in most patients in whom a complication of acute cholecystitis such as empyema, emphysematous cholecystitis, or perforation is suspected or confirmed. Patients with uncomplicated acute cholecystitis should undergo early elective laparoscopic cholecystectomy, ideally within 48–72 hours after diagnosis. The complication rate is not increased in patients undergoing early as opposed to delayed (>6 weeks after diagnosis) cholecystectomy. Unsuspected common bile duct stones are left behind in approximately 1–5% of cholecystectomy patients. The incidence of common bile duct stones increases with increasing age of the patient, so that up to 25% of elderly patients may have calculi in the common duct at the time of cholecystectomy.
- VII-107. The answer is C.** (*Chap. 339*) IgG4-associated cholangitis is a recently described biliary disease of unknown etiology that presents with biochemical and cholangiographic features indistinguishable from primary sclerosing cholangitis (PSC), is often associated with autoimmune pancreatitis and other fibrosing conditions, and is characterized by elevated serum IgG4 and infiltration of IgG4-positive plasma cells in bile ducts and liver tissue. All newly diagnosed PSC patients should have a serum IgG4 level checked. In contrast to PSC, IgG4-associated cholangitis is not associated with inflammatory bowel disease and should be suspected if associated with increased serum IgG4 and unexplained pancreatic disease. Viral infections such as HIV can lead to biliary pathology. In patients with AIDS, cholangiopancreatography may demonstrate a broad range of biliary tract changes as well as pancreatic duct obstruction and occasionally pancreatitis. Further, biliary tract lesions in AIDS include infection and cholangiopancreatography changes similar to those of PSC.
- VII-108. The answer is D.** (*Chap. 340*) Ruptured ectopic pregnancy has been associated with a rise in serum amylase but pregnancy in and of itself is not a common cause of elevated amylase. Table VII-108 lists common conditions associated with an elevated amylase level.
- VII-109. The answer is B.** (*Chap. 340*) Endoscopic ultrasound (EUS) and magnetic resonance cholangiopancreatography (MRCP) have replaced endoscopic retrograde cholangiopancreatography (ERCP) for diagnostic purposes. EUS allows one to obtain information about the pancreatic duct as well as the parenchyma and has few procedure-related complications associated with it, in contrast to the 5–10% of post-ERCP pancreatitis observed. MRI and MRCP are now being used to view the bile ducts, pancreatic duct, and the pancreas parenchyma in both acute pancreatitis and chronic pancreatitis. EUS has been studied as a diagnostic modality for chronic pancreatitis. Criteria for abnormalities on EUS in severe chronic pancreatic disease have been developed. There is general agreement that the presence of five or more of the nine criteria listed in Table VII-109 is highly predictive of chronic pancreatitis. Abdominal CT is not necessary to make the diagnosis of acute pancreatitis but can detect pancreatic necrosis. Plane radiographs have no real role in the diagnostic evaluation of pancreatic disease. Abdominal ultrasound can be helpful in the initial workup of abdominal pain and acute pancreatitis, but image quality can be limited by the presence of overlying bowel gas.

TABLE VII-108 Causes of Hyperamylasemia and Hyperamylasuria

Pancreatic Disease	
I. Pancreatitis	
A. Acute	
B. Chronic: ductal obstruction	
C. Complications of pancreatitis	
1. Pancreatic pseudocyst	
2. Ascites caused by pancreatic duct disruption	
3. Pancreatic necrosis	
II. Pancreatic trauma	
III. Pancreatic carcinoma	
Nonpancreatic Disorders	
I. Renal insufficiency	
II. Salivary gland lesions	
A. Mumps	
B. Calculus	
C. Irradiation sialadenitis	
D. Maxillofacial surgery	
III. "Tumor" hyperamylasemia	
A. Carcinoma of the lung, esophagus, breast, or ovary	
IV. Macroamylasemia	
V. Burns	
VI. Diabetic ketoacidosis	
VII. Pregnancy	
VIII. Renal transplantation	
IX. Cerebral trauma	
X. Drugs: opiates	
Other Abdominal Disorders	
I. Biliary tract disease: cholecystitis, choledocholithiasis	
II. Intraabdominal disease	
A. Perforated or penetrating peptic ulcer	
B. Intestinal obstruction or inflammation	
C. Ruptured ectopic pregnancy	
D. Peritonitis	
E. Aortic aneurysm	
F. Postoperative hyperamylasemia	

TABLE VII-109 Endoscopic Ultrasonographic Criteria for Chronic Pancreatitis (Total Criteria = 9)

Ductal	Parenchymal
Stones	Echogenic strands
Hyperechoic main duct margins	Echogenic foci
Main duct irregularity	Lobular contour
Main duct dilatation	Cyst
Visible side branches	

VII-110. The answer is A. (*Chap. 341*) All pancreatic enzymes have pH optima in the alkaline range. The proteolytic pancreatic enzymes are secreted as inactive zymogen precursors. Enterokinase, an enzyme found in the duodenal mucosa, cleaves the lysine-isoleucine bond of trypsinogen to form trypsin. Trypsin then activates the other proteolytic zymogens and phospholipase A2 in a cascade phenomenon. The stimulatory neurotransmitters are acetylcholine and gastrin-releasing peptides. These neurotransmitters activate calcium-dependent secondary messenger systems, resulting in the release of zymogens into the pancreas duct. In contrast to other species, there are no cholecystokinin (CCK) receptors on acinar cells in humans. CCK in physiologic concentrations stimulates pancreatic secretion by stimulating afferent vagal and intrapancreatic nerves.

- VII-111. The answer is A.** (*Chap. 341*) In combination with a consistent clinical story and radiographic findings, serum amylase and lipase values threefold or more above normal virtually clinch the diagnosis of acute pancreatitis. Gut perforation, ischemia, and infarction should be excluded. Serum lipase is the preferred test and has a higher specificity than serum amylase. There is no correlation between the severity of pancreatitis and the degree of serum lipase and amylase elevations. After 3–7 days, even with continuing evidence of pancreatitis, total serum amylase values tend to return toward normal. However, pancreatic isoamylase and lipase levels may remain elevated for 7–14 days. Elevation of serum amylase is not specific for acute pancreatitis; notably, patients with metabolic acidemia (e.g., diabetic ketoacidosis) may have spurious elevation of serum amylase without pancreatitis. Hypocalcemia occurs in approximately 25% of cases of acute pancreatitis, whereas hypercalcemia is not a feature.
- VII-112. The answer is A.** (*Chap. 341*) The most common cause of acute pancreatitis in the United States is gallstones causing common bile duct obstruction. Although bile duct obstruction may be demonstrated on technetium hepatobiliary iminodiacetic acid scan, right upper quadrant ultrasound is preferred for ease, demonstration of gallstones in the gallbladder, and demonstration of obstructed bile duct. Alcohol is the second most common cause of acute pancreatitis, followed by complications of endoscopic retrograde cholangiopancreatography. Hypertriglyceridemia accounts for 1–4% of cases of acute pancreatitis with triglyceride levels usually >1000 mg/dL. Other potential common causes include trauma, surgery, drugs such as valproic acid, anti-HIV medications, estrogens, and sphincter of Oddi dysfunction. Additionally, a number of rare causes have been described. The most judicious first step in evaluation of acute pancreatitis is to test for gallstones and pursue more rare causes after the most common cause has been ruled out.
- VII-113. The answer is B.** (*Chap. 341*) Blue-red-purple or green-brown discoloration of the flanks (Turner sign) reflects tissue catabolism of hemoglobin from severe necrotizing pancreatitis with hemorrhage. A faint blue discoloration around the umbilicus (Cullen sign) may also occur as the result of hemoperitoneum. While a ruptured ectopic pregnancy could present with hemoperitoneum, the patient's age and other clinical history makes alcoholic pancreatitis a much more likely diagnosis. Pancreatic calcification on abdominal plain films would suggest possible chronic pancreatitis. Endoscopic ultrasound plus magnetic resonance cholangiopancreatography have become common in the diagnosis of pancreatic disease but would probably not be indicated at this time given that the patient's history and physical examination indicated likely severe necrotizing pancreatitis from alcohol abuse.
- VII-114. The answer is B.** (*Chap. 341*) Any severe acute pain in the abdomen or back should suggest the possibility of acute pancreatitis. The diagnosis is established by two of the following three criteria: (1) typical abdominal pain in the epigastrium that may radiate to the back, (2) threefold or greater elevation in serum lipase and/or amylase, and (3) confirmatory findings of acute pancreatitis on cross-sectional abdominal imaging. Many patients with acute pancreatitis have concomitant hypoxemia and may even develop acute respiratory distress syndrome, but hypoxemia is not a diagnostic criterion for acute pancreatitis.
- VII-115. The answer is A.** (*Chap. 341*) An elevated hematocrit at presentation in patients with acute pancreatitis is associated with a worse outcome in patients as it reflects significant hemoconcentration from vascular leakage and third spacing. A failure of the hematocrit to drop after the first 12–24 hours of fluid resuscitation is also a poor prognostic sign and could indicate the need for more intensive monitoring in the intensive care unit. This patient's drop in hematocrit most likely represents a favorable response to initial fluid resuscitation.

- VII-116. The answer is A.** (*Chap. 341*) The initial phase of acute pancreatitis is characterized by intrapancreatic digestive enzyme activation and acinar cell injury. Trypsin activation appears to be mediated by lysosomal hydrolases such as cathepsin B that become colocalized with digestive enzymes in intracellular organelles; it is currently believed that acinar cell injury is the consequence of trypsin activation. The second phase of pancreatitis involves the activation, chemoattraction, and sequestration of leukocytes and macrophages in the pancreas, resulting in an enhanced intrapancreatic inflammatory reaction. Neutrophil depletion induced by prior administration of an antineutrophil serum has been shown to reduce the severity of experimentally induced pancreatitis. There is also evidence to support the concept that neutrophils can activate trypsinogen. Thus, intrapancreatic acinar cell activation of trypsinogen could be a two-step process (i.e., an early neutrophil-independent and a later neutrophil-dependent phase). The third phase of pancreatitis is due to the effects of activated proteolytic enzymes and cytokines, released by the inflamed pancreas, on distant organs. Activated proteolytic enzymes, especially trypsin, not only digest pancreatic and peripancreatic tissues but also activate other enzymes such as elastase and phospholipase A₂. The active enzymes and cytokines then digest cellular membranes and cause proteolysis, edema, interstitial hemorrhage, vascular damage, coagulation necrosis, fat necrosis, and parenchymal cell necrosis. Cellular injury and death result in the liberation of bradykinin peptides, vasoactive substances, and histamine that can produce vasodilation, increased vascular permeability, and edema with profound effects on many organs. The systemic inflammatory response syndrome and acute respiratory distress syndrome, as well as multiorgan failure, may occur as a result of this cascade of local and distant effects.
- VII-117. The answer is C.** (*Chap. 341*) A low-fat solid diet can be administered to subjects with mild acute pancreatitis after the abdominal pain has resolved. Persistent inflammatory changes in the pancreas may remain for weeks to months after an episode of acute pancreatitis. Similarly, there may be prolonged elevation of amylase and lipase. In this regard, persistent changes on CT or persistent pancreatic enzyme elevation should not discourage clinicians from feeding hungry patients with acute pancreatitis. Although there had been prior concern that feeding patients with pancreatitis may exacerbate pancreatic inflammation, this has not been demonstrated. Enteral feeding maintains gut barrier integrity, limits bacterial translocation, is less expensive, and has fewer complications than total parenteral nutrition. The choice of gastric versus nasojejunal enteral feeding is currently under investigation.
- VII-118. The answer is D.** (*Chap. 341*) Chronic pancreatitis is a common disorder in any patient population with relapsing acute pancreatitis, especially patients with alcohol dependence, pancreas divisum, and cystic fibrosis. The disorder is notable for both endocrine and exocrine dysfunction of the pancreas. Often, diabetes ensues as a result of loss of islet cell function; although insulin-dependent, it is generally not as prone to diabetic ketoacidosis or coma as are other forms of diabetes mellitus. Because pancreatic enzymes are essential to fat digestion, their absence leads to fat malabsorption and steatorrhea. In addition, the fat-soluble vitamins (A, D, E, and K) are not absorbed. Vitamin A deficiency can lead to neuropathy. Vitamin B₁₂, or cobalamin, is often deficient. This deficiency is hypothesized to be due to excessive binding of cobalamin by cobalamin-binding proteins other than intrinsic factors that are normally digested by pancreatic enzymes. Replacement of pancreatic enzymes orally with meals will correct the vitamin deficiencies and steatorrhea. The incidence of pancreatic adenocarcinoma is increased in patients with chronic pancreatitis, with a 20-year cumulative incidence of 4%. Chronic abdominal pain is nearly ubiquitous in this disorder, and narcotic dependence is common. Niacin is a water-soluble vitamin, and absorption is not affected by pancreatic exocrine dysfunction.